Life Nieblas

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Technical Manual

Technical details to replicate the different reforestation and fog collection methodologies tested at Life Nieblas

Editor:
Life Nieblas' partners

Edition Coordination:
Saúl Oliva Cabrera - solicab@gesplan.es
Gustavo Viera Ruiz - gvierui@gesplan.es

Life

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of the European Union

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This document presents a comprehensive summary of the actions, strategies, challenges, and solutions undertaken by the Life Nieblas project team between 2020 and 2024.

The Technical Manual has been developed as a practical and informative guide, providing both general descriptions and step-by-step instructions for the implementation of projects inspired by Life Nieblas. It is intended to support individuals, institutions, and organisations interested in applying fog water collection systems and environmental restoration techniques in similar territorial and climatic contexts.

For those seeking detailed methodologies, technical specifications, or scientific data, we encourage you to explore the complete set of technical reports available on our official website: www.lifenieblas.com/documents

All documentation can be freely accessed and downloaded, offering in-depth resources to guide replication, adaptation, and further innovation in sustainable water management and climate change mitigation.

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Project team:
- CIM VDL (Comuniade Intermunicipal Viseu Dão Lafões), Portugal
- CGC (Cabildo Insular de Gran Canaria), Spain
- CREAF (Centro de Investigación Ecológica y Aplicaciones Forestales), Spain
- ICIA (Instituto Canario de Investigaciones Agrarias (Canarian Crop Research Institute), Spain
- ULL (Universidad de La Laguna), Spain
- Heredad (Heredad de Aguas de Arucas y Firgas), Spain
- ITC (Instituto Tecnológico de Canarias, S.A.), Spain
- UAB (Universidad Autónoma de Barcelona), Spain

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Postal address: C/León y Castillo, 54, bajo, 35003, Las Palmas de Gran Canaria, Spain
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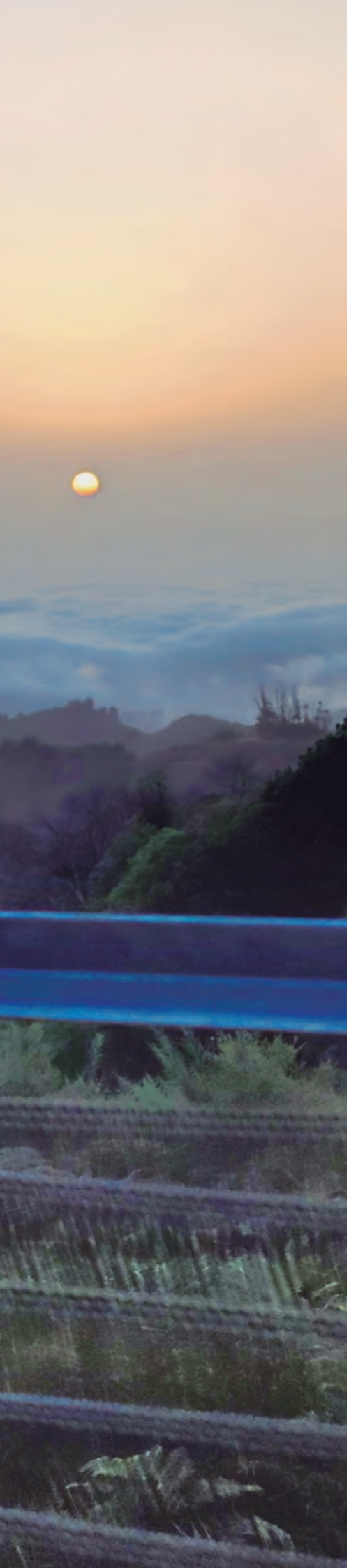

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Presentation

Life Nieblas project, kicks-off in July 2020, taking place in the Canary Islands and Portugal. It covers a total area of 45 hectares, with 35 hectares located in Barranco de la Virgen, Gran Canaria, and 10 hectares between Vouzela and Carregal do Sal, Portugal.

To effectively navigate and make use of this guide, it is important to familiarise yourself with the project's objectives, study methodologies, and results. The guide provides essential information on irrigation systems, fog water collection methods, and reforestation techniques. By following our guide, you can access valuable insights, facilitate the smooth progress of your project, and minimise losses.

Life Nieblas budget is 2.185.777€, with 55% financed by the European Commission's LIFE funds and 33% by Cabildo de Gran Canaria. The project involves various multidisciplinary partners from the Canary Islands, Portugal, and Catalonia, who are responsible for conducting the studies and methodologies described in this manual.

The main objectives of the project include testing different irrigation systems, reforestation and fog water collection methods. The final objective is to adapt these technologies to be exported to different European and non-european regions where fog conditions have the potential to improve water availability and assist in natural habitat recovery. Eventually, finding transferability into the agriculture sector where farmers can take advantage from collecting their own water. Life Nieblas will contribute to climate change mitigation (by reduction of emissions), carbon capture (by natural habitat recovered) and with the development of sustainable technology.

Barranco de la Virgen area in Gran Canaria, which is partially within the Special Area of Conservation (SAC) ES7010038 and the Natura 2000 priority habitat 4050 'Macaronesian heath', the landscape has undergone changes following a major wildfire in 2019. The area now consists of endemic herbaceous and low shrub species, as well as introduced species inherited from historical agricultural activities.

Monteverde forests play a crucial role in providing ecosystem services in the Macaronesian Islands. Their natural distribution is found in the cloud belt influenced by trade winds in the Canary Islands. These forests have the ability to capture fog water recharging aquifers and valley basins contributing to sustaining thousands of endemic species. Restoring the Monteverde would also provide socio-economic benefits, particularly in areas such as archipelagos where potable water resources are limited.

Life Nieblas' reforestation objectives in Gran Canaria are focused in planting up to 13 species within the Monteverde stratification (600-1200m amsl), meeting with the pine forest at higher elevations. El Pinillo area (Barranco de la Virgen, Gran Canaria) aims to plant 20,000 trees using four proposed reforestation methods (traditional, Cocoon, Individual Fog Water Collectors and Autonomous Discharge System).

In Portugal, the project includes two intervention sites within the territory of Comunidade Intermunicipal (CIM) de Viseu Dão Lafões. Vouzela, located within the Vouga-Caramulo Natural Park, and Carregal Do Sal, designated as a Site of Community Importance (SCI) and SAC, have both been affected by forest fires. The extensive presence of eucalyptus (*Eucalyptus globules*) and black acacia (*Acacia dealbata*) plantations, poses a challenge to the growth of endemic species. Restoring oak woodlands in the northwest Iberian Peninsula would provide crucial ecosystem services, preventing soil erosion, improving soil quality, and supporting local populations and the environment.

Reforestations in Portugal include 5,000 trees of three Quercus species. In Vouzela and Carregal do Sal, two of the three innovative reforestation methodologies (Cocoon and individual collectors) are being tested; the Autonomous Discharge System has not been implemented due to technical and logistics reasons.

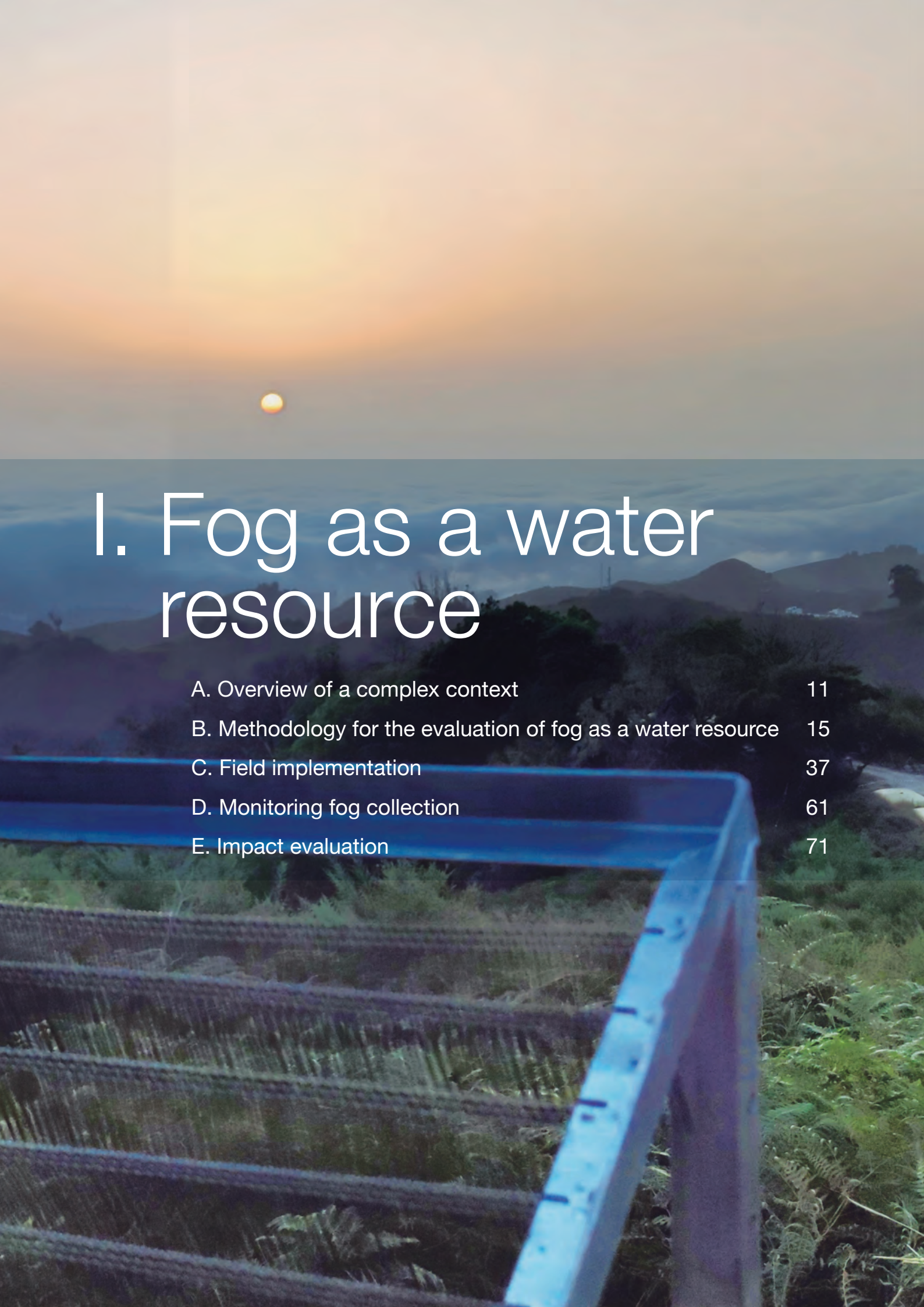
Introduction

Climate change is a major challenge facing our planet, with undeniable consequences for water availability. Reduced rainfall and increased periods of drought are having a significant impact on many regions, compromising the water resources essential to human life, agriculture and the environment.

In this technical manual, we look at the challenges posed by the decline in water availability due to climate change and droughts. We will present **two solutions** that can help tackle these issues in a sustainable way.

- **WATER ACCESS:** The first solution concerns regions where access to water is extremely difficult or even impossible. In such places, we will be presenting the use of fog collectors, an emerging technology that collects and uses atmospheric moisture. This ingenious approach offers a valuable alternative to traditional water sources, harnessing the natural resources available to meet water needs.
- **HABITAT RESTORATION**: The second solution is aimed at deteriorated areas where water soil infiltration is reduced, leading to increased runoffs, erosion related issues and loss of biodiversity. To remedy this situation, we propose reforestation actions. By restoring plant cover, we encourage the natural water cycle and reduce the risk of erosion, allowing nature to regain its balance and water to play its essential role in ecosystems.

This technical manual aims to present these two solutions in detail, providing practical information, application examples and recommendations for their implementation. By adopting these innovative approaches, we contribute to the preservation and sustainable management of water resources, adapting to the challenges posed by climate change and promoting a more resilient future for our planet.



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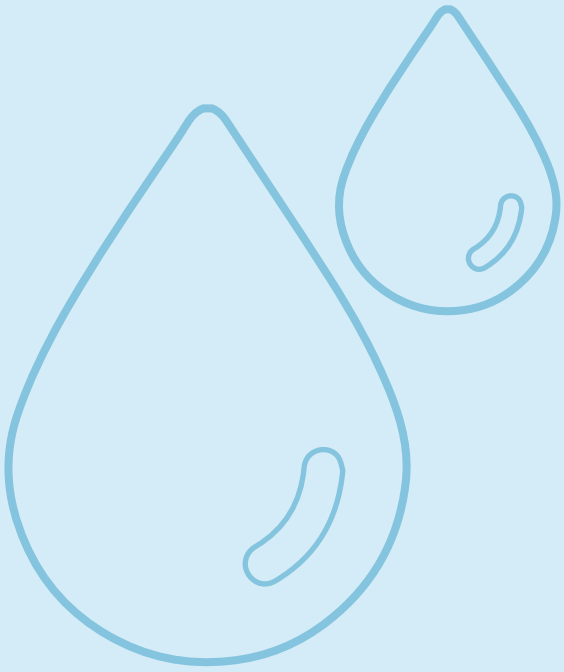
**Fog water collection is not a new technology,
it is past wisdom brought to solve modern issues.**



"Water is the driving force of all nature"

Leonardo da Vinci

A. OVERVIEW OF A COMPLEX CONTEXT

1. FOG: definition and creation process

This part of the technical report focuses on the fundamental concept of fog, exploring the mechanisms by which it is formed, the key factors that contribute to its presence and presenting the associated basic concepts.

Basics concept

Fog is a meteorological phenomenon characterised by the presence of fine water droplets suspended in the air, typically ranging from 1 micrometre to 40 micrometres in size which reduce ground visibility. These droplets are similar in size to cloud droplets, and in fact, fog can be regarded as a cloud with its base at ground level.

Fog forms when relatively calm and moist air encounters specific conditions that promote the condensation of water vapour into tiny liquid particles. The moisture necessary for fog formation can originate from various sources, such as water evaporation from oceans, lakes, rivers, or wet surfaces.

Formation

The formation of fog relies on several factors:

- **Relative humidity:** it's a crucial factor. When the air becomes saturated with moisture, meaning it contains as much water vapour as it can hold at a given temperature. Any further increase in humidity or cooling of the air can trigger fog formation.
- **Wind:** wind also plays a significant role as it determines the dispersion of fog droplets and their movement through the environment. Wind speed is a critical factor in fog collection. The more air passes through a natural or artificial fog collector, the more droplets can be collected.
- **Dew point and saturation:** at a given temperature and pressure, the atmosphere can hold a limited amount of water vapour. When the water content reaches its limit, saturation occurs, and the air temperature is said to reach its dew point. Condensation takes place within a mass of saturated air if there is an increase in vapour tension and/or a decrease in temperature. Under these conditions, water vapour condenses onto surfaces, forming dew.
- **Condensation nuclei:** condensation cannot occur away from the surface unless there are condensation nuclei in the air, which make it possible for drops of water to form. In the absence of sufficient condensation nuclei, the atmosphere becomes slightly supersaturated. These nuclei include particles derived from the soil, vegetation, and ocean, which mix with water droplets and determine the chemical and physical composition of the water.

Fog ascending into Barranco de La Virgen in Valleseco, Gran Canaria.

Type of fog

- **Radiation fog:** it occurs during winter and autumn in temperate latitudes and is associated with the radiative cooling of the lower atmospheric layers during clear and cool nights. Radiation fog forms when there is a mass of warm, humid air over a surface cooled by heat release from the ground during night-time in a clear and relatively calm (wind-free) atmosphere. During calm periods with low liquid water content, radiation fog is less interesting for fog collection.
- **Advection fog:** there are two primary ways in which advection fog forms. Advection refers to the movement of air, and both processes involve air motion. First, the cooling of surface air occurs when warm, humid air moves over a cold surface. If the air is cooled to the dew point, fog is produced. The other type of advection fog, known as high-elevation fog, is formed when winds blow clouds over mountains or hills. When the cloud reaches the ground, it becomes fog, persisting as long as the cloud is forced over the terrain. Both types of advection fog are associated with horizontal air movements, making them suitable for fog collection.
- **Orographic fog:** it forms when warm, humid air ascends a mountain slope, expanding and cooling as it rises. If the air is sufficiently humid, fog will form at the surface. From a distance, this appears as a cap cloud covering the mountain summit.

There are situations where different types of fog are produced at the same place and moment.

«When coastal fog is associated with high pressure systems, a thermal inversion, cold oceanic currents and marine upwelling, there is often a combination of type of fog. In such cases, advection fog tends to predominate, originating as a large stratus cloud deck generated over the ocean. This cloud deck, upon reaching the continent, is intercepted by the terrain. This fog is reinforced by the orographic effect of coastal mountains ranges».

2. Background information on fog collection

The history of fog collectors will be presented here and a brief presentation of the various scientific research studies that have already been carried out will quickly explain the effects of wind and rain on collectors. The problems analysed during their use will also be presented in this section.

Investigation

Plant-based fog collection has been used by various plant species for millions of years. The growth and promotion of endemic and indigenous plants play a crucial role in restoring and enhancing natural water collection mechanisms. One notable example is the foliage of mature trees, which acts as a natural fog collector. The intricate structure of tree branches effectively intercepts fog, allowing the atmospheric water droplets to naturally infiltrate the soil. [Ritter A., (2005)].

Traditional farming practices in regions such as Cape Verde Island, Oman, and the Canary Islands have long recognized the potential of fog water collection. In these areas, farmers have traditionally placed cisterns under specific plant species like agaves, pine trees, olive trees, laurels, and juniper trees to collect water from fog. This practical approach demonstrates the utilisation of natural fog capture mechanisms to secure water resources.

Fog droplets collected by a *Pinus canariensis* (Canarian pine tree) needles.

Chilean Fog Water Collector design

Reproducing this natural phenomenon of fog collection has become possible through various techniques and innovations, with the ultimate goal of providing fresh water to remote and water-scarce regions. The concept of capturing fog using nets or mesh has been implemented worldwide. Fog collectors are vertical surfaces that are installed facing the wind. One country that has made significant advancements in fog collection is Chile. In Chile, fog collectors utilise a specific type of fine-mesh net called "Raschel" mesh to effectively capture water droplets suspended in fog. The fog droplets, collected on the mesh, gravitate downwards and flow into storage containers through the force of gravity. This method allows for the collection of significant amounts of water from foggy environments.

Initially, fog collectors resembled ship sails, with structures supported by two side pillars that offered resistance to wind and allowed for effective fog capture. However, this design proved to be unstable and vulnerable to strong winds, requiring further improvements and modifications to enhance the durability and efficiency of fog collection systems.

GLODA, A.; GUASP, R. Espejo; BALADÓN, A. Acosta. Fog collectors in tropical areas. En Proceedings of the symposium on precipitation and evaporation. 1993. p. 20-4.

CERTINI, Giacomo, et al. Fog collection as a strategy to sequester carbon in drylands. Science of the Total Environment, 2019, vol. 657, p. 391-400.

FWC in THE Canary Islands

In the Canary Islands, ongoing research has being conducted since 1996 to study fog water collection at Teno Rural Park, located in Tenerife Island. The primary objective of these studies is to evaluate the feasibility of obtaining water for various local uses, including domestic, agricultural, livestock, and forestry purposes, through the implementation of fog collectors (FCs). These fog collectors employ the accumulation of water on tensile nets, typically made of materials like polypropylene. As the fog droplets accumulate on the nets, they form water droplets that descend and collect in a gutter system at the lower part of the fog collector structure. (JAÉN, M. V., 2002)

Multiple models of fog collectors have been developed, each tailored to the specific wind and fog conditions in the area of operation. These models exhibit diverse characteristics and efficiency in capturing fog water, allowing for more targeted and optimised fog collection.

The water collected from fog can serve various purposes, including the irrigation of plants during their initial year of development after being planted. Fog water collection projects offer promising solutions to water scarcity in arid and remote regions, providing a sustainable and alternative water source for communities and ecosystems that rely on foggy environments.

B. METHODOLOGY FOR THE EVALUATION OF FOG AS A WATER RESOURCE

Before undertaking a fog collection project, it is essential to assess its feasibility in order to ensure its success. This thorough assessment is based on a number of key criteria, including the meteorological context of the site, the availability of the necessary equipment and the feasibility of the project in terms of time and cost.

This methodical guide in fog collection aims to provide a structured approach to assessing the feasibility of such a project. By following the steps and using the appropriate tools, it will be possible to determine the technical, material and economic viability of a fog collection project, thus contributing to informed decision-making and the success of the project.

1. Evaluation of a potential fog collection site

The first step in this assessment is to analyse the meteorological context of the site being considered for the fog collection project. It is essential to determine whether the site has the right conditions for fog formation, as this will determine the site's potential ability to generate water using this method. Specific climatic data, such as relative humidity, ambient temperature, prevalent wind direction & speed, visibility and fog patterns, must be collected and analysed to assess the viability of the project in a certain location.

Field analysis

First, an evaluation on the area potential for water collection as a hydrological and environmental resource is needed. The following are the necessary analysis:

1. The study of **vegetation** can be done with aerial pictures. Different scales can be used. In the preliminary study, a scale of 1:50.000 may be suitable, since the vegetation cover can be identified. But in more detailed study, a scale 1:25.000 should be used.
2. **Köppen Classification**: The Köppen climate classification divides climates into five main climate groups, with each group being divided based on patterns of seasonal precipitation and temperature. These main climate groups are based on the types of vegetation occurring in a given climate classification region. In addition to identifying climates, the system can be used to analyse ecosystem conditions and identify the main types of vegetation within climates.

Köpen climate classification scheme symbols description table

1st	2nd	3rd
A (Tropical)	f (Rainforest) m (Monsoon) w (Savanna, dry winter) s (Savanna, dry summer)	
B (Dry)	W (Arid Desert) S (Semi-Arid or steppe)	h (Hot) k (Cold)
C (Temperate)	w (Dry winter) f (No dry season) s (Dry summer)	a (Hot summer) b (Warm summer) c (Cold summer)
D (Continental)	w (Dry winter) f (No dry season) s (Dry summer)	a (Hot summer) b (Warm summer) c (Cold summer) d (Very cold winter)
E (Polar)		T (Tundra) F (Ice cap)

3. **Climate analysis**: using meteorological data and previous studies in the region, you need to determine the various characteristics of the selected site, such as : Highest average monthly maximum temperature (ºC) / Lowest average monthly minimum temperature (ºC) / Dew point (ºC) Total annual precipitation (mm/y) / Rainy days per year/ Dry period (months)/ Potential Evapotranspiration (mm/y) / Rainfall and temperature changes per month/ predominant wind current, direction and speed/ Cloud base (ft/m) and thickness (ft/m).

Thanks to all this information, you can analyse the field and obtain some conclusions about sources of water, quantity, probability of horizontal precipitation and what the dry period may look like. It allows you to have a 'big-picture' idea about the different requirements at your location (e.g. field work and intervention, reforestation strategy, water needs, etc.).

Fog capacity

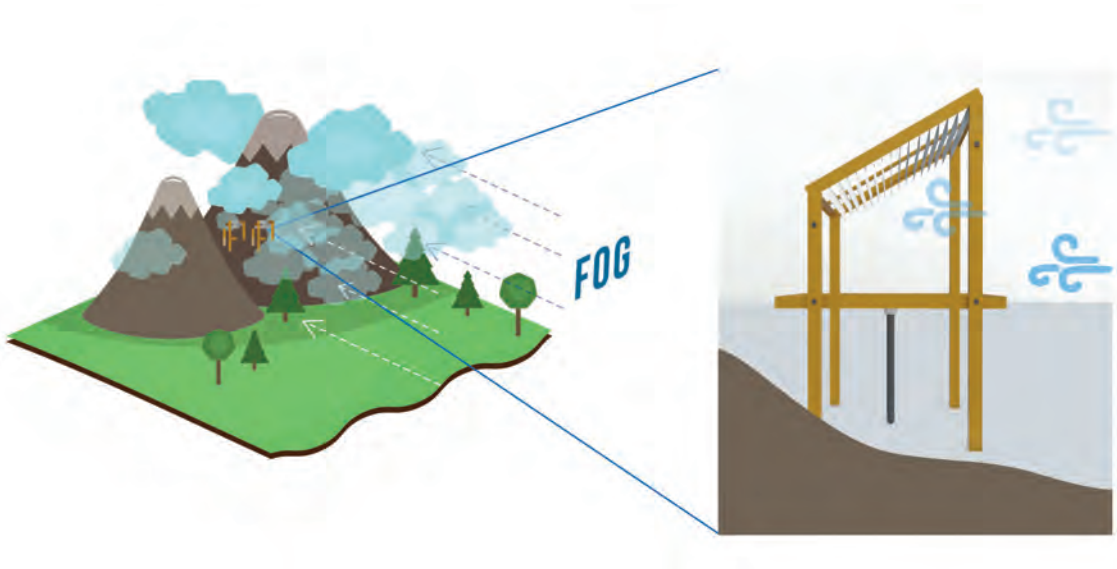
The first step is to analyse the preliminary condition for the presence of fog. Use a map and factors such as location of the site in the island/continent (coast or inland), the presence of anticyclone and cold ocean currents, mountains and wind direction that may influence in fog presence (Schemenauer R., Cereceda P. y Osses P. (2017)).

- **Presence and duration of fog**:
  - **Frequency of fog.** Is fog a regular occurrence in the region? On average, how many days a year is it likely to be foggy? Are some months of the year more affected than others? Are there months when there is no fog?
  - **Visibility during a fog episode.** How good is visibility during a fog episode? Is visibility considerably reduced? When visibility falls below 1000m fog is said to be present.
  - **Presence of condensation.** Does hair or clothing become wet during a fog episode? This may indicate the presence of water droplets suspended in the air.
  - **Presence of water droplets.** Can you see water droplets falling from vegetation during a fog episode? This can be a sign of accumulation and the presence of fog.

- **Presence of mountains**. This is vital, since advection and orographic fog are often found at high elevations. At low elevations fog is normally not intercepted since fog will simply pass over it. The amount of water varies from one place to another. A satisfactory altitude to work at is two-thirds of the cloud/fog thickness, measuring from the base. This portion will normally have the greatest liquid water content.
- **Wind**. The wind factor must be taken into account. Here are some aspects to consider:
  - **Average wind speed.** It is important to know the average wind speed in the area. Stronger winds can favour the movement of fog and affect its presence and concentration.
  - **Prevailing wind direction.** It is also important to determine the prevailing wind direction in the area. When the longitudinal axis of the mountains is perpendicular to the direction of the prevailing winds bringing fog, chances of collecting water increase significantly.
- **Mountain orientation**. Using a topographic map, the orientation of mountains can be determined. When the longitudinal axis is perpendicular to the direction of the prevailing wind bringing fog, fog water collection potential increases. To do this, simply join the highest peaks to define a longitudinal axis. In coastal regions, winds from the sea are generally the most favourable for collecting fog. On the map, the axis should be indicated by a line and the prevailing wind by an arrow. Places where the axis and the wind are perpendicular are the best.

Once all this information has been collected, it is possible to draw an initial conclusion about the region's capacity for fog and its potential for collecting water. A mountain location, regularly exposed to the prevailing wind and with a high frequency of fog, offers better possibilities for fog water collection.

The amount of fog water collected depends on **fog liquid water content** (i.e.**fog droplet size and number)**, **wind speed**, collector's **efficiency** (i.e. capacity to collect water per m^2) and **area** of the site (Regalado y Ritter, 2017; 2019). It should be emphasised that wind plays a major role in determining the yield from fog collectors. Within limits, higher wind speed means higher yields of water. An ideal range of speeds would be from 7 to 36 km/h. It also strongly influences how much rain can be collected by a rainfall catchment system. It needs a vertical component and a proper orientation to achieve the maximum yield. For this reason, fog water collectors are also excellent rain and drizzle collectors.



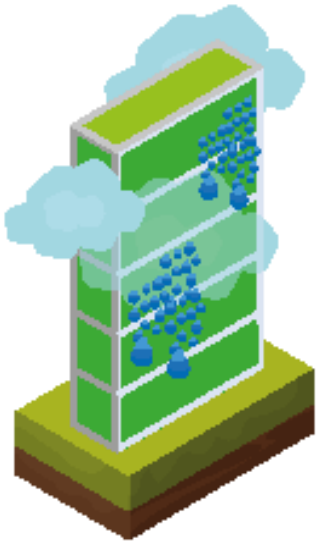
2. Materials and technology presentation

Before embarking on a fog collection project, it is crucial to thoroughly prepare and consider various factors that will contribute to its success. One important aspect to take into account is the selection of appropriate equipment for fog water collection, as it plays a significant role in the success of the project. This includes evaluating the availability of fog collection technologies, assessing their suitability for the specific project needs, and considering storage and distribution systems for the collected water. A comprehensive analysis of these elements is essential in determining whether the project can be effectively implemented with the available material resources.

When planning a fog collection project, several factors must be considered to choose the most suitable equipment. Each type of equipment has its own characteristics, costs, implementation timelines, and maintenance requirements. Therefore, careful consideration is necessary to determine which equipment best fits your project. Taking this into account maximises the chances of project success.

Traditional Fog collector

To overcome the problems encountered on the Chilean 2D collectors, a private company has designed a model with a three-dimensional structure with deep foundations and the ability to withstand winds of up to 100 km/h. We call them 'traditional fog collectors'.

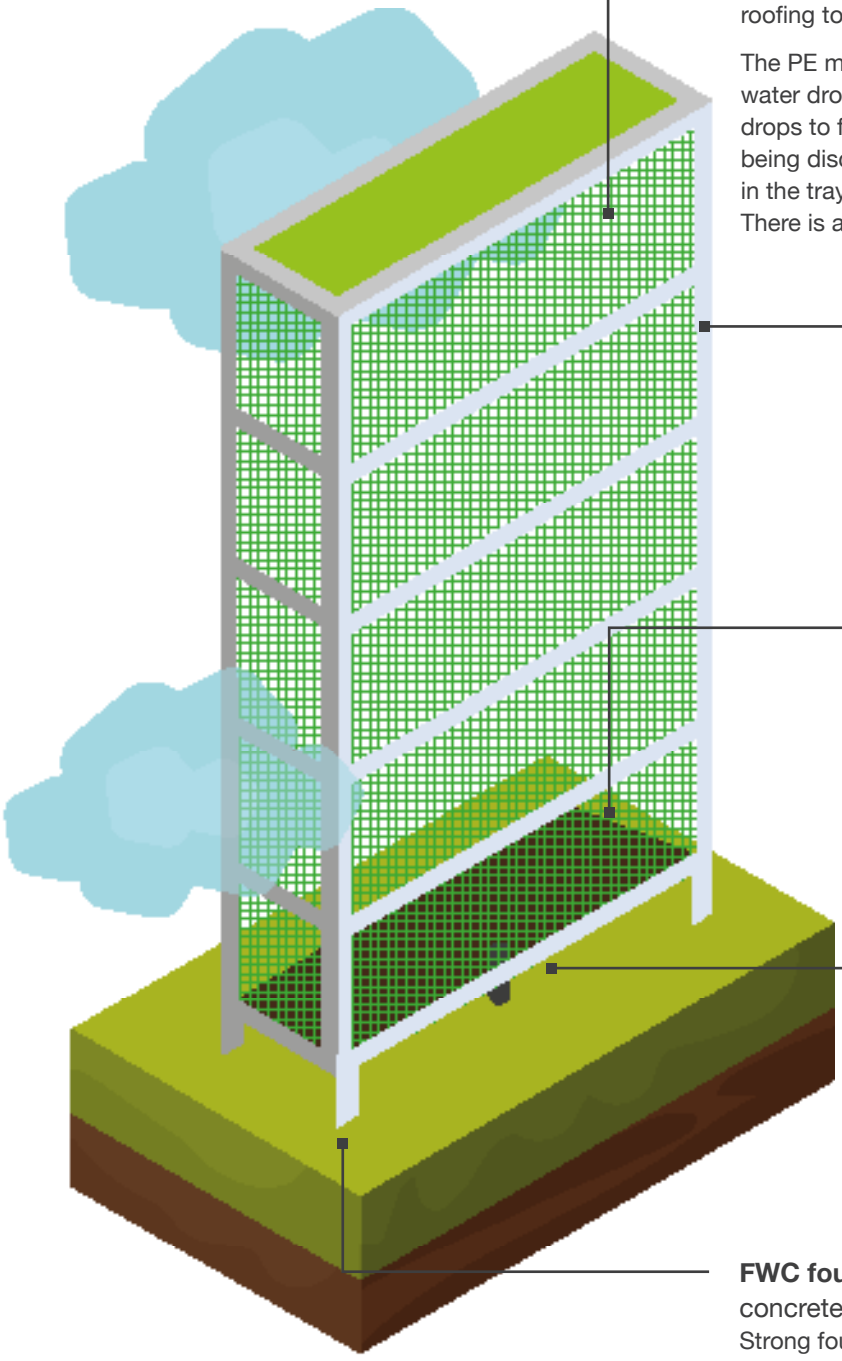


This traditional fog collector is a rectangular structure, with different layers of PE mesh. This structure measures 4x2x1 m and weights 400 kg excluding counting foundation which is usually about 2.5 m^3 of concrete. The materials used for the structure are steel and iron wrapped with 42 m^2 of 3x3 mm PE mesh. However, fog collectors' useful impact surface area is considered to be only 8m^2 which is the predominant fog collection side (i.e. front side of the collector).

Design of the FWCs and specifications were developed by the company. The different structure assembling parts were made by the company and then transported to the site by them. The figure below, shows a brief summary of FWC's main parts.

Traditional Fog collector

Structure



Anti UV monofilament mesh screen. 44,7 m² total
Made from high density polyethylene plastic (HDPE) and specially treated against ultraviolet (UV) rays. This mesh is typically used in greenhouses roofing to prevent pests from accessing the crops.

The PE mesh presents an obstacle against which water droplets will impact, causing the larger water drops to form within the interlaced fabric eventually being discharged by gravity to be collected in the tray and driven to the storage facilities. There is a total of 42m² of mesh per collector.

Galvanised steel structure. Capable of resisting 100 km/h winds.
Is the skeletal metallic structure that carries the mesh. It is 4m high, 2m wide and 0.8m deep.

Water collection tray. Galvanised steel.
This is the container where water droplets are collected and transferred to the pipeline network and conducted to the general storage tanks.

Drainage system. This system transfers and distributes water collected, transporting it to a storage tank or directly to its point of use. A detailed explanation of the drain system is given in section C, Water Network.

FWC foundation. Underground concrete foundation.
Strong foundations ensure that the FWC is firmly anchored underground, preventing it from falling when the structure suffers tension stress generated by the wind.

Diagram of a traditional fog collector at the end-stage of its installation. They are 4m in height, 2m in length and 0.8m depth.

Site consideration

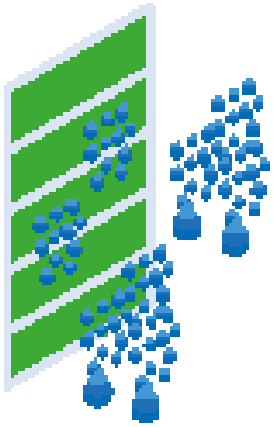
The area around the collector must be relatively clear to allow the free flow of wind current. Obstacles should be avoided. An obstacle could be either natural or manmade, and higher than one metre above ground level. This includes trees, buildings, walls, and soil and/or rock formations.

The presence of wind is essential; therefore the area selected must offer moderate wind speed variations.

Fog collection capacity

The impact surface of a collector is considered to be the side perpendicular to the predominant wind current which is the surface collecting the most amount of water. Even if there are several layers of mesh, only the foremost layer is considered to be the impact surface.

- Surface impact for one FWC : 8m^2
- After 3 years of water collection monitoring, the project has concluded that, on average, a FWC has a water collection capacity of 292L/m^2/year.



Number of FWC

When determining the number of fog collectors (FWCs) required for a project, it is important to consider the fog collection capacity of the specific site. There is no universal rule regarding the exact number of collectors to install, as this depends on a number of variable factors such as weather conditions, site characteristics, wind direction and strength, as well as climatic variations from one year to the next.

292L per square metre of impact surface can serve as a useful reference for estimating fog collection capacity. However, it is important to note that this average value can vary considerably depending on the factors mentioned above. Here is the performance comparison with 2 projects replicating the collection:

Water collected compared from 16th Sep 2022 - 24th Oct 2023

Osorio - 5 FWC - 968amsl	El Pinillo - 11 FWC - 1200amsl	Fuente Fría - 6 FWC - 1600amsl
(10700 L) - 267 L/m^2	(27128 L) - 308 L/m^2	(46000 L) - 958 L/m^2

To start determining the number of collectors needed, it is essential to calculate the quantity of water required for the project. This estimate will depend on the specific needs of the project, such as water demand for irrigation, domestic consumption, or other planned uses. Once you have a general idea of the amount of water required, the results obtained by the Life Nieblas project can be used as an initial reference point.

Ideally, if this has never been done on the site or in the region, it is recommended that pilot studies or preliminary assessments be carried out on the site to specifically measure fog collection capacity. These studies provide accurate data on the amount of water collected over a given period of time, which can be used to refine the calculation of the number of collectors required in the project.

End-of-life

- **Mesh.** If the mesh is intact you can keep it on the structure to collect further. However, PE could degrade due to heat and/or tension stress by wind and may require replacement. On the other hand in a relocation scenario, this mesh could be easily removed and relocated.
- **Structure.** Similarly to the mesh, the structure can be completely removed and re-used for another project. This material does not degrade easily so it will last for many years.
- **Foundation.** Foundations on the other hand are fixed underground anchors and cannot be relocated. However if collectors were to be kept at their original location, those foundations will last forever. Contrary, relocation is impracticable.

Costs

The cost of a traditional FWC depends on the materials used and the contractors that build them. The costs for Life Nieblas were 4670 Euros per collector.

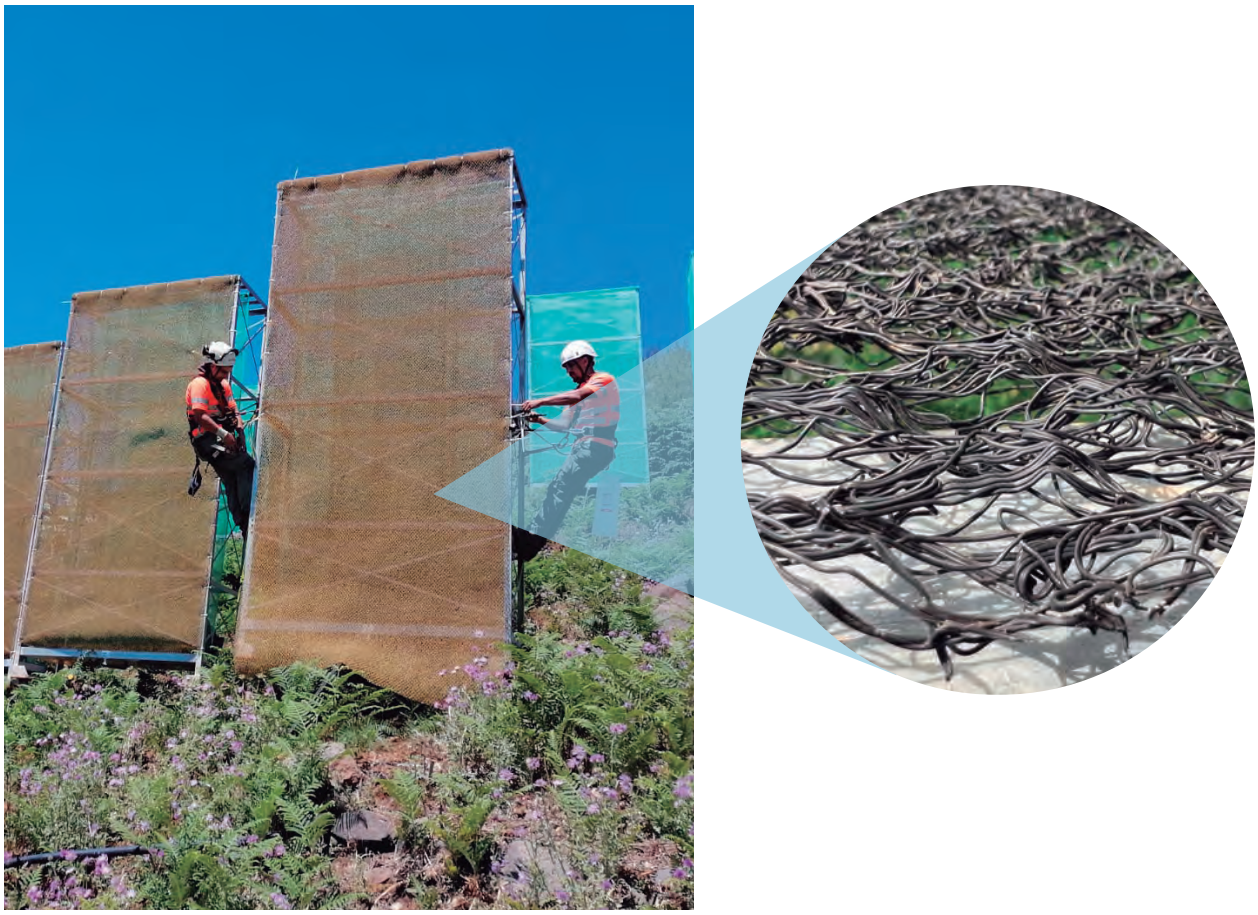
Traditional FWC Performance in Life Nieblas	
Number of collectors:	9 FWC
Size:	8 m^2 of impact surface
Litres collected per m^2:	292 L/m^2/year
Weight:	400 kg
Carbon footprint:	1953 kg CO_2 eq.
Price:	4670 euros

Volumetric- FWC

A variant introduced to the FWC design after technical advice was the use of a volumetric mesh commonly used in construction works to prevent slope erosion. This material was found to be more efficient at capturing fog droplets than the PE mesh originally used in FWCs. Such new collector prototypes have been defined as volumetric fog water collectors (V-FWC).

Structure

The structure is identical to the traditional FWC except for the mesh. The mesh is only installed in the front face of the collector, and the following layer is made up of 2D PE mesh. In these prototypes, the whole FWC has been stripped down from their wrapping PE layers. Only one middle layer of PE mesh remains behind the volumetric mesh.



Fog collection capacity

The impact surface of a collector is considered to be the side perpendicular to the predominant wind current which is the surface collecting the most amount of water. Even if there are several layers of mesh, only the foremost layer is considered to be the impact surface.

- Surface impact for one Volumetric FWC : 8m^2.
- After 3 years of water collection monitoring, the project has concluded that, on average, a V-FWC has a water collection capacity of: 470L/m^2/year.

End-of-life

- **Mesh.** If the mesh is intact you can keep it on the structure to collect further. However, volumetric mesh degrades easily mainly due tension stress generated by the wind and may require replacement. On the other hand in a relocation scenario, this mesh could be easily removed and relocated.
- **Structure.** Similarly to the mesh, the structure can be completely removed and re-used for another project. This material does not degrade easily so it will last for many years.
- **Foundation.** Foundations on the other hand are fixed underground anchors and cannot be relocated. However if collectors were to be kept at their original location, those foundations will last forever. Contrary, relocation is impracticable.

Costs

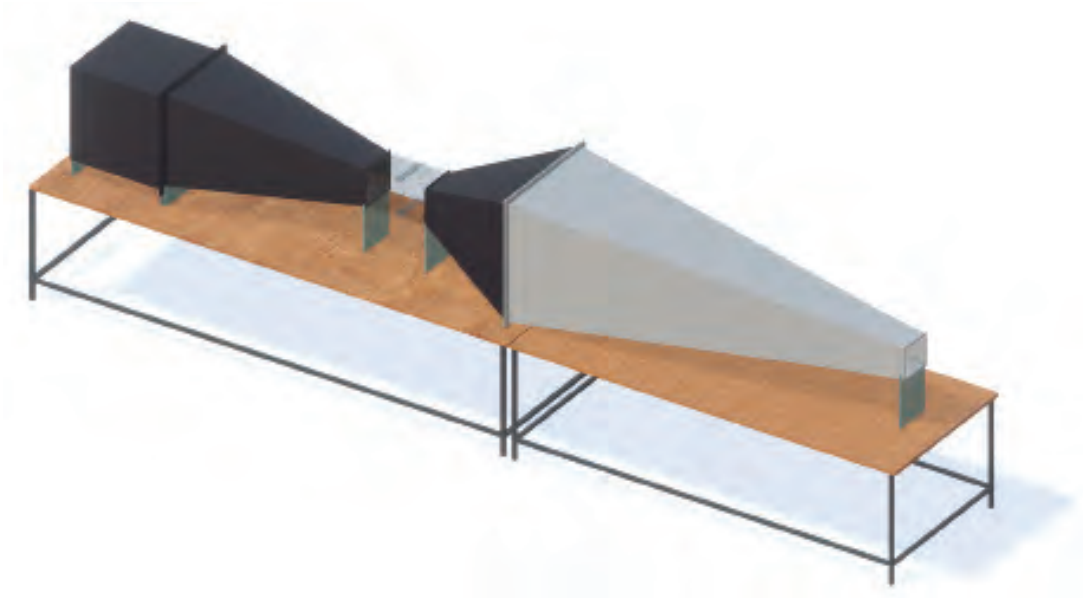
The structure price is identical to the traditional collector, with the addition of the price of the volumetric around 2.25 to 3.25 euros per square metre.

Volumetric FWC Performance in Life Nieblas	
Number of collectors:	3 V-FWC
Size:	8 m^2 surface of impact
Litres collected per m^2:	470L/m^2/year
Weight:	400 kg
Carbon footprint:	1953 kg CO_2 eq
Price:	4700 euros

Innovative FWC

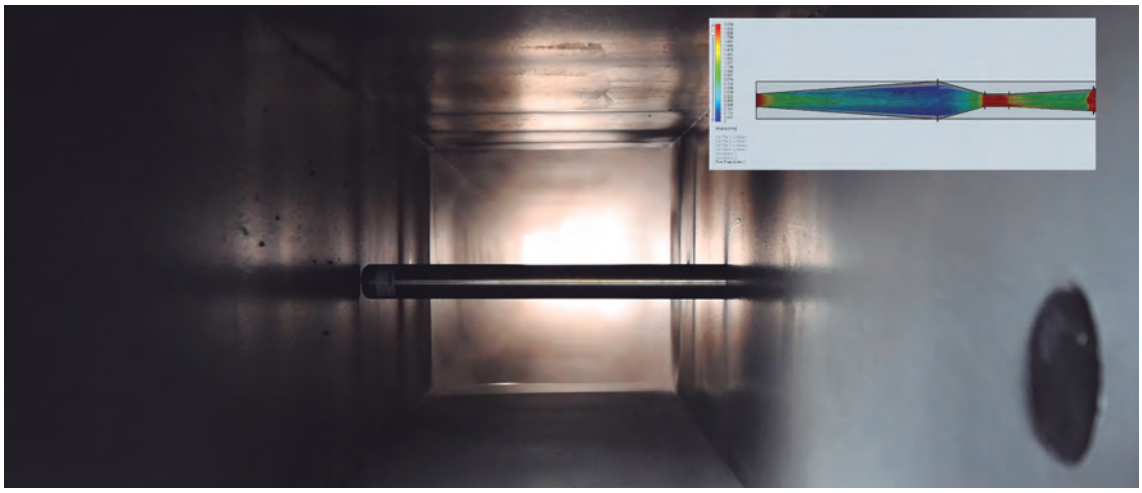
Despite the advantages of fog water collector (FWC), there are certain limitations associated with traditional fog collectors. These drawbacks include the vulnerability of the collectors to strong winds, their relatively large size and weight, and the potential for clogging and re-entrainment of collected water droplets. To address these issues, researchers have been exploring innovative approaches to improve fog collection efficiency and overcome the limitations of traditional designs.

Wind tunnel



Wind tunnel: fog water collector development testing

The research effort involves the use of a wind tunnel to simulate fog and optimise the performance of fog water collectors. Under controlled laboratory conditions, researchers have developed a new model based on the acicular appearance of Laurisilva species and the arrangement of the Namibian beetle (*Stenocara gracilipes*) when they collect atmospheric water droplets positioning their hardened wings against the prevailing winds . This design takes the aerodynamic advantages of the boundary layer to enhance trapping microdroplets. If you want to find out more about the wind tunnel and how to build one, follow this link: https://www.youtube.com/watch?v=0p5jiYBmuZ0



Through experimentation, researchers have determined the optimum number of spikes, the ideal spacing between each spike, and other parameters that contribute to improved fog collection.

«The concept behind this new model involves the passage of fog droplets through filaments, similar to pine needles, where they become trapped, accumulate, and fall down the filament under the force of gravity».

These collectors are also significantly lighter than the FWC (26 kg per collector), their dimensions 1.7x1x1m are considerably smaller than other methods. This prototype does not require foundations, but is anchored to the ground by stakes and retains resistance to high wind speeds due to its aerodynamic design that allows very high air currents to escape without severely damaging the structures.

Finally, this design has proven to be significantly more efficient at collecting fog water in L/m^2 of exposed surface area. Other advantages of the i-FWC include, ease of assembly and disassembly, ease of transportation and also the design's provision for modular collectors, i.e. i-FWCs can be mounted on top of each other and can be increased in height if required.

The i-FWC boasts several key features that enhance its performance and durability.

- Resistant to wind gusts up to 100 km/h
- Constructed using galvanised steel and polycarbonate materials
- No foundations
- Easy field deployment (modular and portable)
- No clogging

Fog Water Collection capacity

- Surface impact for one i-FWC : 0.63 m^2.
- After 3 years of monitoring water collection, the project has concluded that, on average, an i- FWC has a water collection capacity of: 642L/m^2/year.

Number of FWC

As with the traditional FWCs mentioned above (see details under point B.2.a), there is no universal rule for the number of collectors to be installed. It is therefore advisable to follow a similar approach when determining the number of collectors required for your project.

The first step is to clearly define the project's water requirements. It is important to have a clear understanding of the specific requirements in terms of the amount of water needed to meet the anticipated needs, whether for irrigation, domestic consumption or other intended uses. Once you have an estimate of the amount of water required, it gives you a basis from which to start estimating the number of collectors needed.

If no previous studies have been carried out to obtain fog water collection capacity data for the specific area, it is recommended that tests or pilot studies be carried out at the project site. These tests will provide a fairly good idea about the fog collection capacity of the area, taking into account meteorological conditions, topographical features and other site-specific factors.

The reference of 642 litres per square metre mentioned can be used as an approximate reference point to give an order of scale for the initial planning of the project. However, it is important to note that this value can vary considerably depending on local conditions and site characteristics. It is therefore advisable to rely on specific data obtained from tests and studies carried out on the project site to refine the estimate of collectors required.

Innovative FWC

Structure:

The structure is very different from that of a traditional FWC. Life Nieblas has developed several i-FWC prototypes to be able to adapt to the problems encountered during the initial installation phases. Below, a list of all the components for the latest i-FWC version is shown:

1. **Water collector**: this frame will support the fine spikes arranged in several rows. The spikes help to capture fog water as it passes through, thanks to wind action.
2. **Water collecting tray**: this is the part that collects the water droplets and transfers them into the distribution network.
3. **Two short legs and two large legs**: the short legs are placed at the back of the water collector tray, while the large ones are installed at the front. Their size difference allows the collector to adapt to any slope inclination while keeping the collector levelled, ensuring that droplets fall correctly into the collector.
4. **Anchor spikes**: these are 4 elements of different sizes (2 short and 2 large) that are driven into the ground to stabilise the water collector and prevent it from moving.
5. **U-Bolt Grip**: the stakes and collector's legs are hooked together using a U-shaped bolt screwed into a metal plate. This holds the plates in place and stabilises the collector.
6. **Fixing system**: a fixing system, such as hand screw-on, is used to secure, balance and adapt the plates and water collector. This makes it easier to level the collector and ensures that the water flows down the drain without overflowing.

7 **Drain**: a simple sink drain installation will allow water collected to be discharged into the pipeline network as seen below.

7.1

Strainer: the grid is used to prevent leaves or branches from falling into the drain system.

7.2

Screw: used to join the top and bottom of the drain on either side of the water deposit.

7.3

Strainer sleeve: gaskets are used to prevent water leaks during collection. One gasket is placed to the grid and is placed in the upper part of the Water deposit. Another gasket is placed in the lower part and is fixed to the waterway canal of the drain.

7.4

Inlet: the biggest part of the drain is the inlet; it provides the connection between the hydraulic pipe and the drain.

8 **Distribution system**: it distributes the collected water and discharges into storage water tanks. A detailed explanation of the drain system is given below:

8.1

8.2

8.3

8.4

8.5

8.1 **Pipe adapter**: this part is used to connect the drain to the PVC pipe.

8.2 **The elbow**: connects the hydraulic tube to the end of the pipeline.

8.3 **T-fitting**: it allows connection in the same way as the elbow, but allows links to be made in the middle of the pipeline.

8.4 **Flexible PVC pipe**: this is a flexible hose that is capable of withstanding a wide range of temperature and water pressure changes. It forms the link between the i-FWC and the pipeline. At this intersection the type of pipe installed is important as anticipation to any tension, tore or compression stress the drainage system might suffer due to ambient temperature changes.

8.5 **Principal pipe**: this hose collects the water from all the collectors and directs it to the storage tank.

End-of-life

i-FWCs are completely reusable, just remove the anchor spike in the ground (which leaves no impact) and move the collector. It can be disassembled for easy relocation.

Cost

The cost of an i- FWC depends on the materials used. The cost will vary from 500 Euros to 1000 Euros. Fluctuations in price would depend upon materials used, construction contracts and/or methods and amount of collectors bought (i.e. more collectors could reduce costs per unit).

Modular i-FWC

One advantage of these innovative collectors is that they are also modular. They can be assembled and stacked vertically to collect the same amount of water per impact surface, with a smaller footprint.

A modular collector uses the same i-FWC technology but with a few differences in its structure.

It consists of 4 water collectors, assembled into a larger tray. It has the same capacity per square metre of collection as the i-FWC but this one has an impact surface of 2.52m^2.

Innovative FWC Performance in Life Nieblas	
Number of collectors:	24 i-FWC
Size:	0,63 m^2 surface impact
Litres collected per m^2:	642L/m^2/year
Weight:	26 kg
Carbon footprint:	50 kg CO_2 eq.
Price:	500-1000 euros

Water Tank and pipes

It is necessary to adapt storage capacity to the number of FWC installed in the project.

If monitoring by equipment is expected during the project (i.e. water meters), to compare effectiveness it should be noted that independent water tanks are needed for each collection method.

In addition, it is advisable to connect the different collectors in series to increase water flow, as collection water stream would be significantly low individually.

Size

Storage capacity will vary depending on how the water is used. If the aim is to collect as much water as possible in winter to be used during summer, you would need a larger number of tanks for storage. However, if the objective is daily use, there may not be a need for a large storage capacity.

You must consider the following points in order to avoid any overflow or water scarcity over the years:

- Maximum absolute production in litres from one FWC in one day multiplied by the total number of fog water collectors.
- Maximum number of continuous days without fog water production.
- Average water demand by day and by week.
- Occurrence of rain during some seasons.

1

2

3

Material

Different types of tanks are available, below a few examples:

1 **Flexible tanks:**

- This type of tank can be installed anywhere on a flat surface, but care must be taken as erosion and sediment build-up could displace the tank specially when installed over ground debris of questionable stability.
- Sealed containers with a 70mm diameter filling valve.
- Discharge valve 50mm in diameter with Barcelona-type fixing point.
- They will include protection mat, pick-up case and fixing kit.

2 **Polyethylene tanks:**

- These tanks need a stable location to be installed in.
- Material with UV protection and sufficiently opaque as to prevent algal growth.
- Top door-trap 400mm of minimum diameter.
- Water filling and discharge independent valves 40-50mm diameter.

3 **Double-bund metal tanks:**

- These tanks require concrete bundled floors to prevent them from sinking into the ground due to their weight.

Location

In order to have gravity drive water through the system, the water tanks must be located at a lower elevation than the FWC and higher than the desired watering delivery points. A few metres of elevation difference is enough to avoid the use of pumps or any other water distribution energy sources.

3. Evaluation of project potential

The feasibility of the project must be assessed in terms of time and costs. It is essential to determine whether the time required to set up the project is realistic and whether the associated costs are compatible with the allocated budget. This assessment takes into account the planning, installation, maintenance and operation of the fog collection system. Accurate estimates of human resources, timescales and costs must be established to ensure the economic viability of the project.

RMT

To facilitate this assessment and support decision-makers, a specific tool has been created: the Restoration Management Tool created by LIFE Nieblas. You can gain access by following the link below:

https://lifenieblas.com/open-webtool

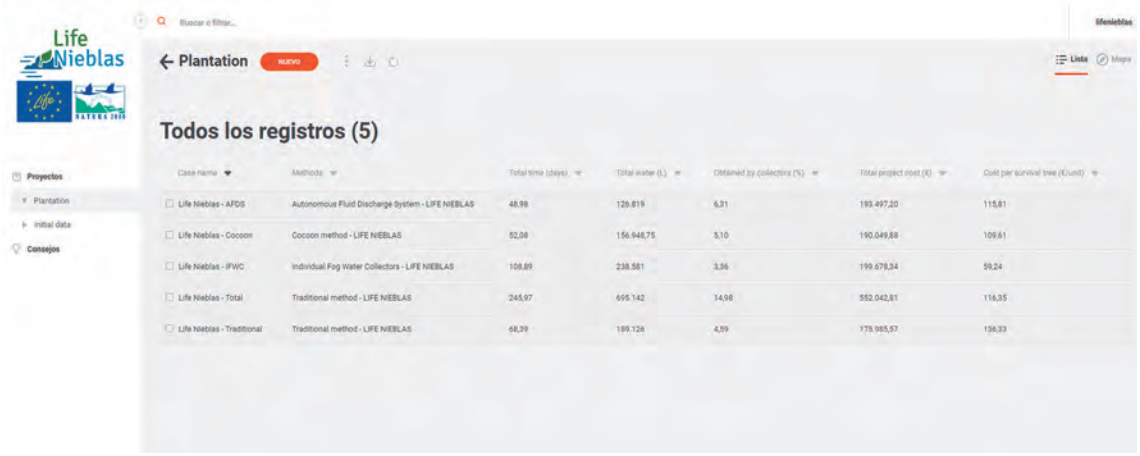
This platform brings together data from different fog collection and reforestation projects, making it possible to estimate the specific needs of a project, assess success factors and helping to make informed decisions throughout the evaluation process.

Based on this initial data, the tool calculates cost estimates, work times, and litres of water collected and to be used overall.

For the same project, different cases can be proposed to compare the results and choose the most optimal one.

Each plantation case consists of:

- **Input data** (specification of each case): Data used to define each plantation case, indicating the project duration, the number of trees to be planted, the number of workers, and the number of fog collectors to be used.
- **Starting data** (data used for calculations): For each plantation case to be calculated, you can select which values from the cost, method, and productivity tables you want to use.
  - Costs: data that allows the calculation of the cost of items used in the plantation, such as collectors (COLLECTOR COST), transportation (TRANSPORTATION COST) and other additional costs (ADDITIONAL COST) like trees, water and hoses.
  - Methods: data used to define the planting method to be used, including information about the cost of materials for that method, planting productivity, irrigation requirements, or survival rate.
  - Productivity: Data used to calculate the required work duration and costs, including information about land preparation time, installation of irrigation systems and collectors, workdays, and wages. It also includes information to calculate the productivity of the collectors based on the impact surface area and water capture capacity for each type of collector used (traditional and innovative).



- **Output data** (results): Based on the input data, the project costs, required work duration, and litres of water used will be calculated.

Time

To determine whether the project is achievable, a number of parameters need to be taken into account. First of all, you need to see if you have enough time to carry out the project. Here are the figures we were able to gather during the project as per implementation time required:

Fog water collectors field installation speed per worker/day

	Traditional FWC	V-FWC	i-FWC	Modular i-FWC
Ground conditionning	200 m²/day			
Manufacturing	Outsourced manufacturing falls out of scope of this analysis			
Foundation	1,25/day	1,25/day	10/day	10/day
Transport	2/day	2/day	11/day	11/day
Assembling	1/day	1/day	2/day	1/day
Tank installation	2 tanks/day			
Pipe installation	120 m/day			

- **Installing the pipeline network and water tanks:** in addition to installing the collectors, water tanks and pipelines need to be installed in order to complete the installation. Implementation times may vary. Preparing the site for storage capabilities is the most time consuming part, requiring up to 2 days per water tank. After that, installation is faster, with an average of 3 to 5 tanks installed per day.
- **Maintenance:** it is important to allow extra time and to free up time for the maintenance of the collectors. You should allow at least 2 days a month for servicing, cleaning and checking the collectors, as well as clearing vegetation debris on the filters to ensure optimum performance.
- **Monitoring time:** in addition, it is necessary to have daily time to monitor the project. Efficiently collecting data as regularly as possible ensures smooth running and maximum performance.

Costs

It is also necessary to calculate the costs related to the project beforehand. Below, you can read a list of costs to be taken into account for the project. An estimate of the costs for Life Nieblas has been proposed but it varies with time and location.

Life Nieblas' development related costs

Costs	Details	Life Nieblas Costs
FWC costs	Equipement	4,500 euros/FWC
	External assistance	
	External transport	
i-FWC costs	Materials	900 euros/FWC
	External transport	
Project transport	Vehicle price	7,500 euros/vehicules
	Number of kilometres	
	Fuel price	
	Fuel consumption	
Pipes system costs	Number of metres	2 euros/m of system
	Price per metre	
	Connection fittings	
Water storage	Amount of water	Between 1,100 and 1,400/tanks
	Tank price	
	External transport	
Personal costs	Number of workers	Calculations derived from particular cases, company conditions and regulation specifications for every case
	Projects duration	
	Gross annual salary	
Monitoring costs	Equipment cost	These costs will have to be analysed based in individual cases
	External assistance	

Assessing the feasibility of a project depends on a number of key factors, such as the time available, the budget allocated and the quantity of water to be collected. By taking into account the number of working days during the duration of the project, it is possible to assess whether it can be completed on time and within budget. If conditions are favourable, the installation phase then can start.

C. FIELD IMPLEMENTATION

This section of the technical manual contains all the information you need to install fog collectors. The installation plans will be detailed step by step, with an emphasis on the challenges to be avoided. Estimates of time required for each stage will be provided, along with all the parameters to be taken into account.

Next, the water storage method will be discussed, specifying the volume required and giving recommendations on the most appropriate type of tank and the ideal location for its installation.

Another aspect to consider is the pipeline network. The different types of pipe available will be explained, along with good installation practices and difficulties to avoid.

Finally, the concept of maintenance will be discussed. The regular steps required to limit potential damage will be explained, as will the risks that this project may face and the preventive measures to be taken to avoid repeating mistakes.

The aim of this section is to provide a comprehensive and detailed guide to assist in the installation of fog collectors, with the emphasis on the safety, efficiency and durability of the system.

1. Infrastructure installation

FWC and V-FWC

These structures measure 4 x 2 x 0.8 m and weigh 400 kg, not including the foundations, which generally represent 2 m^3 of concrete. The structure's material is steel and iron, covered with 45m^2 of 3x3mm PE mesh. However, its useful impact surface is considered to be only 8m^2, which is the predominant fog collection side (Ritter et al. 2005). These fog water collectors discharge directly into general storage tanks.

Materials

- Pickaxe/hoe
- Wire
- Cord
- Welding material
- Tight seals to attach the mesh to the poles
- Mesh (traditional or volumetric)

Manufacturing phase

The manufacturing phase is carried out by the company and is included in the price of the structure. The FWC is delivered with all parts already cut and ready to assemble.

Assembly phase

At least 4 people are recommended for the assembly process:

STEP 1. Ground works

Before beginning assembly of the fog collectors (FWC), it is essential to prepare the site by carrying out ground conditioning. This includes vegetation clearing, levelling the ground by removing obstacles and creating a flat surface. The construction area is marked out and prepared for the foundations of the collectors.

STEP 2. Ground digging

Once the ground is ready, the ground excavation stage begins. The locations of the FWCs are carefully chosen according to wind direction and the presence of fog. Steps are taken to ensure that the collectors are optimally positioned and do not interfere with each other.

A cord is used to clearly demarcate the boundaries of the collectors, based on the first foundation installed. This helps to guide the installation of the other collectors.

The workers place the first foundation and adjust the structure so that it is perfectly level.

STEP 3. Foundations installation

Foundations are made from 4 IPN 80 (the IPN profile is a hot-rolled bar with a double T-shaped cross-section) of 75cm by FWC, on top of which an 80cm wide plate is placed.

FWC foundations are made up of the legs of the collector. These elements are positioned in holes previously dug in the ground. It is important to wedge them in place using stones to prevent any movement during the process. To ensure that the legs are correctly installed, you can place the top of the collector tray between both supporting legs (see on the picture above).

Use of concrete

In addition, the trench was excavated and backfilled with cyclopean concrete. Rocks are used to stabilise the foundation structures. These rocks form the core of the foundations; they provide a solid base for the fog collectors. Concrete is then poured into the foundations to solidify the structure and give it additional strength.

STEP 4. Foundation compacting

The foundations are compacted to ensure their long-term stability and strength. This usually involves the use of compaction tools to compress the soil around the foundations, ensuring a solid and durable base.

STEP 5. Structure transport

Due to the nature of the terrain or the specific location of the collectors, it may be necessary to manually transport the FWC structures to their installation site. As the entire structure can weigh up to 400 kg when assembled, this requires careful handling to ensure safe transport.

STEP 6. Assembly phase

Once all the parts of the FWC structures have been brought to their installation location, the assembly stage begins. The various parts, such as posts, cross-part and supports are attached to each other according to the blueprints. Such assembly involves the use of bolts, nuts and screws to ensure a solid and stable connection between the elements.

STEP 7. Positioning the collector

Once the structure has been assembled, the fog collector itself is positioned. The legs of the collector are placed on the foundations, guaranteeing its stability. The top of the collector is placed on an iron bar to maintain an elevated, optimal position for easy manipulation.

STEP 8. Mesh installation

The mesh is installed on the collector structure to capture the droplets. Before the collectors are lifted into the vertical position, the different layers of mesh are attached to the structure while it is still on the ground. This facilitates the installation process. Mesh is tensioned and fixed firmly with plastic joints to prevent any loosening or detachment.

The mesh must be a 3x3mm monofilament in HDPE polyethylene with anti-UV treatment, the 45m^2 of mesh is arranged as follows:

· **Outside:**

- 3m high and 7.80 m wide wrapped around the FWC (double mesh at the back and single mesh on the rest).
- 1 metre high and 7.80 m wide around the FWC (double mesh at the rear and single mesh for the rest).
- 2.10 metres long and 1 metre wide at the top of the FWC (single mesh).

· **Inside:** double mesh in each layer

- 1 metre wide and 3.8 metres high, reinforcement is required at the ends.
- 0.50 metres wide and 3.80 metres high, reinforcement is required at the ends.

· **Summary:**

- One layer measuring 7.80x3m -> 23.40m^2
- One layer measuring 7.80x1m-> 7.70m^2
- One layer measuring 2.10x1m -> 2.10m^2
- Two layers measuring 0.5x3.80m -> 3.80m^2
- Two layers measuring 1.00x3.80m -> 7.60m^2

Materials required for the building of a traditional fog collector

	Structure	Mesh cover 1	Mesh cover 2	Mesh interior	Grand total (m^2)
Height (m)	4	1	4	3,8	
Length (m)	2	2,1	7,8	3	
Depth (m)	0,8				
Total (m^2)		2,1	31,2	11,4	44,7

STEP 9. Raising the structures

Once the structures have been assembled and fitted with the mesh, they are lifted into their final vertical position. This often involves using ropes or other lifting equipment to elevate the collectors and place them on top of the foundation's base. During this process, it is essential to ensure that the collectors are levelled and correctly aligned.

ETAPE 10. Welding

To reinforce the stability of the collectors, foundations are welded to the structures. This prevents the collectors from falling or shifting under the effect of wind or other external forces.

STEP 11. Drain and Hoses installation

At this stage, the outlet pipes and collector drains can be installed. It's better to install a metallic elbow for the connection, as there's less risk of it breaking over time due to stresses involving temperature fluctuations, tension and tore generated by weight and displacement of the pipeline works, etc.

STEP 12. Connecting the pipeline

Fog collectors are then connected in series to a large common pipe. This pipe is used to transfer the water collected from the collectors to a storage tank. The hoses are connected using appropriate fittings to ensure a watertight and secure connection.

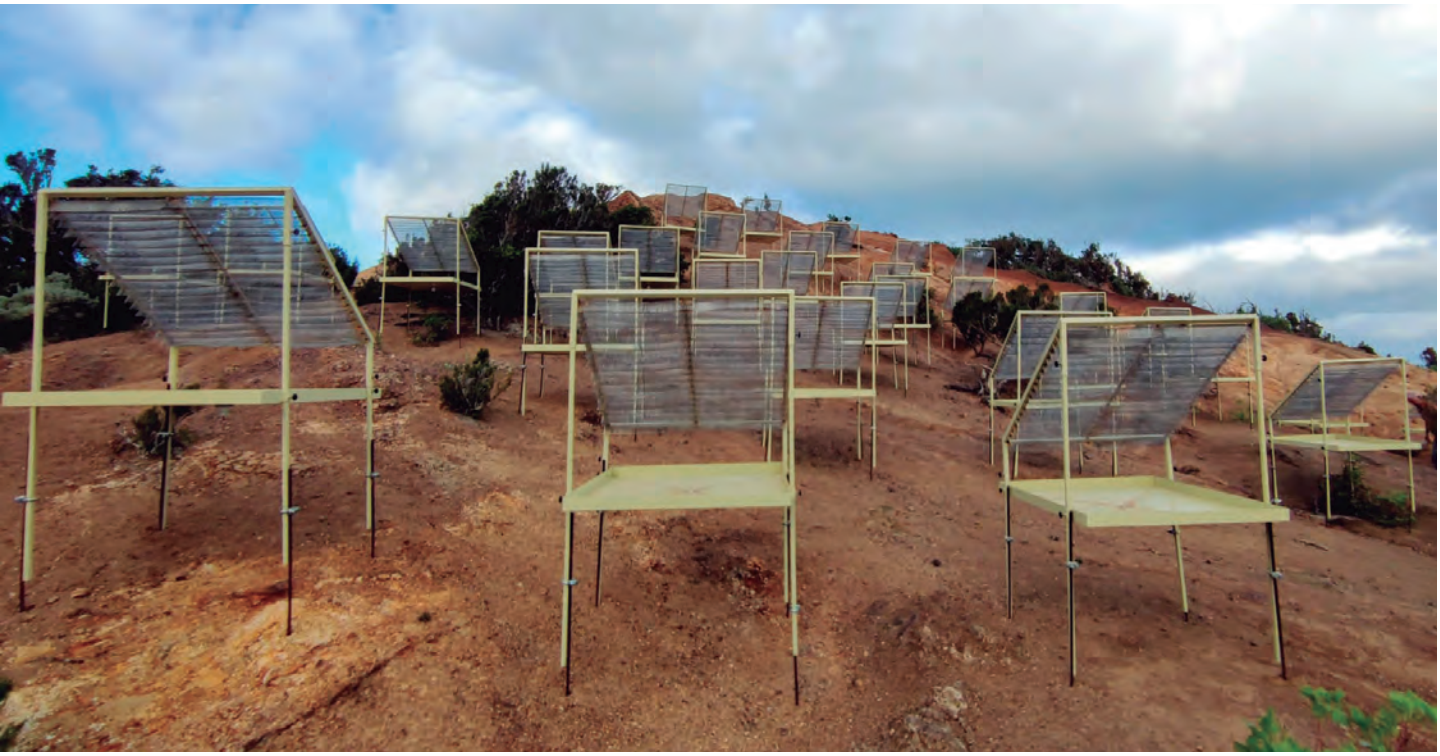
As per the volumetric-fwc

The assembly process for a volumetric is different to that of the FWC in that the structure is stripped from the outer layers all around the collector.

Another way of installing the mesh is by first setting up the structures and later attaching the mesh by means of vertical access techniques as shown below.

Note

For any fog water collector, visual impact should be considered, especially in terms of the traditional and volumetric collectors due to their size. We highly recommend using brown And/or black colours, as seen in the picture above (right). Shades of green colours are bright, while black may and brown are colours that can easily camouflage against vegetation background.



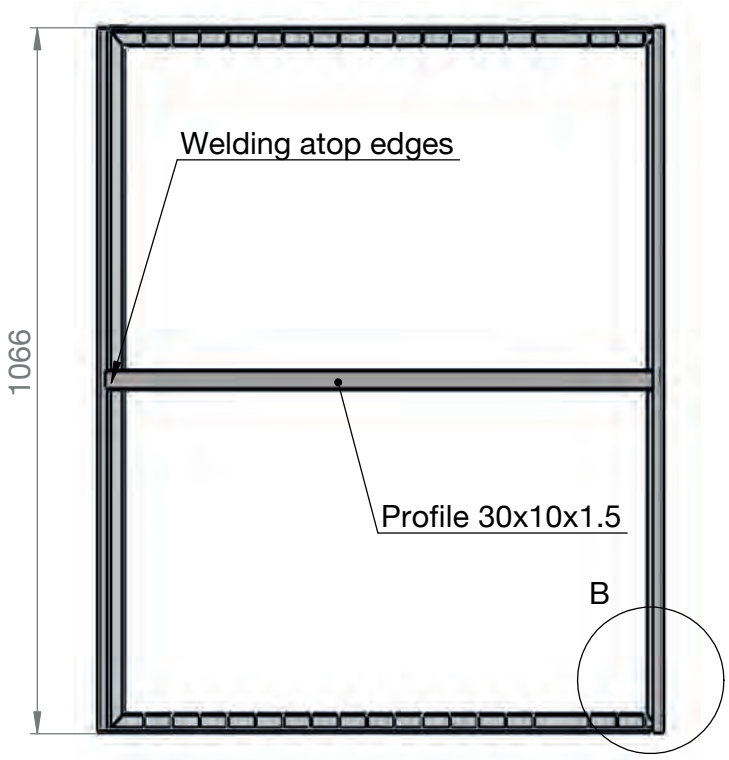
i-FWC (single module)

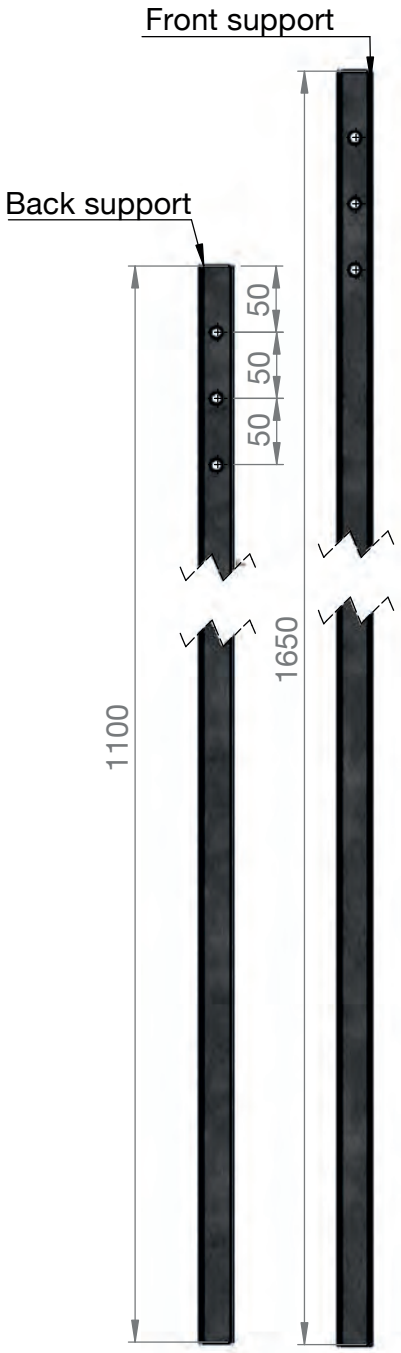
Materials

- 48 M6 x 55 bolts.
- 48 M6 nuts.
- 12 M8 x 55 bolts.
- 12 M8 nuts.
- 2 squared galvanised steel profiles 40 x 40 x 4 mm / 6 metres long.
- Water collecting tray of 1.5x1m.
- 8 U-Bolt Grip.
- Pipe dia 40.
- PVC pipe diam 40.
- Drain diam 40.
- Anti-galvanising paint.
- Anti corrosion coating.
- Stainless steel screening mesh with the following or similar characteristics:
  - Wire diameter: 1.2 mm.
  - Wire horizontal spacing: 1.5 mm.
  - Wire height (vertical spacing): 50 mm (5 cm). 10 cm is also suitable.

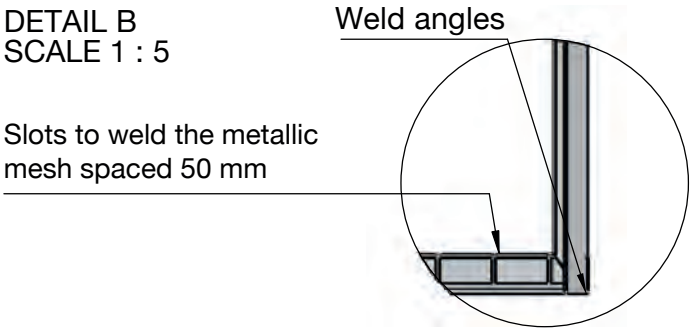
Tools and PPE

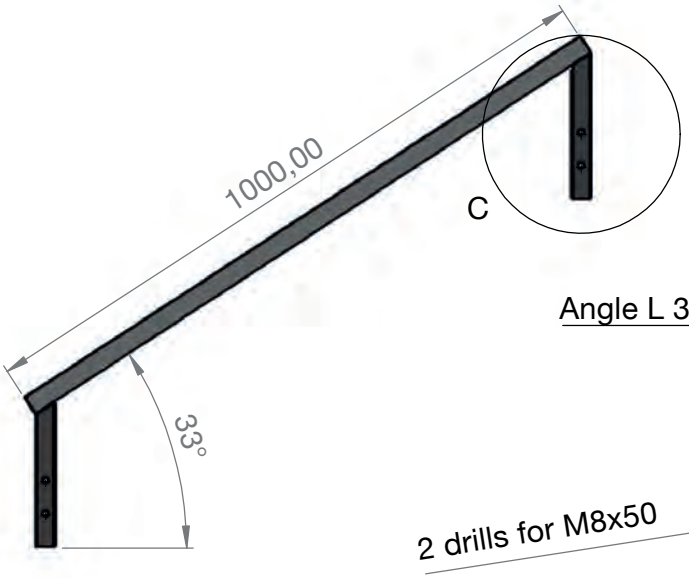
- Wrench.
- Monkey wrench.
- Grinder machine with steel cutting discs.
- Protection clothing and goggles.
- Measuring tape.
- Marker.
- Drill.
- 40mm crown drill for valve kit.
- Drill and screwdriver bits case.
- Any tool to measure angles.
- Welding equipment.
- 2 x Sikaflex 11FC tubes.
- Spirit level.
- Paintbrush.

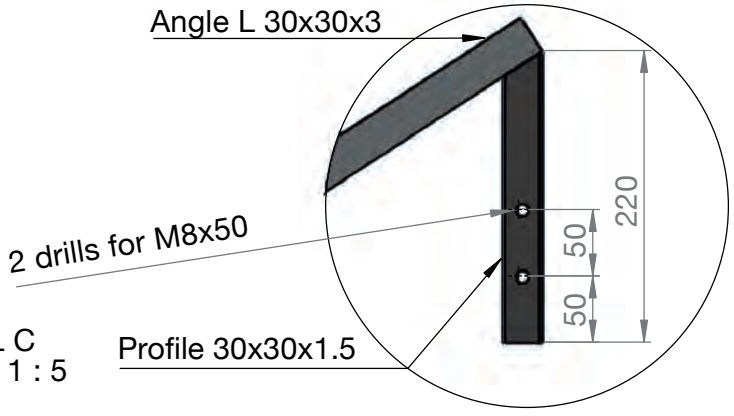




DETAIL B
SCALE 1 : 5

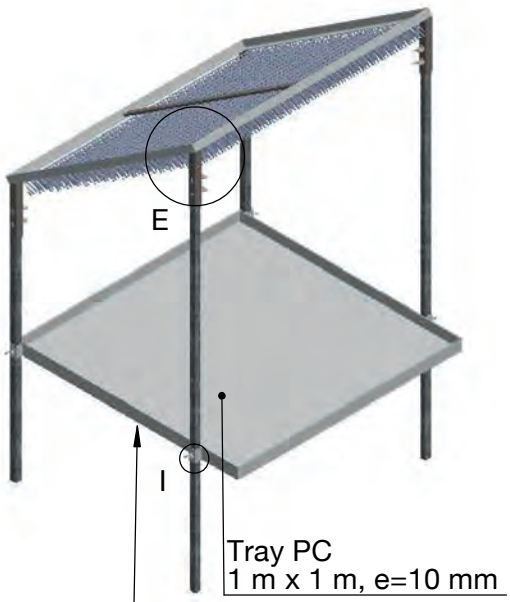




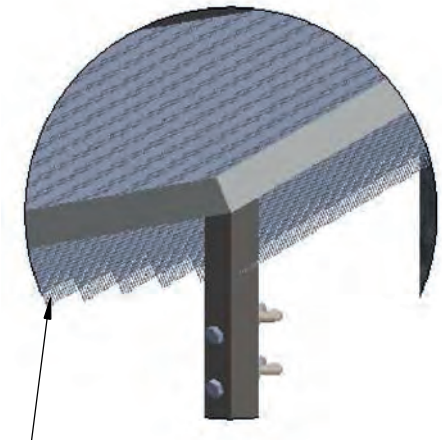


DETAIL C
SCALE 1 : 5

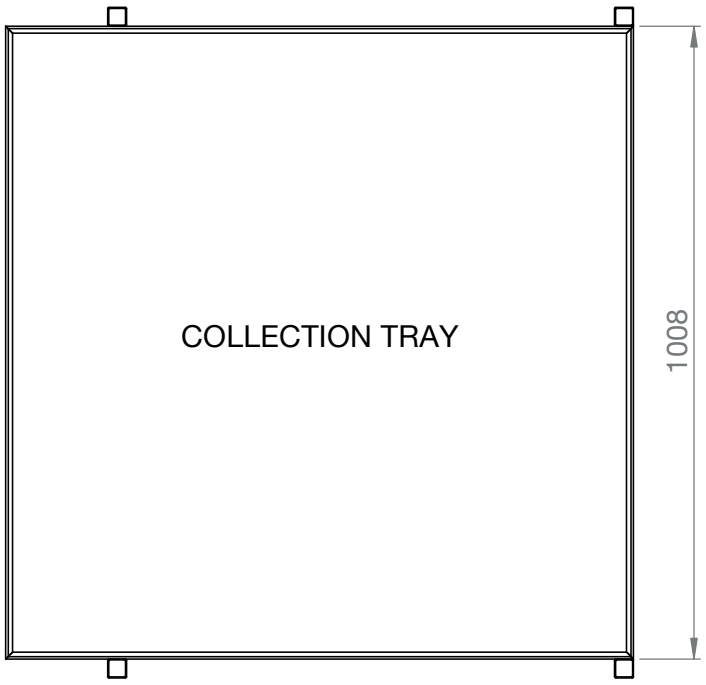
ISOMETRIC VIEW - SCALE 1 : 20



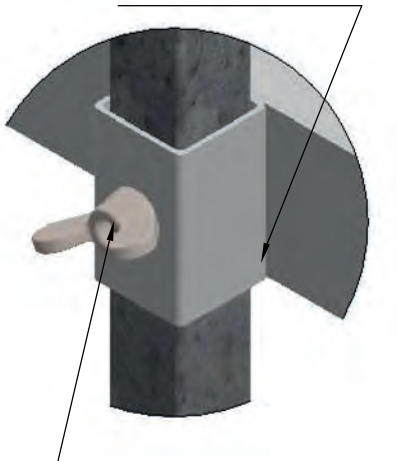
DETAIL E - SCALE 1 : 5

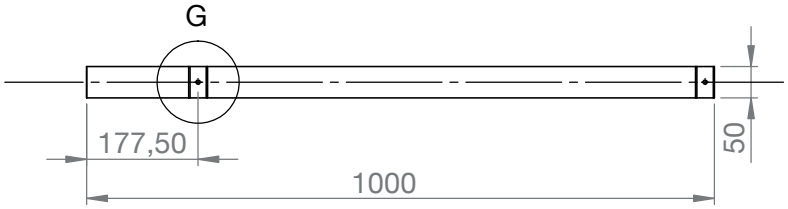


Collectors filaments
- Material: AISI 304
- Length: 50 mm
- Inclination: as per structure
- Filament rows required: 19

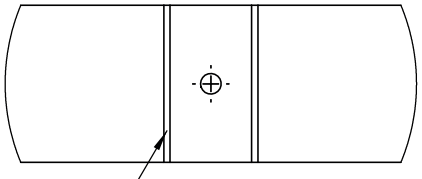


DETAIL I - SCALE 2 : 3



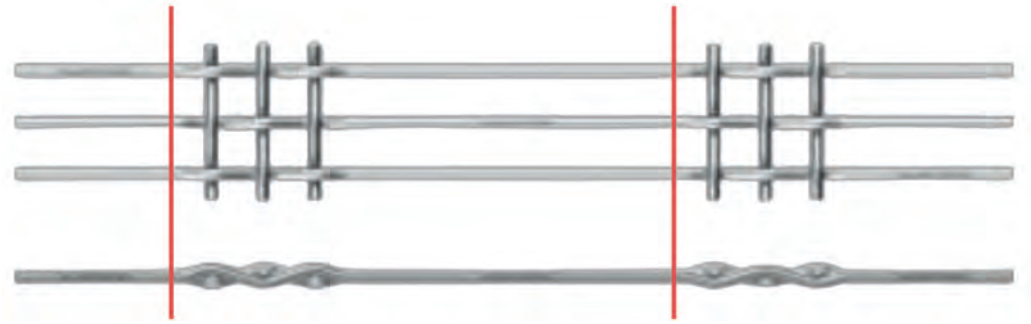


DETAIL G - SCALE 1 : 2



Manufacturing phase

1. Prepare (cut if necessary) the galvanised steel profiles with the dimensions required in the materials list.
2. Once all the tubes and profiles are cut and prepared, drill the holes as shown on the blueprints.
3. Cut the mesh into sections as defined by the red lines below, and group them in piles of 19.



4. Manufacture the top and middle-height module as shown in blueprints and cut in the lateral angles (1m length) 19 slots spaced 5cm between them.
5. Later weld the 5cm long mesh sections into the slots. Also remember to cut slots into the slower part of ISO rectangle 30x10x1.5mm for the mesh to have a middle support for welding. Be careful to weld the mesh so that it is always facing into the wind.

6. Manufacture the perimeter of the water collecting tray
7. Once all the steel parts are manufactured and welded, apply to each one of them (including al the legs and bars) anti corrosion coating for galvanised steel, and when this coating is dry, apply brown paint [1] and let it dry for 24 hours.
8. Weld the mesh sections to the manufactured modules.

1. Brown is a colour that better blends with the background in the natural environment. On the other hand, green or black may result quite visible at simple sight in the distance.

Assembly phase

Based on the list of equipment presented above, here are all the steps involved in building i-FWCs, this presentation is also available in video at this link: https://youtu.be/Eiti4UKpXAE

At least 2 to 4 people are recommended for the assembly process:

STEP 1. Mount the legs

- Place the water collector on the ground with the backside facing up. Begin by inserting the larger legs at the front, pushing them into the designated slots in the corner.
- Similarly, slide the shorter legs at the back of the collector.

STEP 2. Secure the collector's system

- Insert screws into each leg to prevent them from moving when handling or adding other parts. There is one screw per leg.

STEP 3. Install the collecting tray

- Flip the water collection tray so that the tray bottom looks upwards and everything is properly orientated. Centre it and thread the i-FWC legs through the slots on either side of the tray.
- Once all 4 legs are through, push the tray to secure it at the desired level. The closer the water collecting tray is to the spikes, the lower the risk of droplets being carried away by the wind.

- Next, similar to Step 2, insert screws into the water collecting tray to prevent it from moving during handling. There are 4 screws to be inserted, one at each leg. The objective is not to permanently fix the collector but to have it presented into place. Levelling and adjustments will be made later during ground installation.

STEP 4: Install the drain

- First, attach the gasket to the inlet of the drain. Also, assemble the second gasket on top of the drain grille.
- Position the two parts on either side of the water collecting tray (the grille on the upper part and the inlet on the lower part).
- Then, insert the screw from the top of the upper part of the drain to join and secure the two parts together.

STEP 5. Position the collector

- This step involves flipping the collector and positioning it at the ground setup point. The taller legs should be on the lower side of the hill to ensure the collector stands straight.
- There should be no obstacles in front of the collectors higher than ground level (where the collectors are located) to allow fog flowing through without air current being greatly disrupted avoiding generating vortices which directly affect aerodynamic flow and limit flow. In addition, each row of collectors must be installed slightly diagonally within the slope, so that the pipeline is driven downhill and water will flow freely avoiding stagnation.

- Fog water collection spikes should be perpendicular for optimal water collection performance. Pay attention to the exposure of the spikes to air and wind. For good performance, all spikes should be exposed to the air and should try to avoid excessive overlap or be turned into the wrong direction.
- During handling, branches or leaves may get caught in the spikes. Clean them carefully and adjust the spikes manually if necessary.

STEP 6. Install the anchor spike

- This step involves securing the i-FWC to the ground to prevent imbalance or shifting, ensuring efficient water collection while avoiding collectors' collapse.
- Using the collector's frame as a reference by leg placement (i.e. marking the ground), insert the stakes into the ground. Use a larger stake for the taller legs and a shorter stake for the shorter legs.
- If the ground is hard, the use of a hammer is recommended to assist driving the stakes in.

STEP 7. Fix the U-Bolt to the anchor/legs

- Once the stakes are inserted at the desired locations, attach the i-FWC. Each leg is attached to a stake. Secure these legs to the stake to prevent any movement using a U-Bolt Grip.
- Pass the U around both parts, and then screw the plate up against the U-bolt. Afterward, tighten the nuts without over tightening. Repeat this for all 4 legs.

STEP 8. Bring top layer to level by raising/lowering collector's legs

- Before completely tightening the leg's U-bolt nuts, use a spirit level to ensure the i-FWC is levelled. Take measurements on the frame with the water collection spikes and along the 4 legs.
- Once you are sure the collector is levelled, tighten the U-bolt nuts to their maximum to prevent any movement. Use a wrench to tighten them completely.

STEP 9. Level the tray by sliding it up/down the legs

- Using a spirit level, ensure the area where the drops will fall is perfectly flat. You can use the screws at the 4 corners to raise or lower each corner individually.
- Check the level on all 4 sides of the water collecting tray. Once it is perfectly levelled, fully tighten the screws.

STEP 10. Connect all the drain systems pieces

- Start by assembling the 3 pieces together of the pipe adapter. They will connect the pipes to the drain. Once assembled, screw them onto the lower part of the collector, where the drain is located.

STEP 11. Measure the flexible PVC pipe (hydrotube)

- Prepare the elbows to connect to the principal pipe for collecting water from the different i-FWC.
- Measure the required size of the flexible pipe. The connection should be made at ground level to prevent the drain from pointing downward and damaging the collector.

STEP 12. Cut and install the pipe

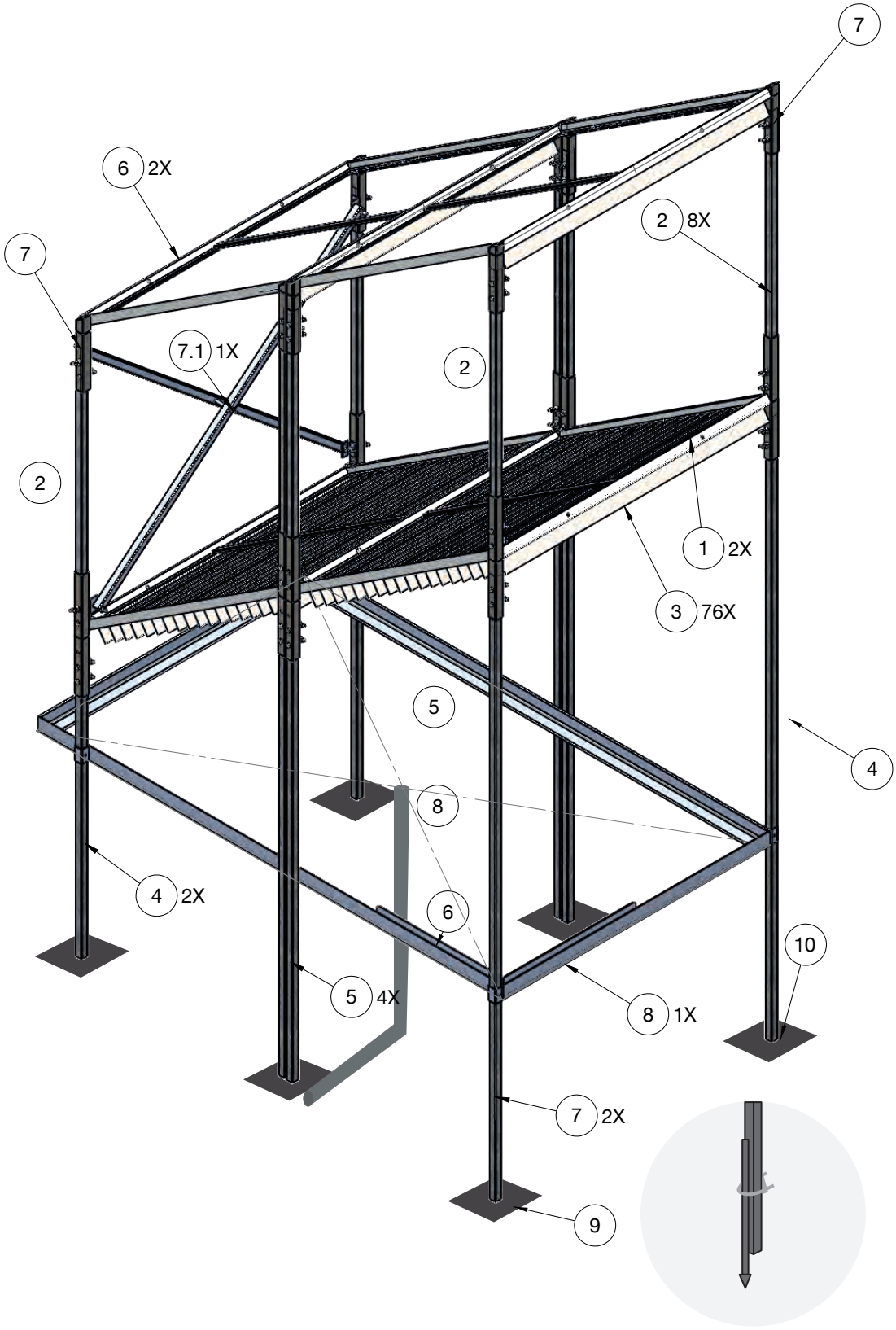
- Once the measurements are taken, cut the flexible pipe to the appropriate size. You can use a saw or a pipe cutter tool (see picture).
- Now, all the pieces are prepared to the correct size, attach the flexible pipe between the water collecting tray and the elbows. Do the same with the elbows.

STEP 13. Connection to pipeline

- If the collector is in the first position in its row, fit the end of the pipeline into the elbow. If the i-FWC is in the middle of the row, you will need to cut the pipeline at the correct position (check the PVC pipe entry point) and integrate both sides to the T-fittings as shown in the picture to the right.
- Then, ensure that all the pieces are tightly secured using a hook wrench.

Modular i-FWC (multi-modular)

Plan ICIA



Material

1 Middle-height module (2 units)

Galvanised steel. Per each unit:

- 1 x 1 m ISO Rectangle 30 x 10 x 1.5mm
- 4 x 42 cm ISO Square 30 x 30 x 1.5 mm
- 2 x 1 m ISO Angle 30 x 30 x 3 mm
- 2 x 1.066 m ISO Angle 30 x 30 x 3 mm

2 Mezzanine bars (8 units)

Galvanised steel.

- 8 x 1 m ISO Square 25 x 25 x 1.5 mm

3 Fog collection mesh (76 units)

Supplied by Nuba screening media/ mallas DMR. Material:

- AISI 304 steel.

Stainless steel screening mesh with the following or similar characteristics

- Wire diameter: 1.2 mm
- Wire horizontal spacing: 1.5 mm
- Wire height: 50 mm (5 cm).
- Provider example: Nuba Screening Media. Mesh used:
  - Width: 1.060 m
  - Length needed (mesh comes as a continuous rolled sheet): 5 m
  - Filament diameter: 1.2 mm
  - Filament height: 50 mm
  - Filament spacing: 1.5 mm

4 Structure´s legs and reinforcement bars

- Rear legs (2 units). Per each unit: 1.10m ISO Square 25 x 25 x 1.5 mm
- Middle legs (4 units). Per each unit: 1.65 m ISO Square 25 x 25 x 1.5 mm
- Front legs (2 units). Per each unit: 2.15m ISO Square 25 x 25 x 1.5 mm
- Reinforcement bars (2 units). Per each unit: 1.35 m ISO Square 25 x 25 x 1.5 mm

5 Water collecting tray (1 unit)

Galvanised steel + PolyCarbonate plate 10 mm thickness

- 2 x 1.85 m ISO Angle 30 x 30 x 3 mm
- 2 x 1.064 m ISO Angle 30 x 30 x 3 mm
- 2 x (50 cm x 6 cm x 3 mm) cut Galvanised Steel Sheet
- 4 x 5 cm ISO Square 25 x 25 x 1.5 mm
- 1 x (1 m x 2 m x 10 mm) Polycarbonate sheet.

6 Reinforcement angles

- 4 x 5 cm ISO Angle 30 x 30 x 3 mm

7 Bolts, nuts, and screws

- M8 x 20 bolts: 4 units
- Hexagon M8 nut: 10 units
- M8 x 50 bolts: 52 units
- M8 Wing Nut (DIN 315): 48 units
- 7.1. M8 Threaded Rod, 12 cm long: 1 unit

8 Hydraulic parts

- 40 mm drain kit:
- Hydraulic accessories necessary to carry the water collected to a water tank diam:40mm (elbows, pipe adapters, T-fittings…)

9 Structure – to – ground fixing means

- 8 x 1 m anchor spikes (Ø12 mm reinforcing bar):

Pike tip: Lance-shaped, formed by welded galvanized steel plate. If it is too expensive to manufacture, alternative spike tips that offer adequate grip on the terrain can be chosen (for example, one or two pieces of triangularly cut sheet welded to the tip of the corrugated iron).

10 Fixing of the spike to the structure

6 mm threaded rod that can surround a 25 mm x 25 mm square profile, anchored by means of a nut to a galvanized steel plate (images below).

Also, a DIN 741 Commercial Grip (U-Bolt Grip) is suitable for the task of anchoring if can be found a size that would surround completely the 25 x 25 mm square profile.

Tools and protection

- Wrench
- Monkey wrench
- Grinder machine with steel cutting discs
- Protection clothing and goggles
- Measuring tape
- Marker
- Drilling machine
- 40 mm crown drill for valve kit
- Drill and drill bits briefcase
- Any tool to measure angles
- Welding equipment
- 2 x Sikaflex 11FC tubes
- Spirit level
- Paintbrush

Manufacturing phase

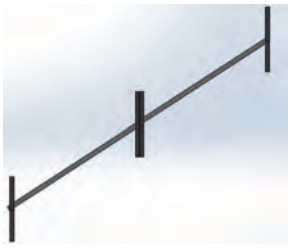
For the multi-modular i-FWC, the manufacturing process is exactly the same as for a single i-FWC. Please, refer to the previous section ‘i-FWC (single module)’.

Assembly phase

STEP 1. Creation of the middle-height module

Join the two middle-height modules with 4 M8 x 20 bolts. Ensure a good tightening between nut, bolt and structure.

Isometric view of middle-height modules assembled.

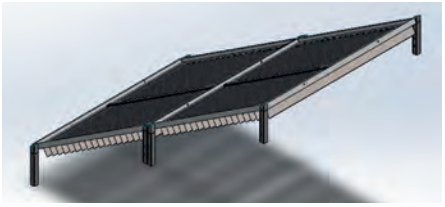


Lateral view of middle-height modules assembled.

Isometric view of one of the M8 x 20 bolts fixing both middle - height modules.

STEP 2. Assembling the top modules

Join the two top modules with 4 M8 x 20 bolts. Also ensure a good tightening between nut, bolt and structure.



Isometric view of top modules assembled.

Lateral view of top modules assembled.

Isometric view of one of the M8 x 20 bolts fixing both top modules.

STEP 3. Add bars

Position and secure (with M8 x 50 bolts and M8 wing nuts) the intermediate bar in the upper insert of the middle-height module.

Perspective of middle - height modules and mezzanine (intermediate) bars mounted.

STEP 4. Join both top modules

Join both top modules with the rest of the assembly by the upper drills of the mezzanine bars (with M8 x 50 bolts and M8 nuts)

Mounted with both top modules.

STEP 5. Add crossbar

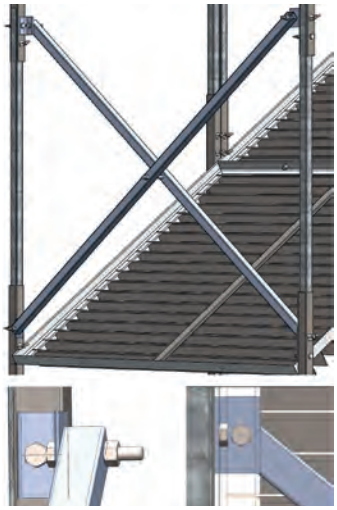
Mount the reinforcement crossbars on the rear part of the structure (with M8 x 50 bolts and M8 standard nuts). Secure the crossbar with the M8 threaded rod at its centre.

STEP 6. Assemble legs

Lay the partially assembled frame on its side and assemble the legs (with M8 x 50 bolts and M8 wing nuts).

STEP 7. Add legs

Set the structure upright and support all the weight on its legs. Insert the legs into the squared inserts on the water collecting tray, which will be resting on the ground. Make sure that the overhang of the collecting tray protrudes behind the structure, and not in front of it. This will help collecting any water droplets being pushed away by the wind.



Detail of structure reinforcement characteristics.

STEP 8. Raise the water plate

Raise the water collecting tray guided by the square inserts on the structure´s legs. Fix a height for the front part of the plate and place one (1) self-tapping M5 screw on each squared insert with the drill. Then, proceed to lower the rear part placing the corner with the sink lower at the lowest point (allowing enough inclination for the collected water stream to flow through). Keep in mind that unevenness of a few centimetres (3 cm to 5 cm) is more than enough. After this, fix the front self-tapping screws.

Important

The water collecting tray should not be placed excessively low or the structure will not be correctly fixed to the ground. This would be due to the available leg's height dedicated to fix the anchoring into the ground. Additionally, lack of underneath space will present difficulties when attaching hydraulic accessories to the drain and discharge system.

STEP 9. Drain

Mount the drain and make a leak test through the collecting tray, to prevent future water losses.

STEP 10. Fix the structure

Fixing the structure to the ground:

a. Dig a hole on the ground for each one of the legs in the structure, leaving enough space to place the anchors – spikes.

b. Place the structure aligned with the ground holes.

c. Place spikes beside each one of the structure's legs and bury them as much as it is possible. Then attach each 'leg-pike' pair with a U-Bolt grip.

d. If necessary, reinforce the structure by tying steel cables and tensors, correctly anchored to the ground by using additional spikes.

STEP 11. Drain system

Mounting the necessary drain system (pipes, accessories, and water storages).

Water network

The different **steps** involved in installing the water distribution network are detailed in the reforestation section C.2.2).

2. Management and maintenance

Regular maintenance

Filter

To prevent clogging caused by dirt from collectors' water discharge, it is recommended to install a filter. Regular removal and cleansing of the filter is necessary.

Clean

The collectors accumulate dirt, branches, leaves, sand, and other debris in the water collection tray. If these particles are not filtered out, they will remain in the tray and become an issue later down the road. Regular cleaning is required for any fog water collector. To clean them, it is advisable to wait until they are dry and then use brushes to collect the waste, air blowers or other cleaning methods. Cleaning may be challenging and care must be paid to avoid damaging the collectors' mesh, but if possible, a vacuum can make the task easier and quicker.

In the same way, the water tanks must be cleaned regularly (depending on the use of water). It must always be covered by a lid to avoid dust, vegetation debris, insects and small animals from getting in. It must be examined on a monthly basis and cleaned as needed. Covering water tanks do not only keep the pipelines clear but also protects local fauna from drowning[2].

2. To provide the local fauna with water we recommend placing small plates regularly filled with water by the water tanks, preventing any local fauna from trying to access the water tanks and getting trapped.

Vegetation clearing

As previously mentioned, it is crucial to have an area clear of obstacles in front of the fog collectors. Having an undisturbed air current will keep water collection aerodynamically effective. Hence, a regular maintenance of the clearway in the collection area is highly recommended. As seen in the picture on the left, vegetation is kept to a minimum typically not exceeding FWC's tray height.

Risks and advice

Clogged water outlet

Despite regular maintenance, filters, and cleaning being done, a blockage could occur, preventing water from flowing through the pipelines. To diagnose discharge point blockages, regularly check (during rainfall episodes) to ensure water is being discharged correctly. Furthermore, if issues are encountered, we recommend detaching the pipes from the collection tray and pouring some water through to diagnose the flowing status of the system. If water does not flow as expected, both pipe and drain should be cleaned using a drain cleaning cable until normal discharge operation is recovered.

Breaking pipes

Regular inspection to all pipelines ensuring there are no leaks is essential to maintain a healthy water discharge network. In addition, minimise structural stress on the pipes (torsion and/or tension stress at junctions) and use rigid materials in sections where the presence of potential hazards such as rocks, pedestrians, vehicles, etc are observed.

Tanks location choice and models

Attention should be specially paid when choosing your water storage tanks location and what kind of container you install in certain places. For instance, over uneven terrain, flexible water tanks may have a higher risk of sliding and/or losing grip under the effect of heavy rain coupled to the weight shifting of the contained water mass. In the pictures to the right, an example of a near miss has been recorded, where a storm encouraged the formation of small streams under the flexible tank. This resulted in the tank sliding downhill and bursting. Instead, smaller and solid water tanks were installed away from the terrain edge. If you require the installation of flexible tanks, make sure to safely anchor them to the ground or trees in the nearby area. Take into account that inaccessible places (where these tanks are useful) are often in poor conditions to find solid even ground. Hence, water tanks require some sort of anchoring.

Metal tanks have a risk of rusting. It is preferable to choose tanks made of appropriate materials for water storage. If no other option is available, anti-rusting paint can be used to address the issue.

Cracked, punctured or broken tanks

Leaks from different nature can occur in water tanks. Regularly inspect the condition of the tanks and their surroundings to ensure there is no risk of damage. Solid terrain clear of rocky outcrops is the best location for water storage installation. When water tanks are larger than $2m^3$ the risk of sinking on loosened terrain is a possibility. Especially after rainy periods, the

infiltration of water coupled with the heavy weight of tanks would encourage uneven pack down of the terrain resulting in water tanks imbalance, hence putting structural stress over certain parts of the tanks. In terms of flexible tanks, a protective underlay is recommended to soften the ground and prevent any protuberance from puncturing the water tanks. If a leak occurs, silicone sealant can be used in solid tanks. Empty the tank, apply silicone to the crack and let dry. For flexible tanks a simple patch can be fixed to the punctured area. If the rip is too large in flexible containers, professional refurbishment may be required.

Rope access and H&S

Something else to be cautious about when working with fog water collectors is the risk involved in vertical access and the human factor. Traditional FWCs can reach a height of 4 metres, making any mesh or structural maintenance works potentially dangerous. Technical rope and vertical access training courses are recommended in order to safely work at heights before engaging in any activities at the top of the collectors.

D. MONITORING FOG COLLECTION

To ensure that the project is optimised, it is essential to carry out regular monitoring of the parameters throughout its duration. As a first step, meteorological monitoring is recommended to diagnose the occurrence of fog and to better understand collectors' effectiveness. Conclusions drawn from the project will be presented in this section, highlighting the parameters that influence optimum performance. Information such as favourable weather conditions and the factors that have the greatest impact on performance will be discussed.

In addition, the monitoring of data collection will be presented. The tools used and the types of analysis carried out will be described, making it possible to compare the results with other collectors, observe changes in performance and compare them with meteorological data.

This comparative analysis will help to assess the efficiency of the collectors under different conditions and to determine the measures to be taken to improve performance if necessary.

Regular monitoring of the parameters will also make it possible to identify any deviations or problems, thus facilitating the implementation of appropriate corrective measures to ensure the smooth running of the project.

1. Meteorological data

Objective

During the project, it is necessary to measure the potential for fog collection that the FWCs have on a given area. To achieve this a motorisation of the ambient meteorological conditions and the FWC water production is essential.

This means recording meteorological parameters such as temperature, humidity, precipitation and solar irradiation.

Device

To measure this, a meteorological station can be installed. The typical components you might find in such stations:

- **Poles and Mounting Structures:** used to hold and to elevate instruments to necessary heights. They are typically made of durable materials like galvanised steel to withstand weather conditions.
- **Power source:** if access to an electrical power supply is not available, the system must be equipped with solar panels, batteries, and a charge regulator to ensure operation.
- **Instrument Shelter:** this is usually a box made of plastic to protect the electronics and some sensors like temperature and humidity sensors from direct sunlight and precipitation. It is either sealed to be watertight for electronic components or ventilated to allow air flow for accurate sensor readings.

- **Data Logger and Transmission Equipment:** electronic data loggers to record and transmit data automatically to a central system for analysis.
- **Meteorological sensors:** it shall include necessary sensors such as:
  - **Anemometer and Wind Vane:** typically mounted on a mast or pole to measure wind speed and direction. They are placed away from obstacles to ensure that readings are not affected by nearby buildings or trees.
  - **Rain Gauge:** typically consists of a funnel that directs the rainwater into a small balance with two compartments. Each time a specific amount of water accumulates (such as 0.2 mm of precipitation), the compartment tips and empties, triggering a switch that records the amount of precipitation. It must be placed in an open area to avoid blockage or redirection by nearby structures.
  - **Solar Radiation Sensor:** these sensors are placed in open areas to measure the amount and intensity of solar radiation. They must have an unobstructed view of the sky.
  - **Temperature Sensor:** measures ambient air temperature. They must be shielded from direct sunlight and other sources of heat.
  - **Hygrometer (Relative Humidity and Dew Point Sensor):** measures the amount of moisture in the air as relative humidity and calculates the dew point.

  - **Evapotranspiration Sensor:** calculates the rate at which water is transferred from the land to the atmosphere by evaporation from the soil and other surfaces and by transpiration from plants.
  - **Visibility Sensor:** measures the distance over which objects can be clearly seen. The one installed in the project uses an active illumination method to detect light reflected off airborne particles.
- **Ground Sensors:** for measuring soil temperature and moisture at various depths, sensors can be buried in the soil next to the meteorological stations and connected by wire.
- **Field sensors:** for measuring environmental parameters such as temperature and soil moisture, as well as data from many other types of sensors, including water counter readings in nearby areas, two options are available: sensors with internal memory for manual data retrieval or sensors equipped with LPWAN technology.
  - Sensors with internal memory can be used, requiring periodic manual downloading of data. This may not be the best choice for very wide or difficult-to-access areas.
  - Alternatively, when manual data downloading is not feasible or desirable, LPWAN sensors may offer a solution. These sensors have built-in radio transmitters that send data over long distances (up to 10km range) while consuming very little power. Technologies used include LoRaWAN (Long Range Wide Area Network), Sigfox, or NB-IoT (Narrowband IoT).
- **Gateway/Receiver:** If LPWAN sensors are used, a gateway or receiver is required to collect data transmitted by the LPWAN sensors. This gateway may be integrated into the data logger and transmission devices.

In the context of the project, the weather station manufacturer provides to customers a variety of radio communication capable sensors. Those are under IEEE 802.15.4 standard which is the basis for LPWAN technologies like Zigbee and 6LoWPAN. Using this technology soil volumetric water content and water counter readings are collected from the near field.

Equipment used in ‘El Pinillo’:

1. Anemometer and wind vane
2. Visibilimeter sensor
3. Temperature and relative humidity sensor
4. Pyranometer
5. Rain gauge
6. Transmitter for a remote sensor

1. Select a Location

Choose a location that's free from obstructions like buildings and trees. This site should also have a stable ground for setting up.

2. Foundation and Structure Setup

Depending on your equipment, you may need to install a foundational structure as a pole or tripod mounted in concrete to provide a stable base for wind sensors.

3. Install the Sensors

Attach sensors to the structure. Position each sensor according to the specific instructions provided by the manufacturer. Be aware that you may need to place them before setting up the structure.

4. Set Up the Data Logger

Connect the sensors to a data logger following the manufacturer instructions.

5. Connect to Power and Data Networks

Connect the system to the power source (like batteries and solar panels) and set up a connection for data transmisión (through cables or a wireless network).

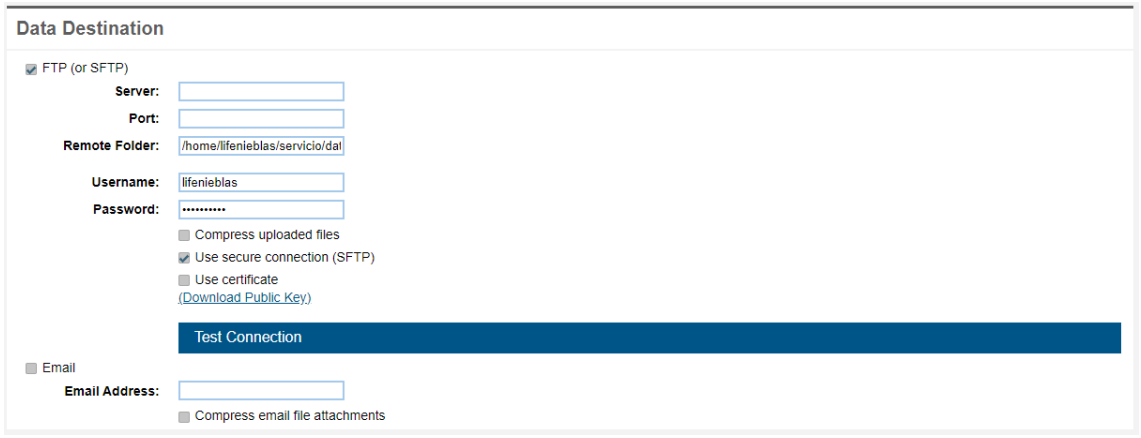
6. Test the System

Run tests to confirm that all parts of the station are functioning correctly and that data is accurately recorded and transmitted. It may be necessary and recommended to perform a calibration as described in the manufacturer's guidelines.

Setting up: video available by following the link: https://www.youtube.com/watch?v=In-1zLefj03A

To manage, safekeep and make use of the data generated by a weather station for further analysis, it is recommended to automatically or regularly import it into a time series database. Such databases, which includes InfluxDB, TimescaleDB or Prometheus are optimised for handling time-stamped data, such as weather measurements. Additionally, employing a dedicated tool for data visualisation, such as Grafana or Tableau, can help to monitor weather patterns and draw insights from the data.

In the Life Nieblas project, the SFPT automatically delivery provided by the manufacturer is used to regularly import into a InfluxDB database and Grafana dashboards are used to explore the data.



To replicate the same data infrastructure in your system you can adapt the docker-compose YAML file that is used in Life Nieblas:

```
version: '3'
services:
  grafana:
    container_name: grafana
    image: grafana/grafana:latest-ubuntu
    ports:
      - 80:3000
    environment:
      GF_RENDERING_SERVER_URL: http://
      renderer:8081/render
      GF_RENDERING_CALLBACK_URL:
      http://grafana:3000/
    volumes:
      - /home/user/docker/grafana/
        data:/var/lib/grafana
      - /home/user/docker/grafana/
        config:/etc/grafana
    restart: unless-stopped
  renderer:
    container_name: renderer
    image: grafana/grafana-
    image-renderer:latest
    ports:
      - 8081
    restart: unless-stopped
  influxdb2:
    container_name: influxdb2
    image: influxdb:latest
    ports:
      - 8086:8086
    volumes:
      - /home/user/docker/influxdb2/
        data:/var/lib/influxdb2
      - /home/user/docker/influxdb2/
        config/:/etc/influxdb2
    environment:
      - DOCKER_INFLUXDB_INIT_MODE=setup
      - DOCKER_INFLUXDB_INIT_
        USERNAME=influxdb_username
      - DOCKER_INFLUXDB_INIT_
        PASSWORD=influxdb_password
      - DOCKER_INFLUXDB_INIT_ORG=org_name
      - DOCKER_INFLUXDB_INIT_
        BUCKET=init_bucket
    restart: unless-stopped
```

Docker is a tool that helps developers pack everything an application needs to run, like the code and the environment settings, into a single package called a container. This makes it easier to move the application between different computers and ensures it works the same way everywhere. Additionally, these packages can be published and shared making it available for others to use. That's the case of InfluxDB and Grafana.

Analysis

The analysis of meteorological data is discussed in the following section with the fog water collection analysis altogether.

2. Water collection monitoring

Water flow

Device

The collection of water flow data by FWCs is an essential process for assessing and analysing the amount of water collected. Although a water metre can be used to measure the amount of water collected, there may be limitations due to the slim stream rate nature of water collected by FWCs. To reduce the margin of error, a small 30-litre Autonomous Discharge System (AFDS) is installed upstream from the water metre.

The AFDS is a device designed to regulate the flow of water at low flow rates. Its operation is described in detail in the next section B.2.2 and its installation is explained in section B.3.2.

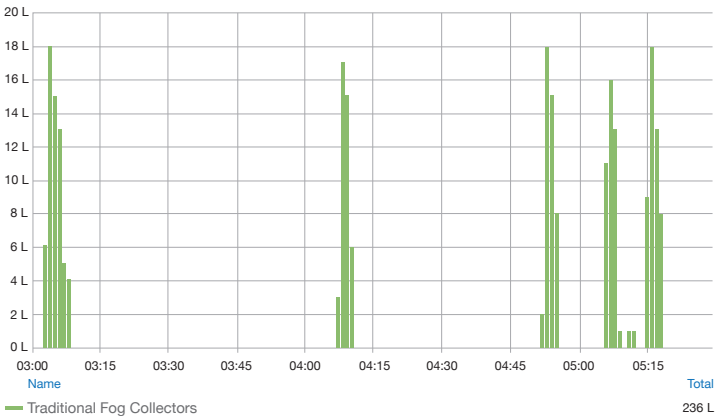
However, it is important to note that not all water inlet points require being equipped with a water metre. In the case of volumetric and innovative FWCs, a visual analysis is carried out to estimate the quantity of water entering daily. This estimate can be made using the graduation of a water tank or by using a scale that is immersed in the tank on a daily basis, as illustrated in the photo above.

Accurate measurements of fog water collection via water meter with the help of a mini AFDS.

Analysis

Once the appropriate water metres have been installed, the daily data collected can be recorded in an Excel spreadsheet or other data management system (for high frequency data acquisition system, time series databases are recommended). This data can be analysed in a number of ways to assess the efficiency of the different water collectors used.

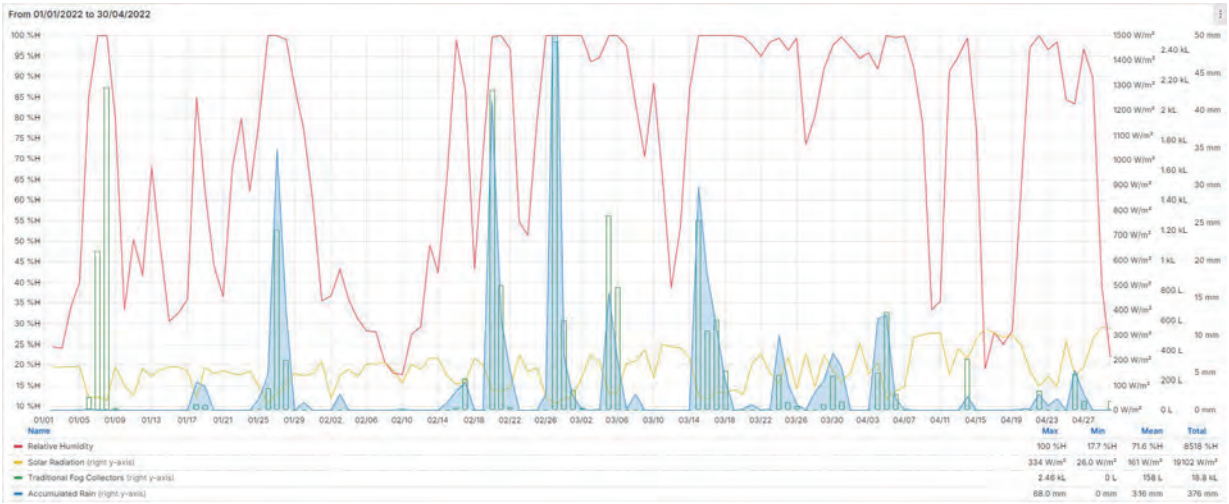
Fog collectors counters from 02/03/2024 to 02/03/2024



Using the daily data collected, it is possible to separate the different types of collectors to assess their respective efficiency. Trend curves can be drawn up by season and month to analyse the performance of the collectors throughout the year.

In addition, to assess collector efficiency, it is possible to calculate the impact surface area of each collector and divide the total quantity of water collected by the total impact surface area. This calculation gives the collector yield in litres per square metre (L/m²), which provides a comparative measure for assessing the efficiency of the different collectors used.

Here are plotted the data collected through the meteorological station from the daily water collected from traditional fog water collectors (using a radio communicating water counter), the daily accumulated rain and the daily average relative humidity, solar irradiance and, for 2024, the visibility:



The graphs show a correlation between high relative humidity, low irradiance, and low visibility with fog collection phenomena. These conditions are consistent with the presence of fog, the main factor for fog capture. It is observed that while these conditions frequently occur during fog collection, they are not solely sufficient. Additionally, when rainfall data is plotted, it shows that precipitation commonly accompanies these conditions, yet water collection can occur with or without rain.

Water collected as measured by water meters

Period	Fog water collection registered by the water meters
January-April 2022	18,80 m^3
May-August 2022	0,37 m^3
September-December 2022	5,66 m^3
January-April 2023	8,60 m^3
May-August 2023	0,56 m^3
September-December 2023	6,30 m^3
January-April 2024	3,46 m^3

Results and conclusion

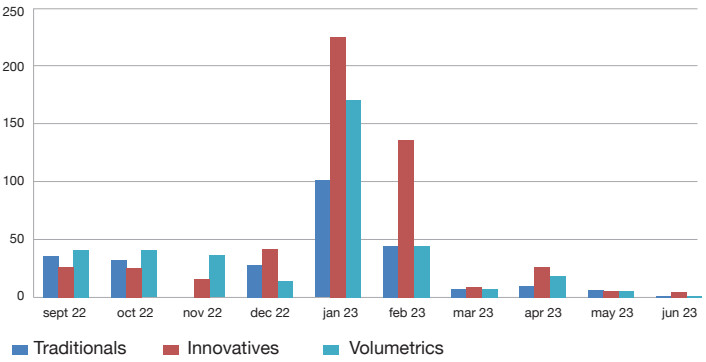
Collector efficiency

	From Sept 20th of 2022 to July 15th of 2023		
Collectors	FWC	i-FWC	V-FWC
Number of collectors	11	15	1
Impact surface (m^2)	8	0,63	8
Total surface (m^2)	88	9,45	8
Total collected (L)	24.339	4.964	3.035
L/m^2	277	525	379

Seasonal trend

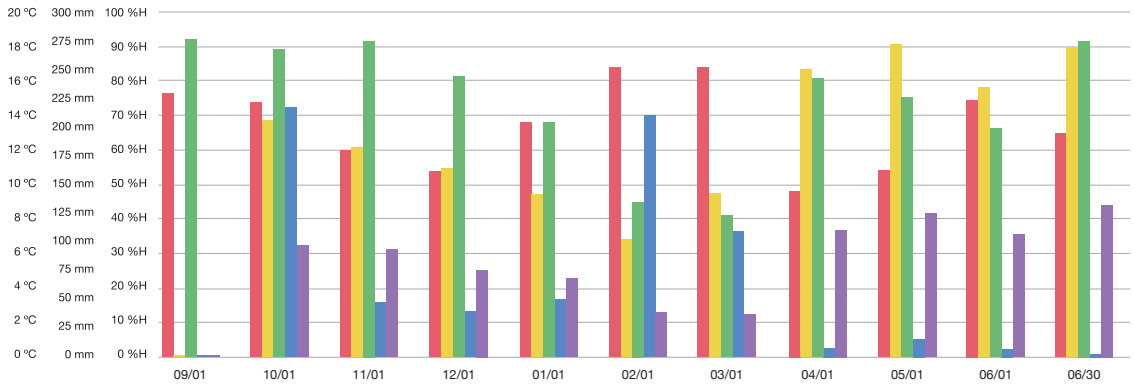
During the Life Nieblas project, daily monitoring of water collection was conducted using three different fog collection methods. The results clearly indicate that the innovative collection methods significantly outperform the traditional ones in terms of efficiency. Additionally, employing a volumetric mesh within the same structure demonstrates even higher efficiency.

Water collected by the collectors from September 2022 (L/m²)



The second graphic reveals a pronounced seasonal trend in the collected water quantities. January and February stand out as the months with the highest water yields, while July and August represent the driest months of the year.

Meteorological data from 01/09/2022 to 30/06/2023



Name	Min	Max	Mean	Total
Relative Humidity	48,2 %H	84,3 %H	67,5 %H	743 %H
Solar Radiation	1 mm	272 mm	179 mm	1.97 m
Temperature	8,16 °C	18,4 °C	15,0 °C	165 °C
Accumulared Rain	0 mm	217 mm	63,9 mm	703 mm
Total Evapotranspiration	0 mm	132 mm	80,4 mm	884 mm

Regarding the meteorological data for the same period, the charts show notable seasonal variability, with peaks in precipitation and collection during the months of heavier rainfall. Relative humidity and temperature exhibit significant fluctuations throughout the months, while total evapotranspiration also varies, showing a correlation with temperature. There is an observed relationship between the peaks in water collection and increases in relative humidity, which is expected since higher humidity facilitates the condensation of microdroplets. Additionally, temperature and solar radiation appear to influence the observed decrease in water collection starting in March.

E. IMPACT EVALUATION

1. Carbon footprint and Life Cycle assessment

Presentation

The main objective of this LCA is to obtain an estimate of the environmental impact generated in the life cycle of the different fog collection systems installed by the project.

A complete stroke is performed, from the cradle to the gate, that is, it is considered the reception of materials necessary for the construction and installation of the different systems and the operation and maintenance of them.

This LCA can contribute to making project decisions. Different results are generated with these methods. To compare this results 2 indicators are used:

- CO_2 emissions annual reduction comparison between the water captured with fog collectors respect to its obtention from water wells and is transport by road to the field
- CO_2 emissions annual reduction comparison between the water captured with fog collectors respect to its obtention from water deposits that do not require machinery for its extraction and its transportation by road to the field.

The necessary data would be:

- The consumption of the tank truck that supplies the tanks when the water is brought from the outside.
- The energy consumption required to pump water from a medium-deep well in the area for subsequent transport by tanker truck.

To assess this case the **SimaPro version 9.1.1** is used, with the **Ecoinvent v3.7.1** database. And the methodology used for the calculation of the carbon footprint has been the **IPCC 2013 GWP 20a V1.03**., which was developed by the Intergovernmental Panel on Climate Change. It contains the climate change factors of IPCC with a timeframe of 20 years. Accordingly, we create different scenarios to calculate the CO_2 emissions saved: 150 and 300 metres deep of the well, transport from the well and transport from a water body.

Results

For the first period of the project (April 2021 - February 2023) the water collected by fog water collectors was **86,500 litres**. With this frame we can calculate the CO_2 with different scenarios.

The conditions used were that the well is 150 or 300 metres deep (average in Gran Canaria) and the transportation is by water tank truck for approximately 20.4 kilometres.

The reference data used to assess the impact of the transport of water are as follows:

- Travel distance: 25km (Well in Arucas – Finca el Pinillo – Valleseco) = 50 km daily (round trip).
- Well water pumping, the following conditions are taken into consideration:
  - Q=10l/s =32 m^3/h
  - Well deep =150 – 300 meters
  - Power pump = 26 kW
  - In this scenario to pump each 1,000 litres is needed 0.78 – 1.56 kWh

Also, if it is considered we can take into account the visits to the area with the operators to receive the water tank truck. With these data the results till February 2023 the results are estimated:

CO_2 emissions reduction

Indicators	CO_2 emissions reduction till February 2023
CO_2 emissions reduction comparison between water collected by fog water collectors (86,5 m^3) in respect to obtaining water through wells and its transport by road to the field.	601 kg of CO_2
CO_2 emissions annual reduction comparison between the water captured with fog collectors respect to its transportation by road to the field (without well pumping).	545,2 kg of CO_2

Project indicators related to climate objectives and CO_2 emissions with 86,5 m^3 fog water collected and using LCA method.

In fact, during this first period of the project more emissions have been saved, since all the water that has not been collected from mist has been pumped from the spring located in the area. To calculate this we evaluate the savings regarding using the spring water. We can see the real savings in CO_2 emissions:

CO_2 emissions reduction

Indicators	CO_2 emissions reduction till February 2023
CO_2 emissions reduction comparison between water collected by fog water collectors (86,5 m^3) and by local spring (303,5 m^3) respect to obtaining water through wells and its transport by road to the field.	601 kg + 2,120 kg – 38,8 kg[1]= **2,682.2 kg of CO_2**
CO_2 emissions annual reduction comparison between the water captured with fog collectors and by local spring respect to its transportation by road to the field (without well pumping)	545.2 kg +1,760 kg – 38,8 kg = **2,266.4 kg of CO_2**

LIFE NIEBLAS project indicators related to climate objectives and CO_2 emissions with 86,5 m^3 fog water collected and 303,5 m^3 spring water, using LCA method.

Conclusion

After analysis of the data, some conclusions of the study can be established.

- The use of metal materials (galvanised steel, iron bars, etc.) in the construction of the elements that constitute the irrigation systems represents a significant increase in the equivalent CO_2 emissions associated with the life cycle of the installations.
- More than 96 % of the emissions associated with traditional FWC correspond to the metal elements of which they are composed. In the case of IFWC, it accounts for 98% and 76% in the i-FWC. Any innovation involving replacing metallic materials with others will significantly reduce the installation's carbon footprint.

2. Social impact

Presentation

Monitoring the socioeconomic impact of the project is as important as monitoring the environmental impact. The effect of the project on end-users will be essential to evaluate the overall benefits. The socioeconomic perception of the actions in all areas will be monitored during the project's lifetime on the local economy and population. The continuous evaluation of the social and economic effects of water collection actions on the local population will allow the project to act in case of deviation which could have consequences on the project's objectives and the expected results.

The following indicators are proposed:

- Job generated directly by the project (personnel hired for the tasks execution).
- Job directly affected by the project (beneficiaries pre-existent personnel with tasks assigned).
- Job indirectly affected by the project (total personnel of each beneficiary).
- Number of companies and professionals directly hired as an external assistance by the project.
- Number of jobs created or affected by replications of the different water collection methods use during the project.
- People are impacted during talks, workshops or different events.
- Farmers and workers who will receive direct information of the project.
- Farmers or any people who will receive water collected during the project.

To bring all this information together, several graphs, tables, maps and charts will be produced in order to visually represent the socioeconomic activity that the project generated.

II. Habitat restoration

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B. Methodology for assessing restoration feasibility 79

C. Field implementation 111

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E. Impact evaluation 179

Sustainable habitat restoration is key when advocating for both nature recovery and the mitigation of climate change.

“It’s surely our responsibility to do everything within our power to create a planet that provides a home not just for us, but for all life on Earth.”

Sir David Attenborough

A. OVERVIEW

1. What are habitats?

Following the Rio Earth Summit (1992) and subsequent ministerial conferences on forest protection, the European Union adopted Directive 92/43/EEC on the conservation of natural habitats and wild fauna and flora in 1992 (Council of Europe, 1992). It recognizes biodiversity and its preservation as an essential objective of general interest. It stipulates that Member States commit to contribute to the European objectives for biodiversity conservation through the conservation of natural habitats, wild fauna, and flora.

From this directive, the Natura 2000 network emerged. It is a European network of protected natural sites created to preserve biodiversity and natural habitats in Europe through the conservation of habitats, wild fauna, and flora.

Definition of habitats

According to Natura 2000, habitats are terrestrial or aquatic areas with distinct geographic, abiotic, and biotic characteristics, whether they are natural or semi-natural. A habitat is synonymous with an ecosystem, comprising an ecologically homogeneous area with vegetation, associated fauna, and a station compartment defined by specific climatic conditions, soil, and parental material. These habitats are essential for biodiversity conservation and ecosystem functioning.

2. Presentation of the importance of restoration, what does it bring?

Habitat Degradation

Unfortunately, natural habitats face significant challenges and are currently in a degraded state. Recent scientific research has demonstrated such degradation, which manifests as habitat destruction by pollution, climate change, land use changes and excessive resource exploitation among other reasons. These factors have led to biodiversity loss, a decline in ecosystem variety, and ecological imbalance.

The consequences of this degradation are manifold. Habitat degradation could directly impact ecosystem services provision provided to humans by nature, from improved water quality and food production to climate regulation, including also non-material benefits such as recreational benefits. Additionally, natural habitats, such as wooded areas, play a crucial role in fire regulation. When degraded, they become more vulnerable to wildfires, endangering ecosystems and surrounding communities. Furthermore, habitat degradation contributes to desertification, compromising the fertility and integrity of terrestrial ecosystems.

Importance of habitat restoration

The restoration of natural habitats is of crucial importance in reversing these negative trends. By restoring degraded habitats, we can restore ecological balance, prevent biodiversity loss, and promote ecosystem resilience. Habitat restoration efforts may include reforestation, wetland rehabilitation, protection and conservation of coral reefs, as well as measures to reduce pollution and promote sustainable resource use.

Habitat restoration also strengthens fire resistance, prevents soil erosion, and facilitates the regeneration of damaged ecosystems. Furthermore, it contributes to the preservation of essential ecosystem services such as water purification, climate regulation, and the provision of habitats for a diverse range of plant and animal species (MARILL, Robert 1992).

B. METHODOLOGY FOR ASSESSING RESTORATION FEASIBILITY

Before undertaking a restoration project, it is essential to carry out a preliminary diagnosis. This diagnosis must start with determining the issues regarding habitat degradation which could be many. The most common degradation problems are landscape modification, aridification, forest fires, overgrazing and soil erosion by water and wind. After determining the problem, the most important thing is to analyse the immediate and underlying causes of the problem, the drivers of land degradation processes (a.e. arable land abandonment, pollution, waste dumping, poor land use planning, unsustainable farming, bad management practices). This analysis of the problem and the drivers of degradation should include as much as possible a consultative process with the main stakeholders, especially those mainly involved in the management of the area like landowners, farmers and area managers. Obviously, land property is one of the most important factors when planning a restoration project and it is crucial to have landowners on board, especially when those are private, as it happens in some rural areas where the property is extremely fragmented.

After determining the problem and drivers of degradation, a justification of why restoration is appropriate and likely to be successful is needed. For this it is important to again involve stakeholders in the definition of restoration objectives and outcomes, and after this objective is clear, analyse which actions must be taken. Restoration could sometimes imply no action since an action in the wrong moment and the wrong place could be pernicious, especially in areas affected by forest fires.

There are various methods for restoring, but some are more appropriate than others, depending on the location of the project. First of all, it is necessary to analyse the site, examining the type of soil, its quality, the vegetation cover and the relief. It is also important to analyse the flora and fauna present on the site to ensure that reforestation is appropriate. This part of the guide will explain how to carry out such analysis and what parameters to consider.

Various reforestation techniques will then be presented. For each technique, a technical description of the tool will be provided, as well as an explanation of the best environmental conditions for its use. The materials required, their origin, design and cost will also be described.

Next, a section will be devoted to the selection of appropriate plants for the project. The plant selection criteria will be explained, considering the parameters analysed above.

Finally, a section will be devoted to the feasibility of the project in terms of time and materials. It is essential to ensure that there is enough time available, to take into account the surface area required and to understand the investment that this type of project requires. In this respect, the Life Nieblas project has developed a platform, already presented in an earlier part of the guide, which enables reforestation criteria to be added. This is a decision-making tool that makes it easier to assess the feasibility of the project.

1. Site assessment

Climate and pluviometry

Climate analysis is an essential part of any reforestation project. Understanding the climatic characteristics of a given region helps to determine which types of trees and plants are suited to that specific environment. There are several factors to consider when analysing the climate for a reforestation project.

First, it's important to consider heat and drought in summer. Heat activates the biological functions of plants, promoting the absorption of mineral elements present in the soil. However, if there is a shortage of water, plants react by limiting their activity, notably by closing the stomata that regulate evapotranspiration from the leaves. This halts plant growth and can lead to death, as heat and drought accelerate the depletion of water resources in the soil. Hydric stress would be one of the main limiting factors when planning a plantation, since it would condition the survival of the seedlings especially during the first years of establishment.

The presence of steep slopes, specially in human-made geomorphologies, like those existing in restored mines following the slope-berm model, could carry serious erosion problems.

Another factor to consider is the presence of heavy rainfall in the cold season. This precipitation cannot be fully exploited by deciduous species, as water can only be absorbed by the roots if the leaves are present to evapotranspire it. Similarly, evergreen species can be affected by cold, which paralyses their biological activity.

It is also important to take account of local variations. Altitude, for example, leads to changes in rainfall and average temperature. Regions at higher altitudes can benefit from higher rainfall and lower average temperatures. In addition, cold, drying winds can have a different impact in different regions.

Meteorological indicators proposed in D1 section will be helpful to take decisions about the species selection. Those indicators will be also interesting during the implementation and monitoring part of the project, to better interpret the results of the monitoring of the plantations.

Topography, relief

The relief of the area is another important factor that could condition reforestation. Mountainous areas with steep slopes could heavily difficult the execution of restoration actions. The presence of rocky walls could make it impossible to access certain areas. It is important to have a detailed cartography of the area in order to detect critical areas where restoration actions must be avoided or must be afforded with special equipment in order to ensure the safety of the workers.

Moreover, geomorphology will also condition the decision about species to plant. Ridges and summit areas, with poor soils will have different vegetation than slopes, and at the same time those would have different vegetation than deep valley areas, with deeper soils and usually with more water available. In steeped slopes soil stability must be guaranteed before planting in or-

der to avoid erosion problems in the future. When erosion problems or instabilities are detected sowing with fast growing herbaceous species could be planned, despite the use of invasive or alien species must be completely avoided. In heavily anthropized areas, the presence of human-made structures conditioning the topography must be taken into account (a.e. dry stone walls, terraces, old mines). Sun exposure is another factor to consider. Regions facing north tend to be colder and wetter, while those facing south tend to be warmer and drier.

To have a detailed topography (scale 1:5000) of the area will help in the decision process. Moreover, orthophotography from previous years (before and after the degradation event) could help in the decision process about species and specific planting actions.

Soil

To know the tendency in soil quality, a detailed physicochemical characterization is required before planting and at the end of the project. The first characterization will be useful for the decision making about the planting areas, the species to plant and the planting methods to use in each area. After plantation, this data will help in the discussion of vegetation development results. The second characterization will be focused in determining C storage in soil; a complete characterization is proposed to fully benefit from the sampling campaign.

First of all, it is important to identify the presence of different soil types. For this we can use existing soil maps, despite this information usually is not available at the required scale (1:25.000). However, general soil maps of the area and geological maps could help to identify the main soil units. In each soil unit at least one soil profile must be described, identifying and analysing the main horizons. This work must be done by an edaphologist or a pedologist due to the specific knowledge needed.

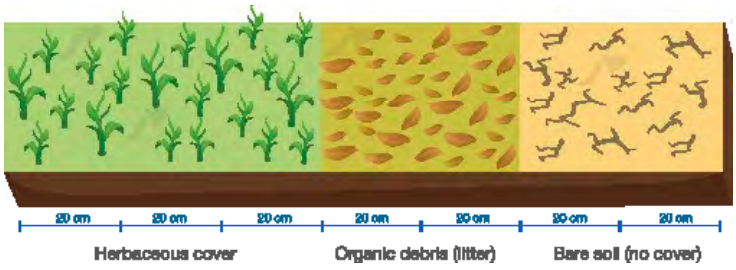
Soil profiles in Carregal do Sal (left) and Vouzela (right) plantations (Portugal).

Vegetation characterization

The growth of spontaneous and planted vegetation is an indicator of the restoration of the site. The evaluation of the structure and composition of this vegetation will allow determining the degree of regeneration (passive restoration) of the degraded habitats.

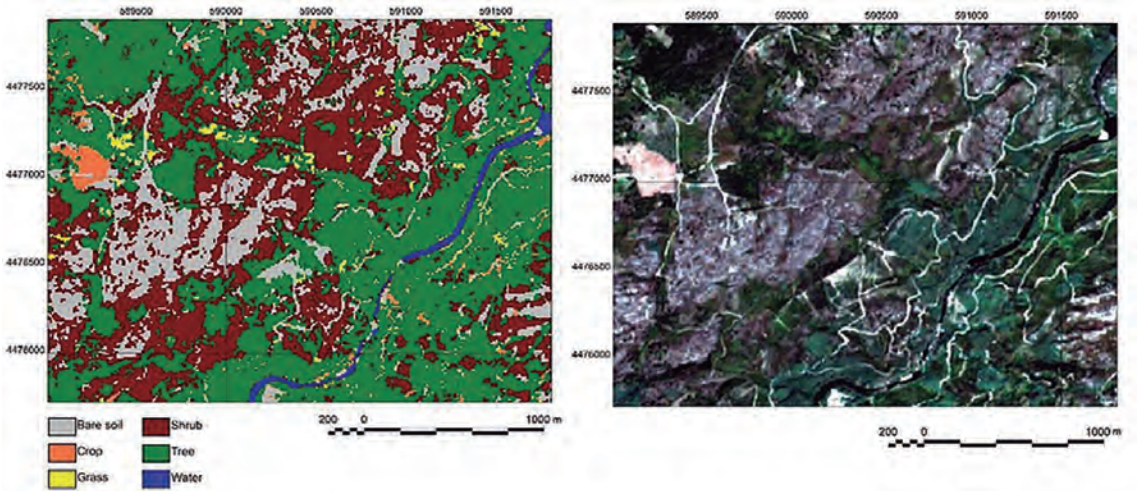
Structure will be evaluated quantifying cover types in each site or subsite (differencing site-specific variables: flat vs slopes, soil types, etc.). It is necessary to distinguish between herbaceous cover, woody species cover (shrubs and trees), other soil cover types (plant remains and rocks) and bare soil (no cover). The method proposed is the point transect (figure below), where the evaluator records the cover type every 20 cm in a 25 cm transect, measuring the height when herbaceous or woody vegetation exist. A form can be found in the report annex on action A.2 Environmental, social and economic indicators selection. Monitoring and evolution protocol at https://lifenieblas.com/documents. Vegetation height measures must be noted in the cells, and a tick must be put when plant remains or rocks are present.

A minimum of 3 transects per hectare must be done to have representative measurements. In slopes, transects must be done parallel and perpendicular to the slopes. Transect must be fixed with metal bars to repeat measures at the same place. It is recommendable to sketch a map with transects localization or localise transect vertex using a differential GPS or similar.



Different soil covers (herbaceous, litter, bare soil) in an evaluation transect.

This vegetation characterization could be done also using remote sensing techniques. Available satellite imagery could be used to determine the main land covers using vegetation indices. If it is possible, the use of drones could be very useful to obtain high resolution images that will significantly improve the monitoring. More information about the use of remote sensing in monitoring degradation and restoration processes could be found in LIFE NewLife4Drylands project (www.newlife4drylands.eu).



Land cover classification in Carregal do Sal area (Portugal) using Knn algorithm in Sentinel2 images (2021).

Eucalyptus resprouts appeared in a planting area after clearing, Vouzela (Portugal).

To determine vegetation composition, all plant species must be identified in each site or subsite (according to site specific variability: flat vs slopes, soil types, etc.). The evaluator could take advantage of the transects to do this despite a general walk for the site or subsite will be needed to detect all the plants (not all the existing plants will appear in the transects). For each plant, abundance estimation must be recorded, using this pattern ranks: testimonial, low frequency (<25% soil cover), high frequency (25%-75% soil cover) and dominant (75% soil cover). A form can be found in the report annex on action A.2 Environmental, social and economic indicators selection. Monitoring and evolution protocol at https://lifenieblas.com/documents.

The presence of known pests and disease must be recorded and can be valuable information for determining the target habitat or at least guide the selection of the planted species. The plantation of species heavily affected by plant diseases in the surroundings of the plantation area could contribute to the expansion of the pest.

Restoration projects must avoid interfering in passive restoration projects unless the reference system differs from the succession community (a.e. substitution for other species more resilient to forest fires, elimination of alien or invasive species). Also, it is important to avoid impacts in endangered or protected species and habitats. In the case that invasive species have been identified, an eradication plan must be prepared. In case that the presence of invasive species is significant in the area, the execution of the eradication plan must be guaranteed previously to the plantation phase, in order to avoid negative effects on the plantations of the treatments for eradicating alien species (a.e. herbicide application).

Fauna

Apart from vegetation it is very important to consider also the fauna present in the area, specially this that can cause severe damages to the plantation like rabbits, goats or wild boards. Actions must be taken in order to guarantee that the plantations would be efficiently protected from this fauna. There are different options ranging from dispositives for run away (repellents, sound cannons, reflectants, etc.) to different types of fences (hunting fences, electric fences, individual fences, etc.). In any case it is important to report the presence of this type of fauna and its abundance in order to prevent a contingency plan. This fact could heavily condition the plantation and in some cases could be a factor for avoiding planting interventions if the problem is not previously solved.

Land use and management practices

Knowledge of land use and management practices prior to the impact can provide valuable information for the planning of restoration actions. In fact, this knowledge should be traced back to situations prior to the impacted system, as it may give clues about alternative reference systems that may be as or more interesting than the pre-impact one. An example of this can be given in long-term abandoned farmland.The presence of previous plantations or crops must be recorded. This could be done using available orthoimages. However, the most valuable information will flourish in participatory processes and interviews where the local population could inform about the previous situation before the impact. During these participatory processes and interviews, management practices must also be recorded. An example of the importance of those management practices is the use of fire in alpine meadows, that could condition the evolution of soil properties; or clearing and grazing historical practices in mediterranean forest, that could give information about potential reference sites for restoration.

Logistic factors

Considering logistic factors like site access for vehicles, access to water, fertilisation available or maintenance needs are also crucial when planning a restoration project. The feasibility to apply some methodologies could be heavily conditioned by the existence of paths or roads that guarantee that the transportation of all the goods and services linked to the restoration project could be done in normal vehicles (cars, vans, trucks). In case that special vehicles are needed (helicopter) or that the transportation must be done by food this will heavily impact the project budget.

Access to water will also condition irrigation methods, dotation, and frequency. However, in our case the use of fog water collection technologies could significantly reduce the water related costs in isolated areas where water is not available. In any case, from fog or natural sources or external it is important to consider the amount of water available to correctly plan settlement, maintenance, and support waterings. Also, it is important to consider the effect in trapping the run-off linked to some planting technologies and techniques (a.e. Cocoon, micro basin).

In some cases, adding organic amendments at the planting hole could be recommendable to increase the nourishment of the plant, promoting the biological activity and increasing the water holding capacity in the surrounding of the root system. This could be done by applying compost or other commercial products, although in rural areas manure could be available in large quantities and it could be also used. It must be also interesting to use organic amendments for restoring soil properties and promoting revegetation in poorly revegetated slopes, sometimes affected by erosion or instabilities. In any case, the use of products or organic wastes accomplishing with the quality parameters must be guaranteed (low pollutants level, no impurities, stable products) and excessive doses avoided (1-2 L per planting hole, 50 Mg/ha for surface applications).

2. Reforestation methodology presentation

Reforestation is an operation that involves creating wooded areas or forests that have been removed or destroyed in the past. It makes it possible to establish a forest population that meets various objectives, such as protecting soils and ecosystems, stabilising pollution, or sequestering carbon, but can also be used for timber production, creating a place for leisure activities, etc. It is therefore essential to ask the following question: what are the precise objectives of the project? Among these objectives, the rapid growth of trees to fulfil their function and free themselves from the lower stratum vulnerable to fire, as well as permanent vigour for better resistance to climatic hazards, pests and fire, are points to consider.

To achieve the objectives set, two main conditions must be met:

1. **Selection of suitable species**: It is essential to choose the species that are best suited to the site. This will depend mainly on the specific characteristics of the climate and soil. It is important to determine the plant requirements for the project and to consider what will need to be added.
2. **Use of techniques to encourage plant development**: The methods used to introduce plants are crucial to maximising their development. It is important to choose appropriate techniques and methods that will promote plant growth and survival.

When planning a reforestation project, it is crucial to take several factors into account in order to choose the most appropriate method. Each reforestation technique has its own specific characteristics, needs, costs and implementation and maintenance timescales.

Soil type, topography, the extent of spontaneous vegetation and climatic conditions are all determining factors in the choice of the most appropriate reforestation technique.

Before deciding which plants and reforestation method to use, it is important to consider the following point:

- **Availability of resources**: It is essential to consider the availability of the resources needed to implement the chosen reforestation technique. This includes plants, seeds, soil amendment materials, human resources, etc.
- **Water resources**: Consider the availability and quality of water in the area. Some reforestation techniques may require regular access to water for irrigation, which can pose challenges in areas where water resources are limited. Make sure you choose a technique that is adapted to local water conditions.
- **Soil quality**: If the soil has water retention problems or is easily flooded, it is important to choose methods that encourage water absorption or to take specific measures to remedy these problems.

- **Costs**: The budget available for the reforestation project is a determining factor in the choice of technique. Some techniques may be more expensive in terms of the labour, materials or machinery required.
- **Access to the land**: Easy access to the plants for watering and other maintenance operations must be taken into account.
- **Environmental impact**: It is crucial to assess the environmental impact of the reforestation technique envisaged. Use methods that respect the local ecosystem, minimise soil disturbance, preserve biodiversity and use sustainable practices.
- **Restoring habitat and biodiversity**: If reforestation is also aimed at restoring habitat and conserving biodiversity, choose techniques and species combinations that encourage the reintroduction of native species and the creation of diverse habitats to support local flora and fauna.
- **Monitoring and evaluation**: It is important to consider introducing regular monitoring and evaluation of the effectiveness of the reforestation technique used. This will make it possible to adjust practices if necessary and optimise results in the long term.
- **Adaptability to climate change**: Given the ongoing climate changes, it is essential to consider the future resilience of plantations.
- **Seasonality**: Climate plays a decisive role in determining the best time for planting. Newly planted trees grow best when exposed to moderate temperatures and adequate rainfall. Plants need time to root and acclimatise before the intense heat and drought of summer, or the freezing temperatures of winter, arrive. Early spring or the start of the monsoon season is therefore the best time for planting. In regions with harsh winters, it is best to plant before the snow arrives.

The presentation of the different methods explained below will enable you to better determine, depending on your conditions, which method is best suited to your situation.

Traditional reforestation

Presentation

Traditional reforestation is a reforestation method used as a control group in the Life Nieblas project, making it possible to establish a database for comparing the results obtained in the field with other reforestation methodologies. This approach involves planting individual plants in holes measuring 40x40x40 cm, without any additional tillage. A minimum distance of 1.50 metres is maintained between each plant to encourage their growth and development. Irrigation is provided by supplying water to each plant using PE pipes.

One of the main features of traditional reforestation is that no additional equipment is required for its implementation. This method is widely considered to be suitable for reforesting degraded soils, despite its role as a control method in the project. It allows the effects of different reforestation approaches to be assessed and conclusions drawn about their respective effectiveness.

Location

Due to the nature of the soil, these measures must be adapted to the peculiarities of each area. For example, there are areas where the soil is poor or very stony, which avoids the standard 40 cm deep planting hole.

Amount of water

Traditional reforestation requires regular irrigation in the first year to ensure the survival and growth of the plants. Irrigation water is delivered individually to the base of each plant using PE pipes. The amount of water required will depend on the specifics of the site and the water requirements of the plant species selected. It is essential to keep a close eye on soil moisture and to provide an adequate amount of water to promote healthy growth.

Cost

Traditional reforestation does not require any significant additional costs. This method is mainly based on the use of locally available plants and land, as well as simple irrigation equipment.

End-of-life

Traditional reforestation does not generate any additional waste or material requiring end-of-life management or specific recycling. As no additional material is used in this method, end-of-life management is limited to the management of mature plants with the possibly the collection of their seeds for future reforestation projects.

Traditional Performance in Life Nieblas	
Planted trees:	3002 trees
Survival rate:	33%
Cost per surviving tree:	136 euros
Need of water per trees:	53 L
Carbon footprint:	with FWC: 12,2 kg CO_2 eq. with tank truck: 0,4 kg CO_2 eq.

Cocoons

Presentation

The Cocoon system is a self-contained and biodegradable nurturing system that is installed individually burrowed under every single tree, providing not only a humid and cool environment but also weather and herbivore protection. Cocoon's deposits can contain 20L of water that will slowly dissolve into the soil with very little evaporation rate since they are insulated underground.

This irrigation method was developed by a previous Life project (The Life Green Link) which was also tested in Gran Canaria at high drought risk areas in the south of the island. One (and perhaps most relevant) advantage of this system is that they do not require constant monitoring or regular irrigation, reducing recurring human action and economic investment (less water usage, no irrigation systems, etc.).

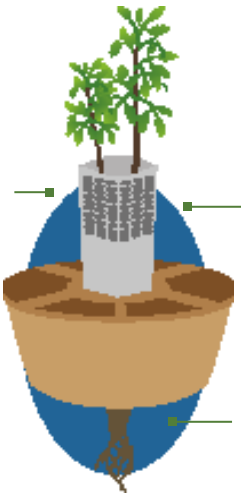
Additionally, the biodegradable materials they are built in are cheaper than synthetic ones used previously in similar systems. Another advantage of the Cocoons is that through its natural degradation, its structure components i.e. crop residues, pastures and other organic compounds will provide a sustained source of nutrients throughout the first couple of years of the seedlings development, increasing success rate and encouraging forest full establishment.

Material

How it works

The Cocoon is filled with **20 liters of water** and buried subsurface with the seedling.

The Cocoon **prevents water evaporation** and **weed growth.**



Seedlings are protected from harsh rays, desicating winds and small animals.

Controlled seepage of water straight to the roots, encouraging a **deep, wide root system.**

Diagram of a Cocoon system and its advantages.

Location

One advice is to avoid very humid areas, as the equipment is likely to deteriorate faster than in arid areas due to insects and small vertebrate activity. In addition, a very sloping landscape is not advantageous for cocoons, as they run the risk of being planted at an incline and spilling out some of the water.

Example of early degradation of the cocoon:

Amount of water

Reforestation with Cocoons requires irrigation during the first year to ensure the survival and growth of the plants. Irrigation water is delivered individually to each plant using PE pipes.

The plant is initially irrigated on its base and the reservoir is filled with 20L. Afterwards, different irrigations will be carried out if necessary during the first year of growth. It is essential to keep a close eye on soil moisture and to provide an adequate amount of water to promote healthy growth.

Cost

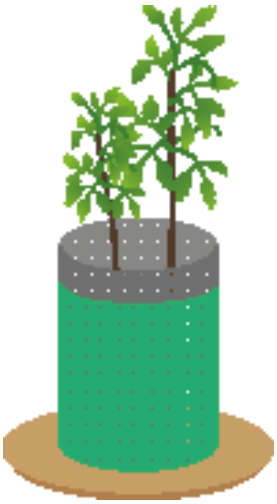
The cocoon is a commercial system, so it can be bought from various companies. It costs between €7 and €8 per unit, without shipping and handling.

For Life Nieblas, the cocoons cost 7.8 each.

End-of-life

No reuse is possible with a cocoon, it is a single use because the cocoon degrades in the soil providing nutrients to the plant.

Cocoon Performance in Life Nieblas	
Planted trees:	1993 trees
Survival rate:	68%
Cost per surviving tree:	109 euros
Need of water per trees:	49 L
Carbon footprint:	with FWC: 12,2 kg CO_2 eq. with tank truck: 1,5 kg CO_2 eq.



Individual fog water collector (IFWC)

Presentation

Technical observations have already indicated the ability of anti-herbivore metal mesh to reach the dew point with the effect of diurnal temperature variation (Life Guguy 2013).

To improve this effect and add robustness to the method, Life Nieblas has used a 25x25mm steel mesh to which a 3x3mm Russell type PE mesh has been superimposed, which is the same material used in traditional fog capture technology. Both layers provide the plants with water by condensation and also by impact (Shemenauer, Cereceda & Osses 2017). In addition, the outer layer provides the seedlings with shade and wind cover in the early stages of development.

To test the effectiveness of adding a second layer, the IFWC plantings were planted 50% with Russell netting and 50% without. But it was found that IFWCs with PE mesh had a much better survival rate than those with just the metallic one.

Material

The collectors are 1.5m high and 44.5cm in diameter. The steel mesh have hole of 25x25mm:

The Russel mesh type PE mesh is only 1m high with 44.5cm in diameter too. The diameter of the holes are 3 x 3mm:

Location

IFWC is a method directly dependent on fog affection. However, any rain event or drizzle will also result in higher humidity concentration within its structure. Additionally this method protects seedlings against excessive isolation and wind gusts. Therefore the IFWC is significantly more versatile than any other method tested in Life Nieblas. Nevertheless, good fog collection conditions should be the primary parameter to consider when implementing this method.

Amount of water

Even though the IFWC condenses water for the plant, this type of reforestation requires regular irrigation in the first year to ensure the survival and growth of the plants. Irrigation water is delivered individually to each plant using PE pipes. The amount of water required will depend on the specifics of the site and the water requirements of the plant species selected. It is essential to keep a close eye on soil moisture and to provide an adequate amount of water to promote healthy growth.

Cost

The cost of the material includes metal mesh and PE mesh. You should allow around 5 euros for metallic mesh and 1 euro for PE mesh for each tree.

i-FWC Performance in Life Nieblas	
Planted trees:	3787 trees
Survival rate:	74%
Cost per surviving tree:	59 euros
Need of water per trees:	30 L
Carbon footprint:	7,5 kg CO_2 eq.

End-of-life

The IFWC can be completely reused. Once the plant has developed sufficiently to draw water from deeper down and its aerial part is high enough to be inaccessible to pests, it is possible to leave the IFWC. In addition the IFWC can start to damage the plant's development if left installed for more than one year or even two years depending on plantation development. Such development must be gauged by the technical team in charge of the reforestations.

As long as the collector is in good condition, it can continue to be used. If the PE mesh is damaged, it can be replaced.

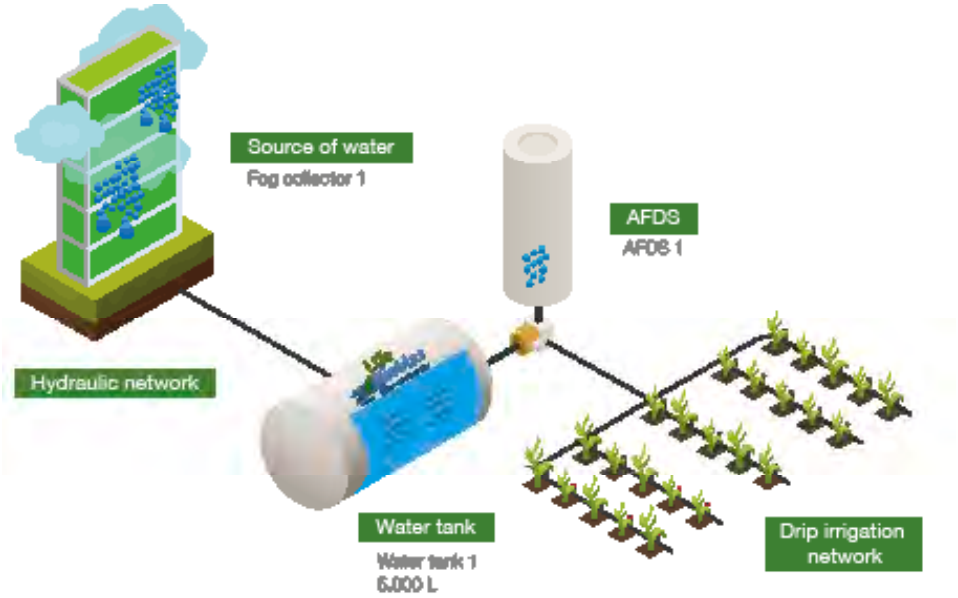
All you have to do is store them in a sheltered place until their next use. On the other hand you could pile them up next to the next reforestation area if such works are taking place immediately after IFWC removal.

Autonomous Fluid Discharge System

Presentation and general system description

The Autonomous Fluid Discharge System (AFDS) aims to grow plants that are difficult to access autonomously, without energy or human intervention. It was invented from the original idea of a farmer called Emilio Navarro Rosario. He has worked on the project collaborating in the development and monitoring of these systems together with ITC staff. As a farmer in remote places, with difficult access and scarcity of water, Emilio had to overcome the difficulties of the environment by making time compatible with other jobs. This intelligent solution has not only saved you time, but has also given you autonomy and a considerable reduction in energy use.

The AFDS reforestation systems will be aimed to supply water through a drip irrigation system. For this purpose, reforestation systems with AFDS have been assigned to supply specific irrigation areas. The general reforestation system with AFDS and drip irrigation network consists of the following main elements:



This reforestation technique or system works as follows:

Water generated on an irregular basis by means of a fog collector, is hydraulically transferred into a water tank where the AFDS will be installed. Once a certain water level is reached, the AFDS Will start running automatically and discharging water from the tank through the drip irrigation network and watering each plant.

Once the tank will be empty, the AFDS returns to its initial state, waiting for the tank to fill up in order to begin the cycle all over again.

The amount of time needed for each irrigation cycle will depend on the amount of water generated by the fog collector (or irregular source of water) and the amount of water to be stored in the tank before the AFDS is activated and starts irrigating and de characteristics of the irrigation network. For this reason, the AFDS system can be adjusted in height, so that irrigation can be carried out with a smaller amount of water. In other words, by regulating the AFDS, the amount of water needed for each irrigation is also adjusted. We suggest the regulation way because, depending on the results obtained from the first tests, and in order to avoid water stress in the plant, during low fog water collection periods, the system will allow us to shorten the irrigation cycles, which will translate in reducing the amount of water supplied.

Concept

AFDS is a device that allows the fluid to be discharged from a tank automatically, controlled and autonomous, without the need for any electrical energy or electronic control device, joining efficiency, autonomy and regularity characteristics without electrical consumption or CO_2 emissions. The device is designed based on the physical laws of buoyancy of the Archimedes principle and the geometric shapes of the Gauss bell.

The AFDS device consists of the following elements:

- Discharge pipe
- Flotation vessel
- Protection shell
- Clamping base

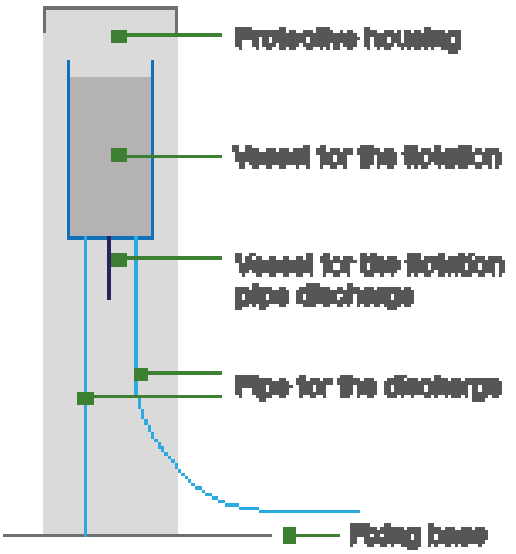


Diagram 2. AFDS schema

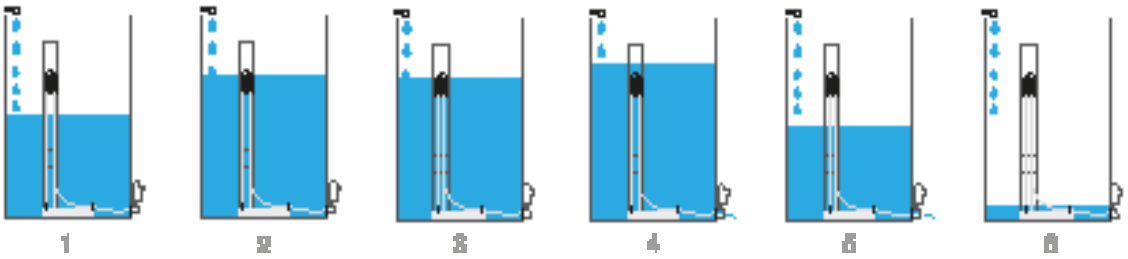


Diagram 3. How does AFDS work?

AFDS operation steps are represented in Diagram 3 How does AFDS work?:

1. Tank filling.
2. Vessel floatation.
3. Maximum level.
4. Start of discharge.
5. Tank discharge.
6. End of discharge.

The AFDS can be installed both inside and outside the tank. The advantages and disadvantages of each option are described below.

Installing an AFDS inside the tank:

Advantages:

- The AFDS is externally invisible.
- Easy to install in tanks with a capacity of less than 1 000 L.
- The system is more compact.
- Less manufacturing materials are needed, as no external support is required.

Disadvantages:

- As it is installed inside the tank, accessibility is worse and maintenance is more costly
- Anchoring systems inside the tank are needed.

Installing an AFDS outside the tank:

Advantages:

- Faster installation.
- Easy installation in tanks with a capacity of more than 1.000 L.
- Adjustable tank water discharge level.
- More accessible maintenance.

Disadvantages:

- The system remains visible and in the open air and affected by temperature changes and dilations that can unsettle the system.
- More manufacturing materials are needed, as it requires an external fixing and level regulation casing.
- More regular maintenance.
- Greater chance of blockage.

Materials and manufacturing

Most of the materials to be used in the manufacture and installation of AFDS systems are both sanitary and pressure Polyvinyl Chloride (PVC) materials, and high-density polyethylene (HDPE) materials, as well as adhesives, silicones, screws, etc., all of which are CE marked.

The most important materials and hydraulic accessories for the manufacture and installation of each AFDS unit are listed in Table 2.

Materials and hydraulic accessories for the manufacture and installation of each AFDS unit

Units	Item
2	Linear meters of sanitary pvc pipe 315 mm
0.5	Linear meters of sanitary pvc pipe 250 mm
2	Linear meters of sanitary pvc pipe 32 mm
0.5	Linear meters of sanitary pvc pipe 250 mm
2	Linear meters of sanitary HDPE DN50 PN10
2	Linear meters of HDPE pipe DN50 PN10
2	PVC sanitary plug 315 mm
1	PVC sanitary plug 250mm
1	Uraplast O-ring 90 mm
2	PVC elbow 32 mm x 90º
1	PVC elbow 32 mm x 45º
2	PVC male plug 32 x 1”
3	PVC female plug 32 x 1”
1	PVC ring 32 mm
1	PVC nipple 1”
1	Metal bulkhead fitting 6mm tube automatic.
4	Linear meter silicone hose Helivil 32mm.
1	Linear meter polyurethane tube 6mm.
8	Rubber ring gasket.
0,2	m2 Tramex GRP.
1	1 micron mesh filter DN32mm.
	Various materials (glues, solvents, screws, silicones, clamps, etc.)

Table 3 is a breakdown of the equipment and accessories necessary for the installation, monitoring and control of each AFDS unit and interconnection with the irrigation network.

Equipment and accessories necessary for the installation, monitoring and control of each AFDS unit

Units	Item
5	Linear meter of HDPE pipe DN32 PN10
4	1” PVC ball valve
8	PE 32-1” link
2	PE 32-1” link
2	PE 32 tee
1	DN32 meter with 1P / 1L pulse sensor Wi-Fi communication
2	1” double effect suction pad, plastic base PN16
2	Glycerine pressure gauge 0-10 bar

Location

In order to determine the irrigation areas, several field inspection visits should be made. The following aspects have to be taken into account for the selection of the definitive areas:

- The slopes to be reforested and their geomorphology.
- The differences in elevation between the discharge points of the AFDS and the lowest point of each irrigation network.
- Distances between each AFDS and the beginning of the network of each irrigation network.
- Distances from each discharge point with flow measurement to atmospheric station and data centralization.

Amount of water

The AFDS system can be adjusted in height, so that irrigation can be carried out with a smaller amount of water. In other words, by regulating the AFDS, the amount of water needed for each irrigation is also adjusted. We suggest way of regulation because, depending on the results obtained from the first tests, and in order to avoid water stress in the plant, during low fog water collection periods, the system will allow us to shorten the irrigation cycles, which will translate in reducing the amount of water supplied.

Costs

The cost for each AFDS, model with 50 mm discharge pipe and output flow between 2,000 (minimum) and 6,500 (maximum) litres/hour, is 855 € (interconnections between system and tank included; field installation and water meter not included).

End-of-life

Tanks, main irrigation lines, AFDS, etc., are fully reusable for other facilities.

Once reforestation is in place and under development, materials can be recovered.

The most difficult element to recover and reuse are secondary irrigation lines with self-regulated drippers. Depending on the complexity of the terrain where they are installed, especially because of the slope, it can be difficult to recover all the materials.

AFDS Performance in Life Nieblas	
Planted trees:	2013 trees
Survival rate:	86%
Cost per surviving tree:	115 euros
Need of water per trees:	75 L
Carbon footprint:	with FWC: 15,1 kg CO_2 eq. with tank truck: 3,8 kg CO_2 eq.

Water network

Water tanks

Presentation

Water storage is essential for a reforestation project, as plants may need to be watered throughout the year. In order to facilitate access to water in the most remote areas, it is necessary to have water tanks and to distribute them strategically. Different types of tank are available to meet these needs.

Material

Consider specific needs when choosing a particular type of water tank:

1. Flexible tank: 12,000 litres.
    a. This type of tank can be installed anywhere on a flat surface, but care must be taken as erosion and sediment build-up could displace the tank.
    b. Sealed containers with a 70mm diameter filling valve.
    c. Discharge valve 50mm in diameter with Barcelona racor.
    d. They will include protection mat, pick-up case and fixing kit.
2. Polyethylene tanks: 5,000 litres / 1,000 litres.
    a. These tanks need a stable location to be installed.
    b. Material with UV protection and sufficiently opaque so as not to allow algal growth.
    c. Top door-trap 400mm of minimum diameter.
    d. Water filling and discharge independent valves 40-50mm diameter.
3. Double-shell metal tanks: 5,000 litres.
    a. These tanks require hard ground to prevent them sinking into the ground due to their weight.

The estimated number of reservoirs required is based on the number of plants present on the project. It is assumed that sufficient water must be available for irrigation during the summer. This means that 12.5 litres of water are needed for each plant planted during the year, over a three-month period, giving a total around 40 litres.

Location

The location of the tanks must be carefully chosen to ensure an adequate supply of water:

Tanks should be placed high up to allow water to flow by gravity, without the need for additional energy.

However, if they are too far from the irrigation areas, it is advisable to place tanks halfway up the field to adapt the water pressure, ensure a continuous flow and reduce the effort required during irrigation.

The location of reservoirs should be consistent with the activity being carried out and the type of land. In areas where irrigation is limited or where there is little vegetation, it is not necessary to have too big tanks. However, installing a small tank for a plot with a big need of water would not be very attractive either, as it would have to be constantly refilled.

It is also advantageous to be able to connect tanks in series to increase storage capacity.

The tanks must be compatible with the project's general irrigation network, so that the stored water can be used efficiently.

Amount of water

It is necessary to calculate the amount of water required during the summer so as not to run out, particularly for the three most important irrigations of the year. Based on the Life Nieblas irrigation regime, an estimate of 40 litres per plant (around 12.5L per irrigation) for the first year is considered. For example, for a project consisting of 3,000 plants planted over the course of the year, water storage of at least 120,000 litres is required. These data may vary depending on the project and water requirements (part of the irrigation regime).

Cost

The price of a tank vary between 1.000 and 1.500 euros each, depending of the materials and the size (for Life Nieblas, flexible tanks of 12.500L : 1.100 euros, and polyethylene tanks of 5000L: 1.400 euros).

Irrigation pipeline

Pipes

The irrigation system is designed to ensure efficient distribution of water throughout the project area. It is divided into several sections, each fulfilling a specific role in the irrigation process. Below is a description of each section and the types of pipe used:

1. **Large Diameter Pipes (40 mm)**: This section is located at the highest points of the project to facilitate the gravity flow of water. Non-flexible 40 mm diameter polyethylene (PE) pipes with a pressure rating of 10 atm are used to transport water to different areas of the project, whether for plant irrigation, filling reservoirs, or any other use. They provide high flow capacity to ensure adequate water distribution throughout the system.

2. **Medium Diameter Pipes (32 mm)**: This section consists of non-flexible pipes that descend to the plots and connect to the 40 cm diameter pipes. These medium diameter pipes are installed at regular intervals to ensure water access anywhere in the project area. They allow for more targeted and controlled water distribution to the plots, ensuring precise irrigation.

3. **Irrigation Pipes (19 mm diameter)**: These pipes are connected to the 32 cm diameter pipes and are used for direct irrigation at the base of plants. They are flexible, making them easy to handle and position around crops. They are not permanently installed and are moved as needed.

   Except for the AFDS plots, the reforested areas will be equipped with irrigation networks which provide enough connections to allow the simultaneous use of 2 pipes on each area. These systems will be installed in a way that gravity generates sufficient pull to reach each plant without significant pressure issues. However, the purpose of this configuration is to facilitate irrigation maintenance and act as a mitigation method in case of severe drought, especially during the summer.

   It's a simple system of cervical pipelines branching off from the main water supply pipeline. These branches are equipped with 19 mm valves connected every 25 m longitudinally. Eventually, water is distributed through 19 mm pipes equipped with watertight fastening systems that assist the field specialised workers (SFW) in connecting and disconnecting the pipes along the branching pipelines.

Connection fittings

They are essential to ensure proper leak-free operation of the irrigation system. They provide a tight connection between different pipes. These fittings include **elbows**, **connectors**, **end pipe tap, T-junctions** (allowing the junction between 3 pipes), **manchons** (which can be used for reservoirs), and irrigation **collars**.

Control fittings

Control fittings include **valves**. Irrigation of the plots, each with different water requirements from the same water source, requires subdivision of the irrigated surface into sectors controlled by valves. These valves are manually and gently operated. The entire system is controlled by various valves that open and close the pipes, thereby directing water to desired locations. These valves play a crucial role in regulating water flow and provide the necessary flexibility to adjust irrigation to the specific needs of different project areas.

Similarly, **pressure regulators** with pressure gauges are used to maintain the desired performance of the irrigation system. They control excessive input pressure and ensure constant output pressure in the pipes.

Plant selection

The selection of plant species for a reforestation project is a crucial step (ALEXANDRIAN, Daniel, 1984). To make a good selection, different information must be collected from the previous analysis; the following factors must be taken into account:

1. Analysis of the site potential: from the precedent analysis, the depth, composition, structure and richness of the soil, as well as collecting climatic data, such as annual rainfall. This analysis helps determine which species are suitable for the site conditions.
2. Environmental or landscape prescription: Some areas may have specific environmental or landscape requirements that limit the use of certain tree species. These prescriptions need to be taken into account during the selection process.
3. Presence of disturbances: Factors or elements that may negatively affect the future of certain sensitive species should be considered. For example, invasive species.

Parameter

Each tree species has its own characteristics and specificities, and it is important to identify species that can meet the objectives of the reforestation project.

- **Availability**: Plants must be easy to buy or to collect.
- **Soil characteristics**: This includes the depth, stoniness, structure, and physicochemical parameters (organic matter, pH, salinity, nutrients content).
- **Climatic data**: Factors such as annual rainfall and temperature are considered to ensure that the selected species are adaptable to the prevailing climate in the area.
- **Environmental or landscape prescriptions**: Any limitations or recommendations regarding the use of certain species are taken into account.
- **Invasive species**: Species that have invasive tendencies and may negatively impact the ecosystem are avoided.
- **Natura 2000 listed habitat**: If the site falls within a designated Natura 2000 area, the selection of species is done in accordance with the conservation objectives of the habitat.
- **Disturbing elements**: Factors such as pests or vermin that may pose a threat to certain species are considered.
- **Native/non-native species**: Preference is often given to native species as they are better adapted to the local ecological conditions and support biodiversity.

- **Human needs**: The species selected must meet human needs such as protection, production, aesthetics and other environmental considerations.
- **Adaptability**: Species should be able to thrive in the prevailing climatic conditions of the region.
- **Soil improvement**: Species should help reduce soil erosion and improve soil fertility.
- **Resistance**: They should be resistant to pests, diseases, and other adverse factors.
- **Cost-effectiveness**: Species that can be easily established with minimal cost are preferred.
- **Growth rate**: Fast-growing species are often favoured for reforestation projects. However, the use of invasive or alien species must be avoided.
- **Ecological compatibility**: The selected species should not significantly alter the existing floristic composition of the surrounding forest or become a weed in the future.
- **Allelopathic effects**: They should not produce substances that inhibit the growth of other plants through allelopathy.
- **Regional and micro-climate suitability**: The species should be capable of adapting to both the regional climate and the specific micro-climate conditions of the plantation site.

By considering these factors and criteria, the selection of appropriate plant species ensures a successful reforestation project that promotes environmental improvement and resilience to climate change.

3. Evaluation of a potential project

RMT

To facilitate this assessment and support decision-makers, a specific tool has been created: the Restauration Management Tool created by LIFE Nieblas. you can access following this link : https://dev.surikat.io/s/

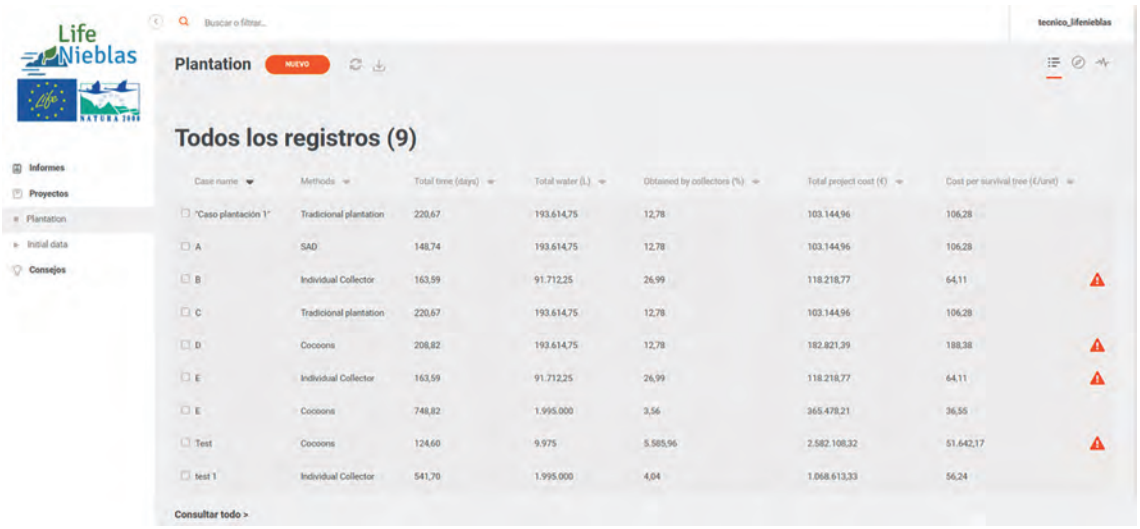
In the same way as the RMT for fog collection, this tool also includes reforestation projects. This platform brings together data from different fog collection and reforestation projects, making it possible to estimate the specific needs of a project, assess success factors and make informed decisions throughout the evaluation process.

Based on this initial data, the tool calculates cost estimates, work times, and litres of water to be used.

For the same project, different cases can be proposed to compare the results and choose the most optimal one.

Each plantation case consists of:

- **Input data** (specification of each case): Data used to define each plantation case, indicating the project duration, the number of trees to be planted, the number of workers, and the number of fog collectors to be used.
- **Starting data** (data used for calculations): For each plantation case to be calculated, you can select which values from the cost, method, and productivity tables you want to use.
  - Costs: Data that allows the calculation of the cost of items used in the plantation, such as collectors (COLLECTOR COST), transportation (TRANSPORTATION COST) and other additional costs (ADDITIONAL COST) like trees, water and hoses.
  - Methods: Data used to define the planting method to be used, including information about the cost of materials for that method, planting productivity, irrigation requirements, or survival rate.
  - Productivity: Data used to calculate the required work duration and costs, including information about land preparation time, installation of irrigation systems and collectors, workdays, and wages. It also includes information to calculate the productivity of the collectors based on the impact surface area and water capture capacity for each type of collector used (traditional and innovative).
- **Output data** (results): Based on the input data, the project costs, required work duration, and litres of water used will be calculated.



Time

To determine whether the project is achievable, a number of parameters need to be taken into account. First of all, you need to see if you have enough time to carry out the project. Here are the figures collected during LIFE NIEBLAS on the time needed to set up the project:

Reforestation time (per worker unit/days)

	Traditional	Cocoon	IFWC	AFDS
Trees transport	250 plants/day			
Ground conditionning	200 m^2/day			
Equipment preparation	0	0	11 unit/day	14 valves/day
Irrigation network installation	120 m/day			
Plantation	20 plants/day	7 plants/day	17 plants/day	22 plants/day
Maintenance irrigation	80 plants/day	90 plants/day	80 plants/day	500 plants/day

Cost

List of costs to be taken into account, which vary according to the location of the project:

Project costs estimations breakdown

Costs	Details	Life Nieblas costs
Tree costs	Nursery	Details methodology presentation
	Seedling	
	Material	
	Methodology	
Water costs	Need of water	From FWC: 0 From tank truck: 12 euros/m^3 From gallery: ~0,5 euros/m^3
	Origin of water	
	Project duration	
	Number of irrigation per years	
	Numbers of trees	
Project transport	Vehicle price	7,500 euros/vehicules
	Number of kilometres	
	Fuel price	
	Fuel consumption	
Pipes system costs	Number of metres	2 euros/m of system
	Price per metre	
	Connection fittings	
Water storage	Amount of water	Between 1,100 and 1,400€/tanks
	Tank price	
	External transport	
Mobilephone or iPad to use monitoring app ~ 400-1000€	Equipment cost	These costs will have to be analysed based in individual cases
	Analysis	
	External assistance	
Personal costs	Number of workers	Calculations derived from particular cases, company conditions and regulation specifications for every case
	Projects duration	
	Gross annual salary	

C. FIELD IMPLEMENTATION

This part of the technical manual covers all the steps to be taken before reforestation work, from seed collection to nursery and soil preparation.

Next, all the information needed to install the different reforestation methods will be presented. They will be detailed step by step, with an emphasis on the mistakes to be avoided. Estimates of the time required for each stage will be provided, along with all the parameters to be taken into account.

For good plant development, a section will focus on irrigation, with the recommended quantity and frequency.

Finally, the concept of upkeep and maintenance will be discussed. The regular steps required to limit potential damage will be explained, as well as the risks that this project may face (insects, erosion, etc.) and the preventive measures to be taken to avoid repeating mistakes.

The aim of this section is to provide a comprehensive and detailed guide to assist with the various reforestation techniques, with the emphasis on the safety, efficiency and sustainability of the system.

1. Plants settlement

Seeds collection

You need to recover natural regeneration plants, taking them from the natural environment as much as possible. To do this, you need to collect the seeds from the forest floor.

The seeds of superior quality are collected from genetically superior trees (seed production area, seed orchards). This considers tree form, size, height, diameter and pest and disease resistance. The seeds are collected from the vigorously growing middle aged to mature trees of the above mentioned characteristics.

When these seeds germinate in the substrate of the parent tree, there is a large quantity at the foot, sometimes up to 600 trees per square metre, so they can not all thrive. You have to take advantage of this opportunity. It is advisable to leave only 100 to 200 to safeguard the process.

Nursery

1. Collect seeds

First of all, you need to collect the seeds. You can buy them, but it's better to collect them in the field (see 'seeds collection' part). In fact, you can choose the parent tree of the young shoot by checking that it is healthy and disease-free.

2. Well-labelled tray

Once the seeds have been collected, they need to be placed in a well-labelled tray for traceability.

3. Prepare the seeds

Bring them in, wash them and treat them appropriately.

4. Germination time

About 3 months, but can vary a lot depending on the plant.

OR

It is possible to collect the young seedlings, already germinated, directly from the foot of the trees.

5. Transplanting. Once they are big enough, start transplanting.

Once the seed has green leaves, use a spade to lift the plant a little so as not to damage the roots. Then remove the plant with its roots and take it to the forest container (this is the final container that will be used for planting in the field).

6. Put them in forest containers

To put them in, start by making a hole with a stick or other tool. Take the plant by the aerial part, putting the roots in so that they don't get twisted. Insert it into the hole and press down on the substrate.

The substrate is a mixture of peat, perlite and coconut fibre with compost.

All that's left is to water them and wait for them to grow.

7. Open air

When they exceed 50 cm, for a period of adaptation before planting them, leave them in the open air.

Ground conditioning

Before undertaking reforestation works, in addition to the installation of safety measures, the distribution of water storage and its associated piping network, sometimes vegetation clearing is also required. It is necessary to remove or manage existing vegetation to make space for seedlings and avoid competition by water and nutrients that would directly impact plantations performance.

Therefore, the technique used involves clearing vegetation in strips with a width of 0.5 to 1 metre, perpendicular to the direction of the slope, while leaving an existing vegetation strip of 0.5 to 1 metre between them. This method allows erosion control through the remaining vegetation strips.

Tools

The machinery used to do this ground conditioning are:

- Brush cutter
- Blade cutter
- Chainsaw

The brush-cutter is used to clear the dry grass and then for bigger plants where strings are not effective such as wood bushes, the blade cutter is more efficient. And when the blade cutter cannot get it, the chainsaw will be helpful to open pathways to continue after with the bush cutters.

2. Plants settlement

Traditional

STEP 1. Choosing the seedlings

- Go to the nursery and select plants according to their species and state of health.
- Choose plants that have had adequate sun exposure, are moderate in height, have thick roots and are adapted to harsh environmental conditions.
- The characteristics of a good plant include well-developed, branched roots, a woody base, attractively coloured leaves and a straight, leafy stem.

STEP 2. Transporting the plants

- Be sure to transport the plants in a cool, shaded environment to avoid heat stress.
- During transport, make sure the plants are stable and upright to prevent them breaking.
- Water them sufficiently to maintain their humidity.

STEP 3. Identifying the planting spot

- Identify the planting spot and clearly mark the spaces between the plants before you start digging.
- Determine the appropriate spacing according to the distance between plants: wider spacing saves seed and plants, reduces planting costs and requires less labour. However, narrower spacing limits weed growth, reduces soil deterioration and encourages natural pruning, but requires more seed and plants, which increases costs and the amount of labour required.

STEP 4. Preparing the planting holes

- To optimise water use, we recommend digging deep holes.
- Use a pick, shovel or if necessary a mechanic drilling auger and hoe to dig holes 20 to 40 cm square and 40cm deep.

STEP 5. Plant distribution

- Place the plants close to the planting holes.
- Make sure they are evenly distributed, avoiding grouping all plants of the same species in the same place.
- To remove the plant from the forest container, proceed slowly, keeping the root ball intact.

STEP 6: Planting the seedlings

- Place each plant straight into the hole, making sure it is at the right height: the collar (the transition point between the stem and the roots) should be slightly below ground level.
- If the plant is too high, dig the hole deeper; if it is too low, add some soil to raise it.
- The roots should not be buried too deeply, as this will restrict access to oxygen and affect the proper growth of the tree.

STEP 7. Filling the hole

- Use the soil previously removed from the hole and mix it with compost or biochar to improve the quality of the soil.
- Fill in the hole gradually, taking care not to bury the crown of the seedling under the soil.
- Leave a small ditch around the tree to encourage water retention during irrigation, so that the water doesn't run off onto the ground.

STEP 8. Waste collection

- Collect all materials used during planting, especially plastics, to avoid polluting the environment.

STEP 9. Initial watering

- After planting, carry out an initial watering of around 10 to 15 litres of water per plant to ensure good initial hydration.

It's important to note that these steps are general and may vary depending on local conditions, tree species and the specific reforestation techniques used in your area.

Cocoon

Materials and tools

- Cocoon
- Pickaxe
- Compost (optional)

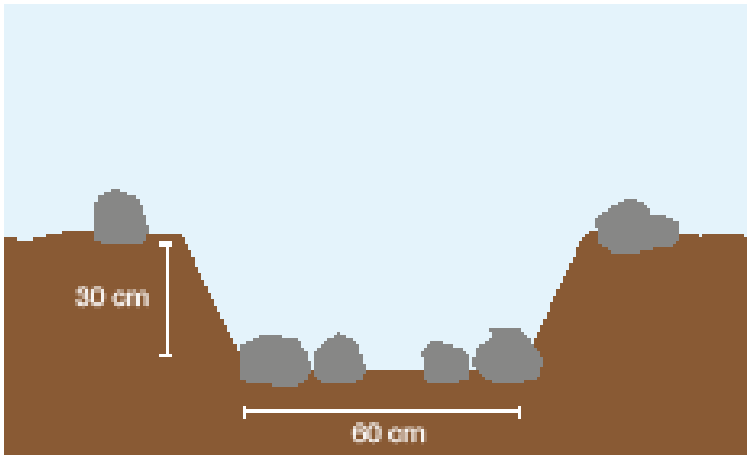
Setting up: video available by following the link: https://www.youtube.com/watch?v=GaVyrI_YviE

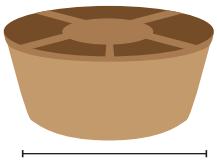
Methods

The first step of plant selection (step 1), transport (step 2) and identification of planting locations (step 3) are identical to those presented above for traditional planting (part B.3.1). The cocoon planting method begins to evolve from the moment the hole is dug:

STEP 1. Digging the hole

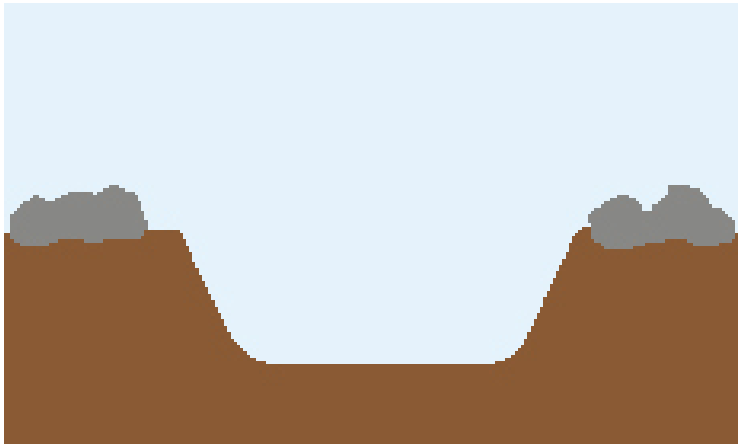
The first step for Cocoon installation is to dig a hole. Bigger the hole, the easier it will be to install the cocoon on it. Due to the size of the cocoon, 30 x 60cm, the hole must be at least 60cm width. A pickaxe can be used to dig.





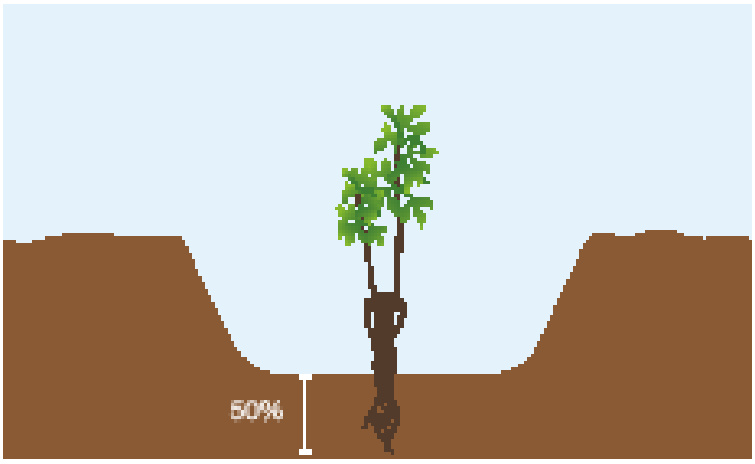
STEP 2. Cleaning the hole

Once the hole is done, you need to remove all stones or debris (roots, sticks, ...) present to avoid damaging the cocoon to allow it to perform its water storage function.



STEP 3. Planting the plant

With the select plant, deep it until around 50% of its root ball in the hole. Then check that the plant is firmly in place and holds in the soil.



STEP 4. Install the Cocoon

Take the basis of the cocoon; pass the plant through the central hole. The top part of the equipment should be in line as much as possible to the ground to avoid any leak of water. Start to add earth all around the cocoon, to stabilise it.

STEP 5. Put the cover

Close correctly and slowly the cocoon with the cover of the equipment.

STEP 6. Fixe the plant on the ground

Then, cover the resting 50% of the root ball in the hole. Add soil until the plant is well established in the soil. Only add soil to the roots, never to the stem.

STEP 7. Filling the Cocoon

The next step is to fill the cocoon with water, the deposit has a capacity of 20L. It's important to leave a margin of two fingers without water in the top, this avoids the contact between the water with the cover of the equipment.

STEP 8. Settlement irrigation

For the plant settlement, to assure plant adaptation and development, you should water directly the seedling with an amount of 10-15L.

STEP 9. Protector installation

Close the cocoon with its cover. After that, you can insert the protector. To do this, slowly push the plant inside. Insert it in the correct direction (with the holes facing upwards), then push it in as far as the line indicated by the arrow.

STEP 10. Fill the hole

Fill the hole with soil (adding an organic amendment or fertiliser if necessary, see section 3) up to the underside of the cover, so that the next time you fill the cocoon it will be much easier to access. This can be done with a pickaxe.

IFWC

Setting up: video available by following the link: https://www.youtube.com/watch?v=aLXDV2lIVao

Material and tools

- Metal mesh 25x25mm 1.5m high
- Plastic mesh with 3x3mm Russell PE mesh, 1m high
- Posts 1.5m high
- Wire
- Cutting pliers
- Hammer
- Pickaxe
- Reel

STEP 1. Cutting the mesh

The first step is to cut the metallic mesh to a length of 140 cm from the rolls. To do this, use the uncoiler and cutting pliers.

STEP 2. Add the PE mesh

Then, if you wish, you can add plastic mesh of the same length, i.e. 140 cm.

STEP 3. Fix the meshes

Using the wire and the wire cutters, the second mesh needs to be fixed to the wire mesh so that it holds throughout the manipulation. It should be positioned at the bottom of the collector so that it touches the ground.

STEP 4. Roll the collector

Tie the two ends together with the wire so that the collector forms a tube.

STEP 5. Pass the stake through the wire mesh

Place the stakes on either side of the mesh so that it is secure and cannot come out, and so that the stakes extend at least 30 cm beyond the bottom of the collector.

In the photo, the collectors don't have plastic mesh, but it's the same system with or without mesh.

STEP 6. Plant the plants

Once the IFWC is ready, you can go to the planting spot to start the reforestation. To select the plants, transport them, dig the hole and plant them, follow the traditional planting procedure explained in section B.3.1. Also consider using organic amendments or fertilisers (see section 3).

STEP 7. Push the collector into the ground

Using a hammer or sledgehammer, drive the collector into the ground, making sure it doesn't move. If necessary, pull the stakes out of the mesh to drive them deeper into the ground.

STEP 8. Plant irrigation

From this point on, the plant can receive its settlement irrigation with a quantity of between 10 and 15 litres.

Autonomous Fluid Discharge System (AFDS)

Manufacturing phase

The steps to follow in the manufacture of each AFDS system are described below:

- Material acquisition needed for the full system manufacturing.
- Manufacturing the internal flotation container.
- Manufacturing the external casing.
- Joining the internal container to the external casing.
- AFDS equipped with connection hoses—input, output and filling hoses.
- Height-adjustable fixing and anchorage system.

Once the AFDS is manufactured, operational tests must be carried out. The steps followed in the manufacture of each AFDS system are described below:

- Materials needed for the full system manufacturing.

- Manufacturing the internal flotation container.

- Manufacturing the external casing.

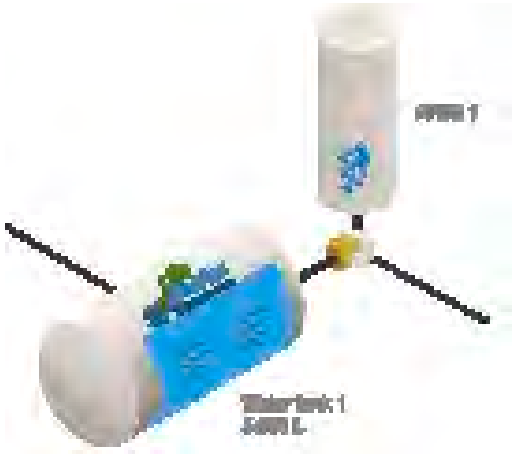
- Joining the internal container to the external casing.

- AFDS equipped with connection hoses—input, output and filling hoses.

- Height-adjustable fixing and anchorage system.

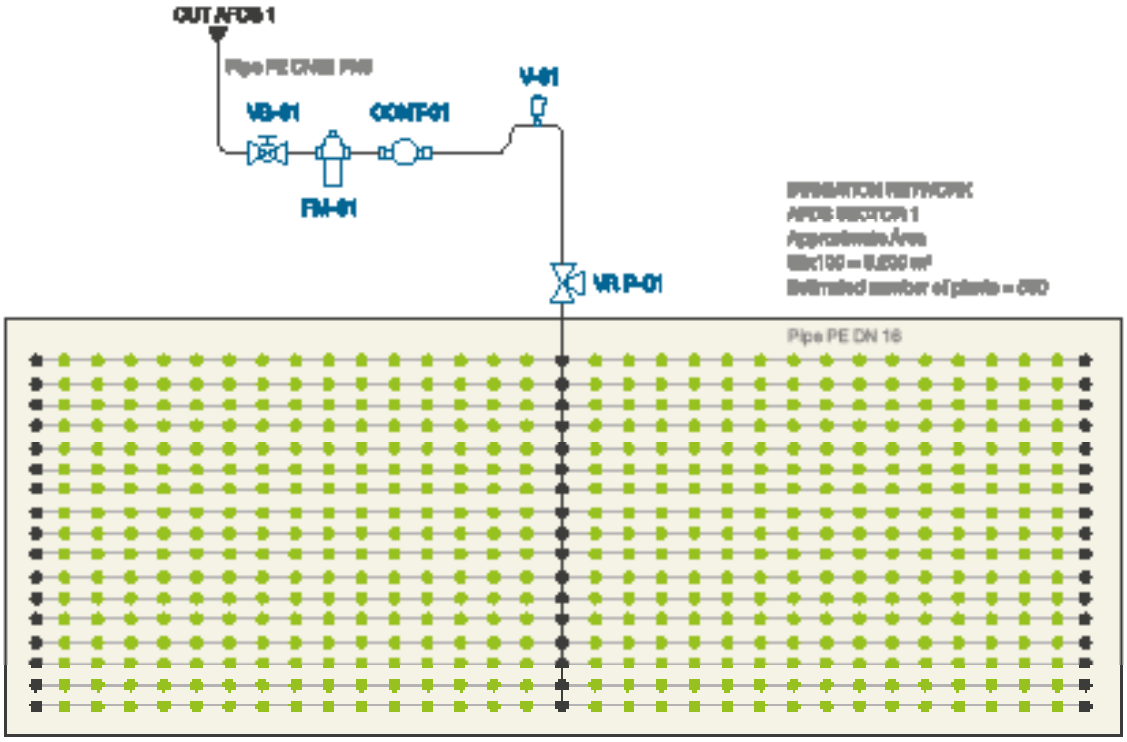
General description of the installation

- The installation consisted on the following field work:
- Field preparation.
- Tank installation and AFDS fixing system.
- Hydraulic installation of the tank (input and output connections, air vents and level indicator).
- Installation of the AFD System inside the anchor pipe and hydraulic connection to the tank.
- Installation of the water meter, with a pulse counter and a communications system linked to the weather station.



Depending on the available land, the irrigation area is divided into sectors (one for each AFDS). The size of each sector depends on the number of plants to be incorporated and the supply of irrigation. For the example of LiFE Nieblas, each sector is approximately 96 meters long and 34 meters wide. This allows you to install 17 hoses per sector with up to 49 drippers each, with a maximum of 833 drippers. In practice, depending on the orographic conditions and obstacles, the planting includes 400 to 500 plants. It is necessary to use self-regulated drippers capable of maintaining a constant flow rate over a certain pressure range.

The water distribution is carried out through a 32 mm low density polyethylene (LDPE) central pipe that feeds the 16 mm LDPE pipes containing the aforementioned drippers. Each branch shall be equipped with a small manual valve.

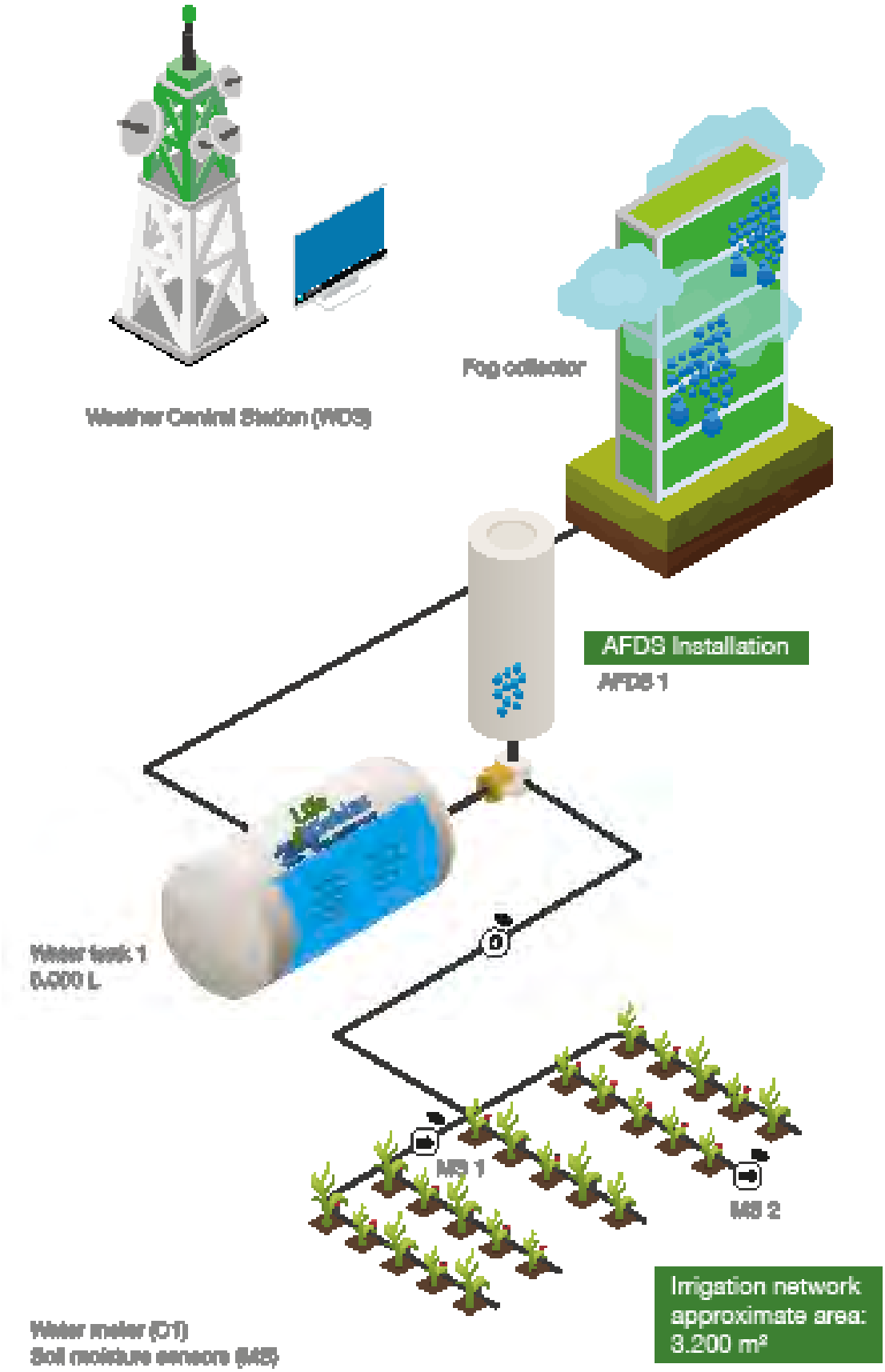


Once the installation of the irrigation network is done, it is advisable to check pressures and flows at different points in the network.

Finally, each AFDS reforestation system shall consist of the following basic elements:

- Water storage tank
- Autonomous fluid discharge system
- Irrigation network

Dimension and preparation of the irrigation network



The workflow to dimension the AFDS and its associated drip irrigation network could be as follows:

1. Determine the total water requirement: The first step is to calculate the total water requirement for all the plants by multiplying the water needs of a single plant by the total number of plants. Each plant's water requirement depends on its type, size, and the local climate conditions.
2. Sectorize the irrigation network (optional): You may divide the system into different sectors or zones, ensuring that the total flow rate demand of the emitters in each sector does not exceed the flow rate capacity of your water supply. As it will be stated, in some cases the sectorization is unavoidable.
3. Dimension the water tank: Then you are able to dimension the water tank and the AFDS level setup point to meet that amount. Depending on the size of the irrigation area, it may be necessary to use more than one AFDS.
4. Establish dripper flow rate: Once known the total irrigation need is time to choose the appropriate flow rate for each dripper (litres per hour) from those available in the market. It should match the plant's hydration needs and the soil's absorption capacity and the flow limitations of the network. Pressure in gravity systems depends directly on the height of the water source above the irrigation lines, so it affects the dripper flow rate. Choose Between Compensating and Non-Compensating Drippers:
    a. Traditional drippers: Their flow rate is directly proportional to the water pressure so require a homogeneous pressure distribution along the pipes as long drippers flow is pressure dependent where alpha is around 0.5:

    $$q = c \cdot p^{\alpha}$$

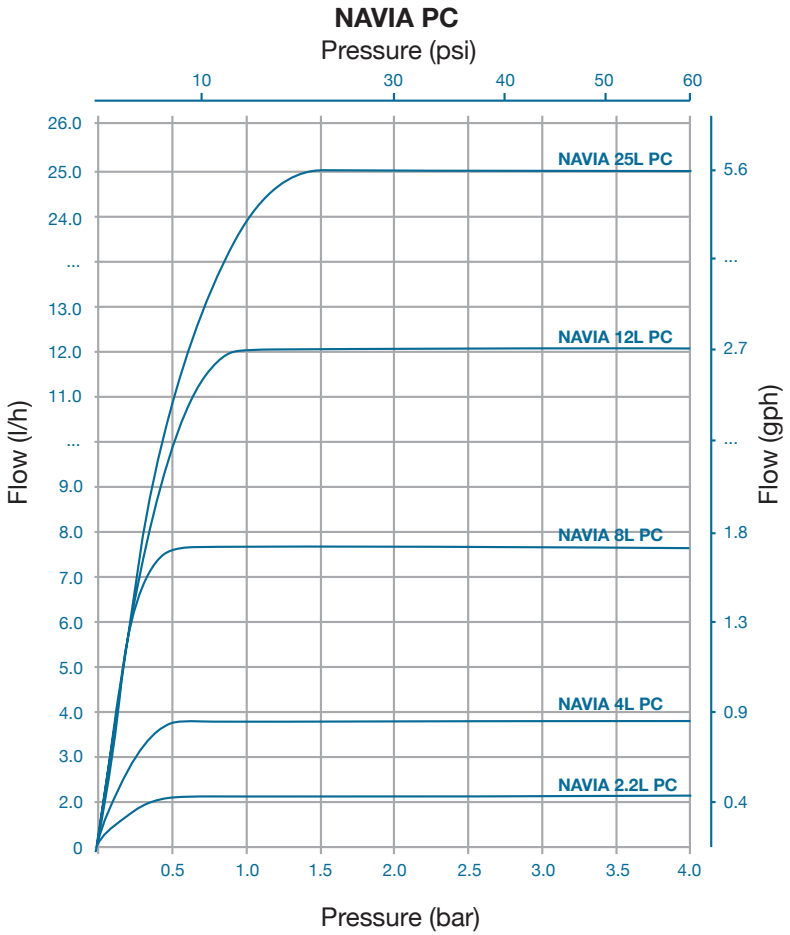
    These are suitable for flat irrigation areas where there are minor pressure changes along the network.
    b. Self-compensating drippers: are able to keep a constant flow rate (alpha = 0) in a certain pressure interval, making them suitable for sloped areas where significant pressure variations occur along the network.

Drippers characteristics

NAVIA models		PC					ND/HND			
		2.2L	4L	8L	12L	25L	2.2L	3L*	4L	7.8L
Nominal flow	(l/h)	2.20 0.58	3.80 1.00	7.80 2.06	12.00 3.17	25.00 6.60	2.20 0.58	3.10 0.82	3.80 1.00	7.80 2.06
Pressure-compensation interval	bar	0.6-3.5 9.0-51			0.8-3.5 12-51	1.25-3.5 19-51	0.6-3.5 9.0-51	1.0-3.5 15-31	0.6-3.5 9.0-51	0.6-3.5 9.0-51
Closing pressure / Opening pressure	bar	- -					0.15-0.3 2.0-4.0	0.35-0.9 5.0-13	0.15-0.3 2.0-4.0	0.35-0.9 5.0-13

Disc filtration recommended ≤ 130 micron

*HND (high pressure no-drain)



Dripper pressure curve

5. Dimension of the water pipes: The size of the main and sub-main pipes should be capable of carrying the water flow required without significant loss of pressure, which can affect the performance of the emitters. If the required diameter results too high it would be necessary to either reduce the dripper flow rate or number of plants associated with the ADFS.

In the case of the LifeNieblas project, the drippers chosen force the irrigation network to work in a range of pressures between 0.6 and 3.5 bar. For the calculations, the relative elevation of every single dripper with respect to the AFDS was estimated using linear interpolation based on the known elevation of six of them (the four in the corners and two in the middle of the longest sides).

The water distribution is made through a central 32 mm agricultural grade low density polyethylene (LDPE) 10 bar nominal pressure pipe that we will call 'main pipe' that feeds the 16 mm LDPE pipe which contains the drippers previously mentioned (branches). Every single branch will be equipped with a small manual valve as the one in the figure below:

Small manual valve for every single branch.

Due to the significant slopes in the area, the elevation difference between drippers from the same network could be more than 20 meters (1.96 bar) , which makes necessary the installation of a pressure regulating valve. A first approach was to reduce pipe diameters to drop enough pressure by friction in order to maintain it in the desired range. However, it was insufficient.

Inplace AFDS and irrigation network field installation

Assembly phase

STEP 1. AFDS Installation

- Tanks installation. Install the water tanks, they will be used to keep and collect the water between two irrigation events.

- AFDS installation. Install the AFDS link to the water tank, the AFDS will be used to release water automatically when it reaches a certain predetermined level.

STEP 2. Installation of the irrigation system

- Main pipe Installation. This pipe serves as the main axis of the irrigation system, supplying water to the different lines. There is one pipe per AFDS and per irrigation zone, running from the selected reforestation area and traversing the field from top to bottom. The diameter of this pipe is 32mm.

- Levelling future lines. Using a cord and a level, reference the central pipe to determine the path of the lines. The lines should follow the pressure stimulation established during the preparation phase. If the pipes do not follow this pressure model, it can lead to pipe bursts or leaks during irrigation. The lines should be straight, slightly sloping to allow water flow. There should be a distance around 2 metres between each line. Use a cord to accurately trace the path of each line before planting.
- Planting. Before installing the pipes, it is preferable to plant the vegetation. Try to align the plants in rows, following the previously traced line. To select the plants, transport them, dig the hole and plant them, follow the traditional planting procedure explained in section B.3.1.
- Installation of lines. Once the plants are in place, you can lay out the irrigation lines, which have a diameter of 16mm. When the line reaches the end, cut it, bend it, and secure the end to prevent water leakage.

- Connecting the lines. Once the lines are installed, connect them to the main pipe of the field using a manual valve that can be opened or closed to control the water flow.

Small manual valve for every single branch.

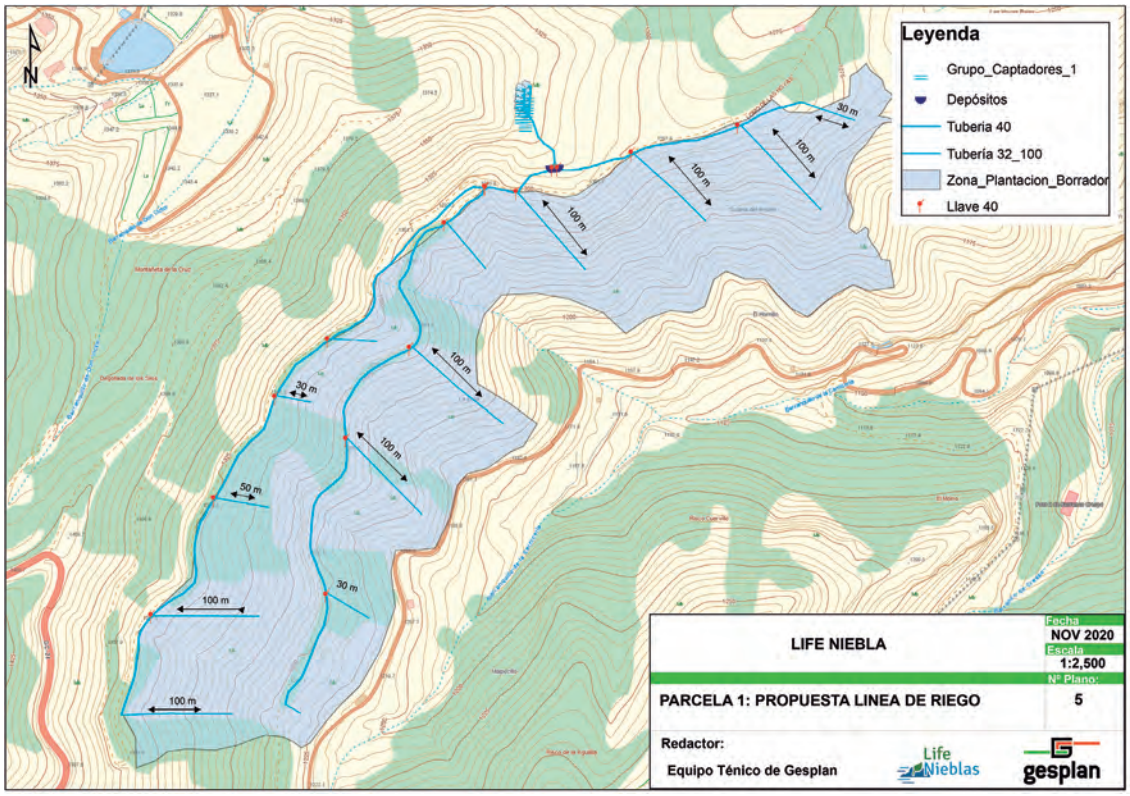
- Fixing the lines. To prevent the lines from shifting over time, secure them to the ground using U-shaped iron bars driven into the soil with a hammer, as shown in the photo.

- Drippers installation. Install drip emitters for each plant, ensuring they are centred and directed towards the plants.. If a plant is located too far from the line, an additional tube can help the dripper to water the desired location. Once all the drippers are installed, the installation phase of the irrigation system is completed.

STEP 3. System activation

- Connecting the main pipe to the AFDS. Connect the irrigation system to the water tanks using hydraulic connections, establishing the link between the two systems. Each tank has its own surface area and main irrigation pipe. It is advisable to clearly label or tag them for identification.
- Water supply (gallery, Fog collector, …). Once the entire system is ready to operate, connect the tanks to the water source, whether it is natural gallery, rain, fog...

Field installation



STEP 1. Draw the plan

Throughout the plan development, it is important to consider the available water source for irrigation, such as a well, a gallery, a fog collector, a municipal water network, or another water supply source. Ensure that the irrigation system is compatible with this source and that the flow rates and pressures are appropriate.

1. Map study and plot zone identification: With map, identify the different zones of the plots where irrigation is needed. This may include nurseries, planting areas, tanks to be filled, etc.
2. Topography analysis: Examine the topography of the area to determine the main axes, slopes, and cliffs. This information is crucial for planning the route of the pipelines and ensuring efficient water distribution.
3. Placement of water reservoirs: Identify the locations of water reservoirs in the highest areas of the surface, preferably on a flat surface. These reservoirs will serve to store and provide the water needed for the irrigation system.
4. Drawing the first pipeline section: Start by sketching the first section of the pipeline. The pipelines should follow the natural slopes of the terrain to create main axes. Ensure easy access to these pipelines for opening and closing valves conveniently.
5. Secondary pipelines: Trace the pipelines that cover all the plots. Use pipes with an appropriate diameter, for example, a diameter of 32, and ensure they run in a straight line. Also, provide a valve at the top to open and close the pipeline. It's important to note the water flow rate required for each area and adjust the pipe diameter accordingly. Ensure that all sections are properly connected to the main pipelines and shut-off valves.

If the pressure is not enough to go down until the end of the pipe, it can be necessary to install a tank at half height. This will facilitate the flow.

6. Placement of irrigation points: Identify the points where the irrigation pipes will be connected to the pipelines.
7. Plan verification: Once the plan is developed, carefully check all the layouts and connections to ensure they meet the specific requirements of the irrigation project. Also, verify that the valves are correctly positioned to allow efficient control of water flow.

STEP 2. Ground conditioning

Before installing the tanks, it is important to prepare the ground. Given the weight of the water when the tank is full, there is a risk of falling. The surface should be as flat as possible, with no rocks to puncture the tanks. If the slope is too steep, a terrace can be created with rocks to ensure stability.

It's also worth analysing the path of the water when it rains to ensure that no sediment accumulates under the tanks. In addition, you need to take care with the composition of the soil to make sure it's not too soft. With the weight of the tank, there is a risk that it will sink into the ground over time.

STEP 3. Tank transport

To transport the tanks to the selected site, it is necessary to identify the best method of transport. This can be by truck, trailer or another method appropriate to the situation. Depending on the location, it may be necessary to use a crane or rope system to position the tanks in their final location.

STEP 4. Tank installation

During their installation, make sure that the reservoirs are placed on a flat, and clear surface, and that all the valves and openings are accessible. If there is a significant difference in elevation or a steep slope, it may be necessary to add a mid-level reservoir to facilitate water flow and irrigation. Ground preparation is also required before installing the reservoirs.

STEP 5. Installation of water tank outlets

Next, install the water outlets from the reservoirs and ensure their connection to the entire irrigation system. This may involve the placement of appropriate pipes and valves to direct water to the reforestation areas.

STEP 6. Installation of main pipes

The main pipes are installed according to the previously established plan. Valves should be installed at regular intervals to allow for shutting off the pipe in case of leaks and to minimise water losses. When pipes need to cross a road, it is necessary to bury them underground to avoid damage.

STEP 7. Installation of 32-Diameter pipes

Then, the 32-diameter pipes are installed in the reforestation plots. They should be straight and sloping downward to ensure proper water flow. These pipes are connected to the 40-diameter main pipe using valves that can be opened and closed as needed.

STEP 8. Installation of 32-Diameter valves

Along the 32-diameter pipes, valves are installed at regular intervals. These valves allow for the connection of watering hoses when in use, providing precise control of irrigation in different reforestation zones of traditional, Individual Fog Water Collectors (IFWC) and Cocoon methodologies. These branches have 19mm valves connected to them every 25m longitudinally. Eventually, water is delivered through 19mm hoses fitted with seal fast systems that help SFW (Specialised Field Workers) connect and disconnect hoses along the branching pipelines.

It is important to note that these steps are a generalisation of the process for installing an irrigation system for a reforestation project. Specific details and technical terms may vary depending on project requirements, site topography, and available equipment.

3. Management and maintenance

Irrigation frequency and quantity

Reforestation helps to limit desertification. However, after planting, the roots of young plants have difficulty in reaching the water in the soil, reducing their survival. Also, given the heat and rapid evaporation of water and the long dry summer period, watering is essential in some parts of the world.

Therefore you need to think about an irrigation regime that is suited to the project, taking into account the type of soil, rainfall and the type of tree planted. You need to think about the different types of irrigation possible and identify the quantity of water required and their frequency (or the parameter triggering irrigation).

During Life Nieblas, there were 4 types of irrigation: nursery (very regular for young plants), maintenance (at regular, continuous intervals and for the most difficult periods), settlement (when the tree is planted) and support (when the tree is under stress).

It is possible to set up different irrigation regimes in the same project, by varying the frequency or period of irrigation, so as to compare the evolution of the trees and their development in order to identify the best yield. Care must be taken to identify each plot according to each irrigation regime, so that subsequent analysis can be carried out.

Here are the choices made for the Life Nieblas project:

- **Nursery**: irrigation the plants before planting them. Using a classic hose, water twice a week all the nursery with a amount around 30L for 100 plants..
- **Settlement**: the water used in every reforestation as the first watering provided to plants right after they have been planted in the field (usually around 10-15L/seedling).
    - **Traditional reforestation method** settlement irrigation 10-15L.
    - **IND** 10-15 settlement irrigation.
    - **AFDS**: 10-15 of settlement.
    - **Cocoon:** settlement irrigation 10-15 and tank filling with 20L.

- **Support** irrigation refers to the amount of water mostly used due to extreme weather conditions (i.e. heat waves, suspended dust periods, lack of 'normal pattern' weather, etc.).

  You need to check the weather regularly, as well as the condition of the plants. If there is no rain in the forecast for a long time, it may be necessary to irrigate the plants to avoid losing the effort invested in reforestation. Before deciding to water the trees, it is advisable to monitor the condition of the plants. If many of them start to show signs of drought (yellow, dry...), irrigation may also become necessary.

    - **Maintenance** refers to the classic irrigation during summer.
    - **Traditional reforestationn method:** 15L watering per plant, once a month from July to September (both inclusive).
    - **SAD:** connected to the fog water collectors, when the tank is full, the irrigation will start, releasing about 15L per plant every 3 weeks.
    - **Cocoon:** 15L watering per plant, once a month from July to September (both inclusive).
    - **IND:** 10L watering per plant, once a month from July to September (both inclusive).

Organic Amendments

The addition of organic amendments and fertilisers into the planting hole must be considered in order to improve soil conditions for planting. The most common option is compost, that could be from different origins and properties, despite other innovative options like biochar or hydrogels could be also interesting in some cases:

Compost

Compost can be added to the substrate at the time of planting. It improves the physical and chemical characteristics of the soil: provides nutrients, improves soil structure and increases water retention capacity, among others. Compost could be from different origins, from farm wastes (manure) to urban ones (kitchen wastes, sewage sludge). It must be produced in controlled conditions in order to reach the quality standards (EU Fertilizer Regulation (EC) No 2019/1009).We recommend adding 1 litre of compost per plant.

Biochar

Studies have shown that biochar amendment can improve soil properties (e.g., improved water holding capacity, soil enzyme activity, nutrient content and retention). However, this heavily depends on the biochar type and soil conditions. The use of biochar at the tree planting is a feasible and environmentally sustainable strategy to improve the success during the tree establishment phase, specially in acid soils. It is especially interesting to combine biochar and compost, in order to better retain the nutrients in compost and reduce nitrogen release. Another interesting issue regarding biochar is that it is constituted by highly stable carbon that can remain in the soil for long periods, which increases the potential in carbon sequestration of the plantation.We recommend adding 0,5 litres of biochar per plant.

Regular maintenance

Regular maintenance of the project involves a number of essential tasks designed to ensure that the plants function properly and develop optimally. Carrying out these maintenance tasks on a regular basis ensures that the project runs smoothly and promotes the growth and health of the plants. It also helps prevent potential problems with the equipment and ensures the long-term sustainability of the project.

Vegetation clearing

In winter, it is important to clear the weeds to give to the plants space and light and reduce competition for water. This will encourage their growth and development.

In summer, it may be beneficial to leave some weeds, as it can provide more moisture and shade, which is good for other plants.

Plants maintenance

As well as irrigation, which is a key part of plant maintenance, it is sometimes necessary to check and monitor plant growth. You need to make sure that the plants are not buried under stones or that nothing is preventing them from developing properly. It is essential to check their condition regularly and take action if necessary to ensure they develop properly.

Materiel maintenance

a) **Individual collector:** The individual collector may fall off, so check it regularly and replace it if necessary. After 1 or 2 years, when the plant has developed sufficiently, it no longer needs the collector. The collector should therefore be checked and removed to allow the plant to develop properly.

b) **AFDS:** The drippers can move, so it's important to check regularly that the water is being distributed to the right places. You should also check for leaks along the pipes and make sure they are not kinked, which could impede the flow of water.

c) **Cocoon:** The plant's protector can become deformed if stones, branches or people walk over it. If the protector is deformed, the water will not be properly contained. If this happens, every effort should be made to restore it to its original shape. The protector can also fall off and hinder or damage the plant. It is therefore important to check regularly that it is properly in place. In highly degraded Cocoons the protector must be retired in order to avoid damages to the seedlings, and also a negative visual effect. Despite being biodegradable, the presence of protectors spread out in the planting area could cause a negative perception of the plantation, despite having good results in the plantations survival and performance (the aesthetic values also count!).

Machinery maintenance

1. To ensure a safe and efficient working space, at the end of the day, it's important to clean the air filter. To do that, normally you just have to remove the plastic cover where the air filter is located; remove and clean it.

2. The blade or rope head also needs to be cleaned and greased regularly.

3. Daily maintenance for the chainsaw involves sharpening of the blade and cleaning of the filter.

Risk and advice

Health and safety

- **Safety equipment.** To work safely and avoid risks, you should always carry a fire extinguisher and a first aid kit with you, and keep it close to the workplace.

  Operators will be provided with personal protective equipment and means of communication (e.g. mobile phone) to alert emergency services.

- **Training and Formation**.
  - First aid formation. This training course of first aid and fire safety will allow the workers to learn the first rescue measures in order to limit the deterioration in any exceptional situation like an accident or a fire.
  - Aerial tree pruning. To avoid accidents in the trees, the workers should do a training course to teach the correct methods to safely cut trees at heights. It is necessary if the area has some mature trees in the area.
  - Vertical access. To avoid the risk of workers falling, a vertical access permit is required for certain slopes.

  This must be taken into account when working in areas with a slope of more than 45º or in very slippery areas. This course teaches you how to work with ropes and rescue techniques. So you can work safely at height. It also teaches you how to install anchors on the ground for rope.

- **Signalisation.** Signs should be provided to warn of hazards to hikers and employees.

Fall of tanks

It is mandatory to monitor the anchoring of the various tanks throughout the project, to ensure that the tanks do not move or sink into the ground. Here, in the photo, sedimentation has accumulated under a tank and pushed it down the slope, causing the loss of 12,500L present.

Leaks and burst

Anywhere on the irrigation network, leaks can occur. The best solution is to monitor and measure the water present. If water loss from reservoirs is unexplained, the source should investigate. When the leak is identified, it is possible to repair with, depending on the material, silicone or glue.

Pest

- Insects: eat plants.
- Rabbit/goat.
- Cocoon insect/snail/mouse feeder if humid environment.

Leak of AFDS

You need to pay attention to the AFDS level pipe, there is a possibility that it leaks, it never loses a lot of litres but every day a little. To limit leakage, it may be necessary to change the seals regularly.

Sinking tanks into the ground

The weight of the tank has an impact on its correct positioning. A tank that is filled and placed on soft ground may, over time, sink into the ground and shift its centre of gravity. This can have an impact on storage efficiency. Over time, this increases the risk of the tank leaking or tipping over. You can apply the tanks to a hard surface as much as possible and, if necessary, add wood or other resistant material underneath.

D. MONITORING REFORESTATION

To ensure that the project is optimised, it is essential that the parameters are monitored regularly throughout.

As a first step, soil monitoring is recommended to better understand the effectiveness of the methods. The conclusions drawn from Life Nieblas will be presented in this section, highlighting the optimum parameters.

In addition, the monitoring of data collection will be presented. The tools used and the types of analysis carried out will be described, making it possible to compare the results with other methods, observe changes in performance and compare them with meteorological data.

This comparative analysis will help to assess the efficiency of the collectors under different conditions and to determine the measures to be taken to improve performance if necessary.

Regular monitoring of the parameters will also make it possible to identify any deviations or problems, thus facilitating the implementation of appropriate corrective measures to ensure the smooth running of the project.

1. Experimental plot

Objective

An experimental plot has been established in a terrace close to Barranco del Andén to better compare the effectiveness of the different treatments and include new ones: DeepDrop system and organic amendment with biochar. The soil is skeletik (high stones content) and similar to the soils of the area.

In order to better compare the effects between treatments (including new ones):

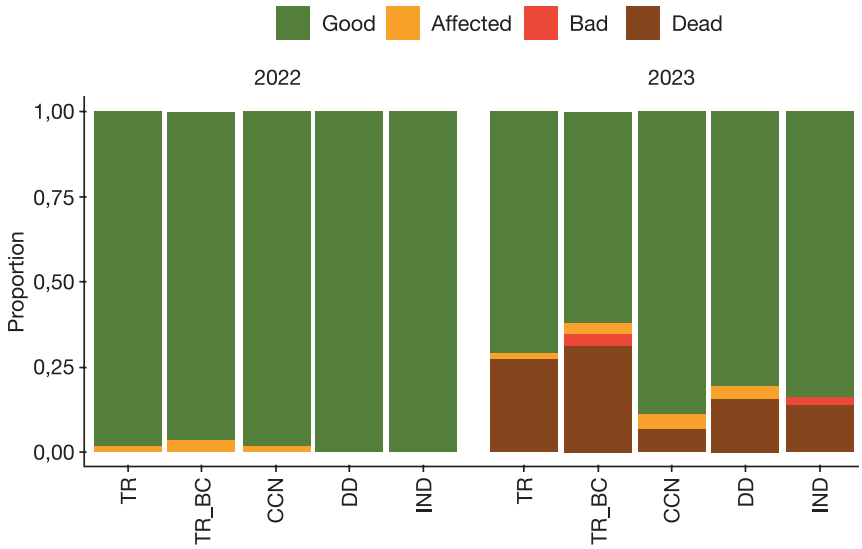
- Cocoon
- IFWC (improved version)
- Traditional
- DeepDrop
- Traditional+biochar

Material

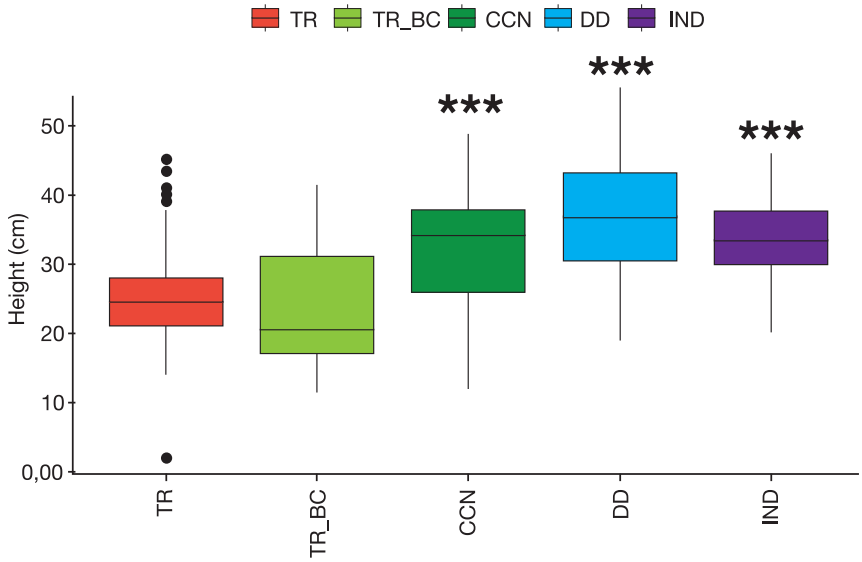
- Most representative/planted species (*Myrica faya*)
- 50 seedlings per treatment
- Compost at 1 l per hole
- Traditional+biochar treatment with two doses (1 kg and 2 kg of biochar per planting hole)
- Simplification of watering regime (Jul-Sep + emergency)

Results

We can see a clear effect of the irrigation and planting systems in the physiological state of the planted seedlings, specially for DeepDrop (DD) and IFWC (IND). However, the amendment with biochar (TR_BC) seems not to have a clear effect on the physiological state. This effect is maintained also when measuring height of the seedlings, with a significantly better growth in Cocoon (CCN), DeepDrop (DD) and IFWC (IND).



Proportion of monitored seedlings (referred to the unit) according to its physiological state in experimental plot, per treatment. Traditional: TR; Traditional+biochar: TR_BC; Cocoon: CCN; DeepDrop:DD; IFWC: IND.



Height of monitored seedlings in experimental plot, per treatment. Traditional: TR; Traditional+biochar: TR_BC; Cocoon: CCN; DeepDrop:DD; IFWC: IND.

2. Irrigation water monitoring

Objective

To monitor water usage within the different reforestation methods as well as the purpose of each irrigation event, technicians have divided water use in the following parameters: Nursery, Traditional1, AFDS1, IFWC1, Cocoon1, and IFWC2 for reforestation methods during period 2021-22. Purpose subdivisions are: settlement, maintenance, support, waste and other. This system allows technicians to better judge the amount of water required per reforestation system but also to find out about possible losses, incidents and other activities that require the use of water.

Material

For traditional hose irrigation, the amount of water present before and after irrigation should be noted, so that the amount of water used can be calculated by subtraction.

In addition, to find out how many plants have been watered, the use of a tally counter is very useful.

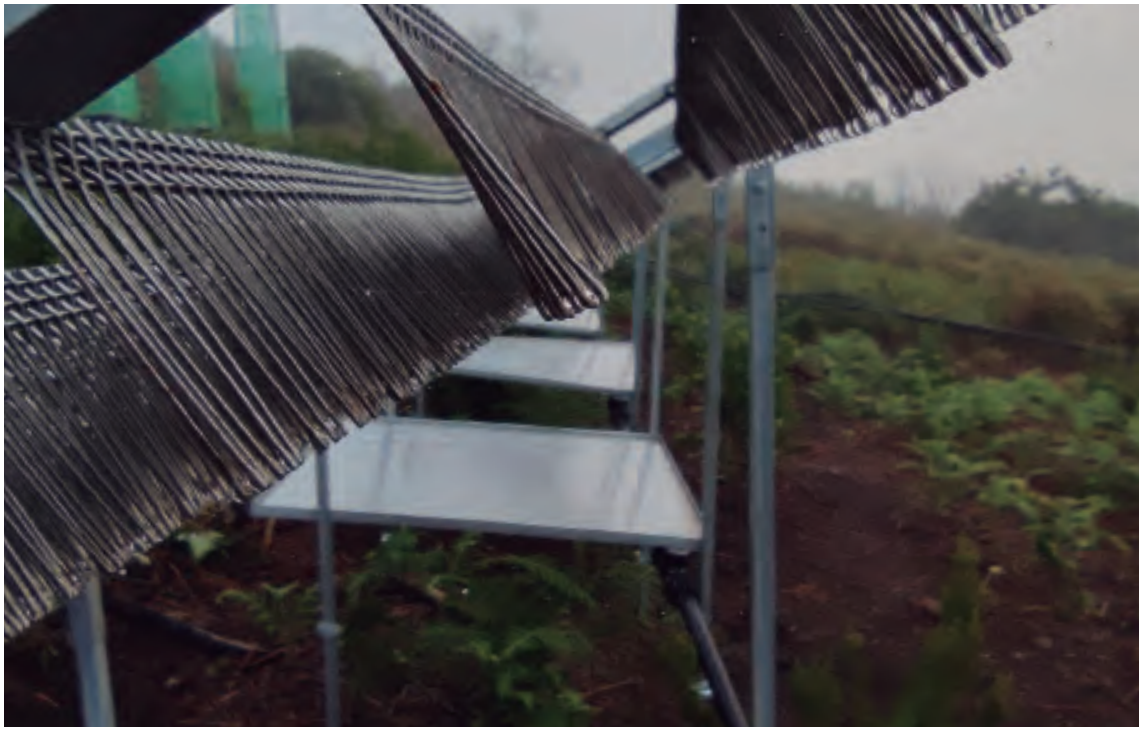
For each AFDS reforestation system, water metres with pulse sensors will be installed, as well as communication via Wi-Fi, with the data collection system located in the meteorological tower. This is to determine the following:

- The number of discharges/irrigations performed in the system.
- The frequency of these irrigations.
- The quantity or volume of water (in m^3) supplied in each irrigation.

The water metres have to achieve a precise measurement result, even with difficult water quality (suspended solids) or in difficult installation situations (outdoors). It must be equipped with communication interface for:

- Electronic pulse.
- Wired M-bus.
- Radio via wireless M-bus.

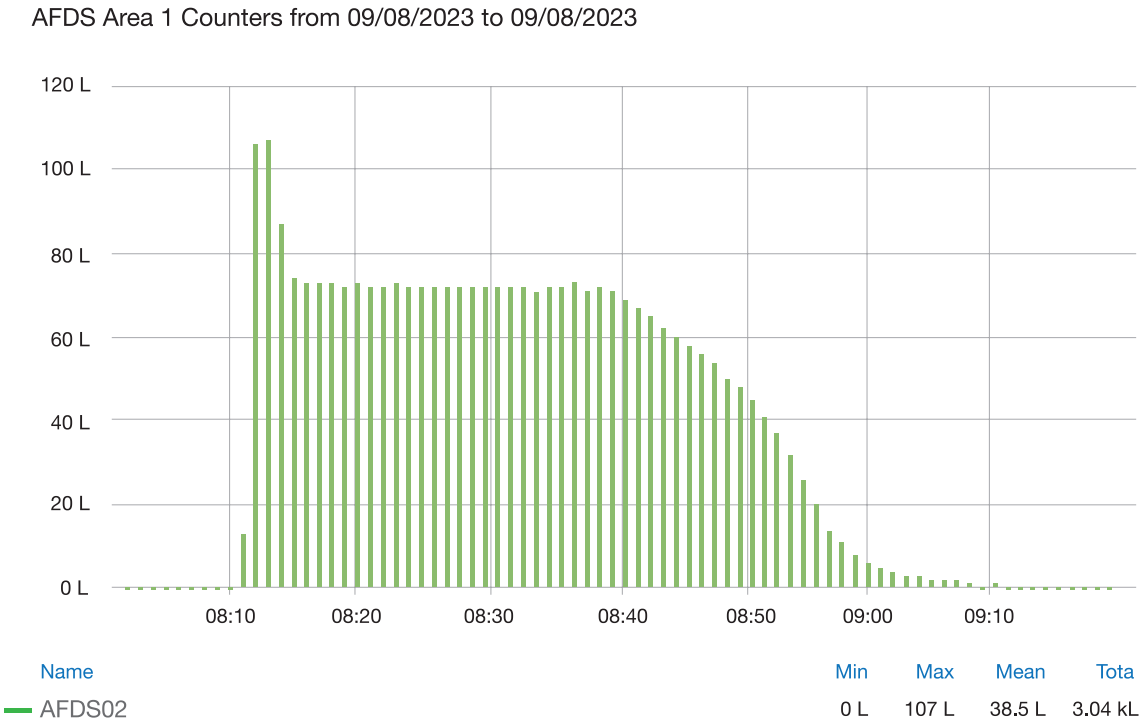
With these characteristics, the water metre selected is the multi-jet dry dial metre MTKD-M DN 32 model from ZENNER International GmbH & Co. KG.



If the data acquisition system is to be automatic, as in the case of Life Nieblas, it is necessary a Wireless Pulse Input Sensor to connect to the counter pulse output.

However, it is advisable to check the amount of water in the AFDS tanks on a daily basis. This will ensure that spills occur regularly and that the quantity of water spilled is known, so that the information provided by the monitoring tools can be confirmed.

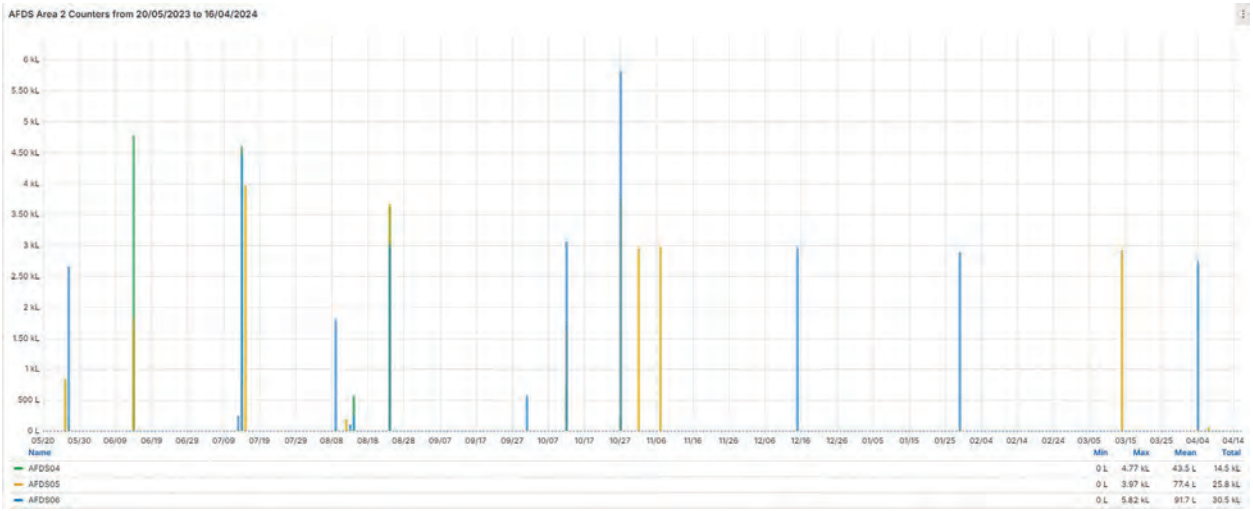
The following graph shows an AFDS discharge process measured through its water counter and recorded in the data acquisition system. In this case it corresponds to the discharge of about 3000 litres of fog water in one of the irrigation sectors:



Parametre

All the information can be recorded manually in datasheets. The types of irrigation are recorded, along with the corresponding number of plants, the date and the amount of water used. This makes it possible to view the amount of water used for each irrigation method and per surviving plant.

As mentioned above, an automatic data acquisition system may also be installed. As well as a data processing system like the one used in Life Nieblas. The following graph shows different irrigation discharges recorded in different sectors with AFDS.



Results and conclusion

Based on the data collected in July 2023, the analysis reveals that the traditional method is the most water-intensive per surviving tree. However, it is evident that all the innovative methods employed in the Life Nieblas project have successfully contributed to reducing water utilisation.

To better understand the extent of water reduction achieved by each method, we compare the amount of water used per surviving tree in the traditional method (133 Litres) with the innovative methods. The results show that IFWC demonstrates the most efficient water usage, with only 35 Litres per tree. The second most water-efficient method is Cocoon, averaging 57 Litres per plant, followed by AFDS, which requires 101 litres per tree.

Reduced water consumption indicator

Once the water consumption indicators are identified, the next step is to analyse the actual reduction in water usage achieved during the Life Nieblas project.

To accomplish this, a comparison can be made between the water consumption of the trees under innovative methods and what it would have been if traditional methods were employed instead. By utilising control groups with traditional methods, it becomes possible to estimate the water needs if all projects were following traditional approaches.

- Specific data for the Life Nieblas project as of July 26th, 2023:

Statistics of methodologies development

Methods	Total (L)	nb of trees planted	L/tree	Survival rate	Surviving trees	L/surviving trees
AFDS	175146	2082	84.1	0.83	1728	101.4
Cocoon	90195	1807	49.9	0.87	1572	57.4
i-FWC	120400	3829	31.4	0.89	3408	35.3
Traditional	175146	3052	57.4	0.43	1312	133.5

With this data you can have the reduction of water consumption per methods:

- IFWC Area: In this area, a total of 3,408 survival trees were planted. By implementing the IFWC method, there was a significant reduction of **334,402** litres of water consumption compared to using the traditional method.
- AFDS Area: The AFDS method was applied to a total of 1,728 survival trees in this area. This led to a reduction of **55,478** litres of water compared to using the traditional method.
- Cocoon Area: Within this area, 1,572 survival trees were planted using the Cocoon method, resulting in a reduction of **119,614** litres of water in comparison to the traditional method.

When summing up the water savings achieved by each method, the Life Nieblas project successfully reduced water consumption by a total of **509,495** litres.

In conclusion, the data clearly demonstrates the effectiveness of the innovative methods employed in the Life Nieblas project. By analysing the reduction in water consumption through various methodologies, it highlights the significant positive impact on water conservation. These findings can serve as valuable insights for future projects and initiatives aimed at sustainable water management and environmental conservation.

3. Soil monitoring

Soil moisture

Objective

To understand and effectively manage water storage in the soil, soil moisture monitoring involves collecting precise data on soil water content. By doing this, informed decision-making can be made about the amount of water needed for plants, irrigation management, and reforestation methods.

Material

There are several types of sensors that can be used to monitor soil moisture levels in reforestation projects. The main categories could be:

- **Capacitance Sensors (TDR and FDR):** Measure changes in capacitance between two plates within the sensor, which vary with the water content in the soil. The capacitance is affected by the permittivity of the soil, which in the soil as any other mixture is directly related to the permittivity of its individual components. Water, having a particularly high permittivity compared to other soil elements, is the main influence in this measurement.
- **Resistance or Resistive Sensors:** These use two electrodes to measure the electrical resistance of the soil, which changes with its moisture content.
- **Tensiometers:** Instead of measuring the total amount of water content in the soil it measures the tension at which water is held in the soil and the force needed to extract it. They can also be either capacitive or resistive.

To perform this monitoring, capacitive sensors (FDR) were chosen. Those are capable of measuring the dielectric permittivity of the soil, which varies proportionally with water content. Capacitive sensors are convenient for measuring soil moisture as they provide accurate results and require minimal maintenance. Furthermore, these sensors are suitable for telemetry.

The model installed in field is the following:

Implementation

It is possible to install a wireless network of soil moisture sensors and connect them to the central weather station using wireless communication equipment. They should be installed at strategic points, including:

- At least 2 per reforestation method areas at two different locations.
- In the surrounding forests to have a reference about the soil behaviour in natural conditions.
- One in the surrounding deforested area for the same purpose.

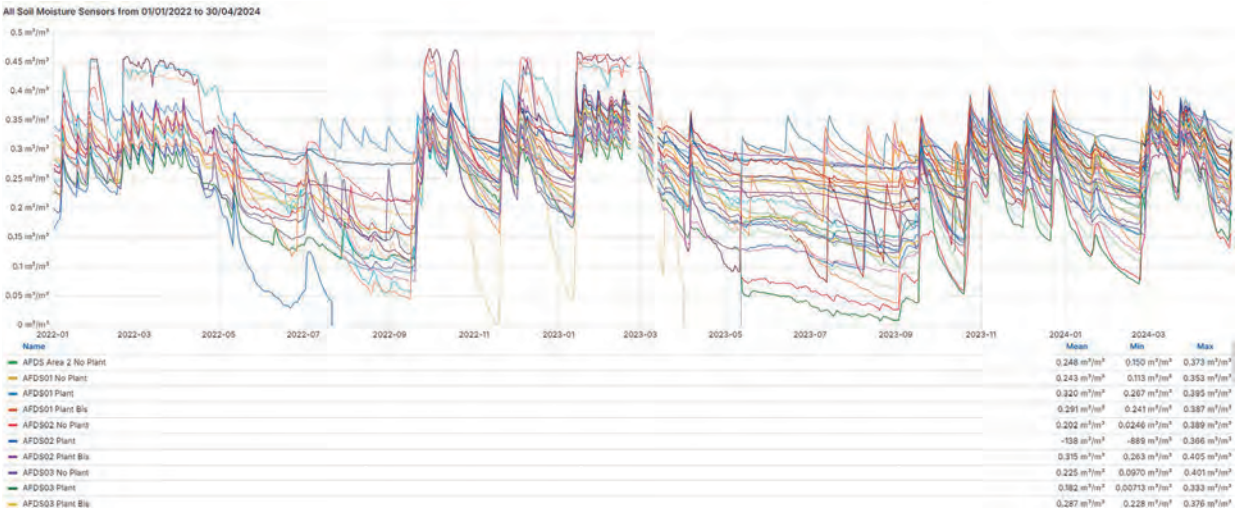
To retain the collected data, it is necessary to integrate them into a data acquisition and recording system. This system should have a data storage capacity of at least one month and be fully self-sufficient in terms of power supply. The sensors transmit moisture information wirelessly to the weather station for online availability. However, if the data download is manual, it is important to clear the memory of the sensors after downloading the data. If the memory is full, the sensors may stop recording new data, which should be avoided.

Regarding installation, sensors were placed vertically in the soil at 30 cm depth which provides an average moisture value along the probe; it could also be installed horizontally so the moisture is measured at the specific depth of the probe. It is important to consider that burying the probe can modify the soil characteristics, potentially affecting the representativeness of the reading.

Despite differences among dielectric methods, none are affected by soil texture, temperature, or salinity. However, these sensors have certain limitations related to frost, soil conductivity, gaps, initial calibration, temperature, and agricultural mechanisation.

At the reforestation area of Finca El Pinillo is a network of 35 soil moisture sensors distributed in the different work areas:

- 4 reference sensors in pine forest, chestnut forest, elm forest and in a non reforested area.
- 3 sensors in the traditional reforestation zone (one of them without associated trees and without irrigation).
- 2 sensors in the reforestation zone with Cocoon (one of them without associated trees and without irrigation).
- 5 sensors in two reforestation zones with individual fog collectors (two of them without associated trees and without irrigation).
- 13 sensors in different reforestation areas with an irrigation system associated with 6 AFDS (four of them without associated tree and irrigation, representative of each irrigation area).
- 8 sensors associated with the experimental plot that includes traditional reforestation, Cocoon, a deep irrigation system, individual collector and traditional reforestation with two doses of Biochar applied to the ground (one of them without associated tree and without irrigation, representative of the area).



When it comes to soil moisture, there are several factors that can affect their readings, even under the same conditions. These factors can include:

1. Soil composition: The composition of the soil can affect the dielectric constant, which is the measure of a material's ability to store electrical energy in an electric field. Soil with a high clay content, for example, will have a higher dielectric constant than soil with a low clay content, which can affect the readings of a capacitive soil moisture sensor.
2. Soil temperature: The temperature of the soil can affect the readings of a capacitive soil moisture sensor, as it can affect the dielectric constant of the soil. Cooler soil will have a lower dielectric constant than warmer soil, which can affect the accuracy of the sensor's readings.
3. Soil compaction: Soil compaction can affect the readings of a capacitive soil moisture sensor, as it can affect the electrical conductivity of the soil. Compacted soil will have a higher electrical conductivity than loose soil, which can affect the sensor's readings.
4. Sensor placement: The placement of the sensor in the soil can affect its readings, as it can affect the area of soil that the sensor is measuring. If the sensor is placed too close to a rock or other obstacle, for example, it may not be measuring the true moisture content of the soil.
5. Sensor calibration: The calibration of the sensor can also affect its readings. If the sensor is not calibrated correctly, it may produce inaccurate readings, even under the same conditions.

Interpretation

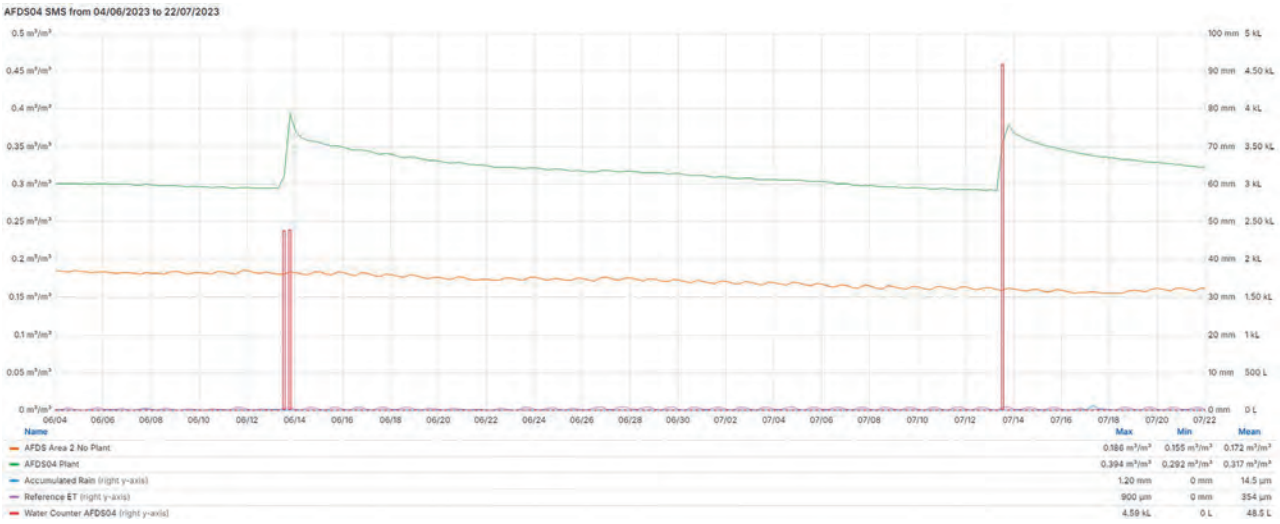
Once data is collected using soil moisture sensors, it is time to analyse and interpret it correctly to extract relevant information. Here are some steps and factors to consider for soil moisture data analysis:

1. Data collection: Ensure that the data is properly recorded and stored in a data management system. Verify that all relevant data is present, such as soil moisture measurements at different locations and time points.
2. Data quality control: Perform data quality checks to identify any errors or outliers. Check for outliers, inconsistent measurements, or data gaps. If issues are identified, recalibration of sensors or error correction may be necessary.
3. Correlation with other parameters: Analyse soil moisture data by correlating it with other relevant parameters such as rainfall, temperature, sunlight, etc. This helps understand the relationships between soil moisture and weather conditions or irrigation and planting practices.
4. Study temporal variations: Analyse temporal variations in soil moisture over time to identify trends and patterns. Look for seasonal patterns, short-term fluctuations, or exceptional events that could impact soil moisture.

5. Spatial analysis: Analyse soil moisture data based on its location within the study area. Identify spatial variations in soil moisture and look for areas with particular conditions, such as drier or wetter zones, areas with drainage issues, etc.

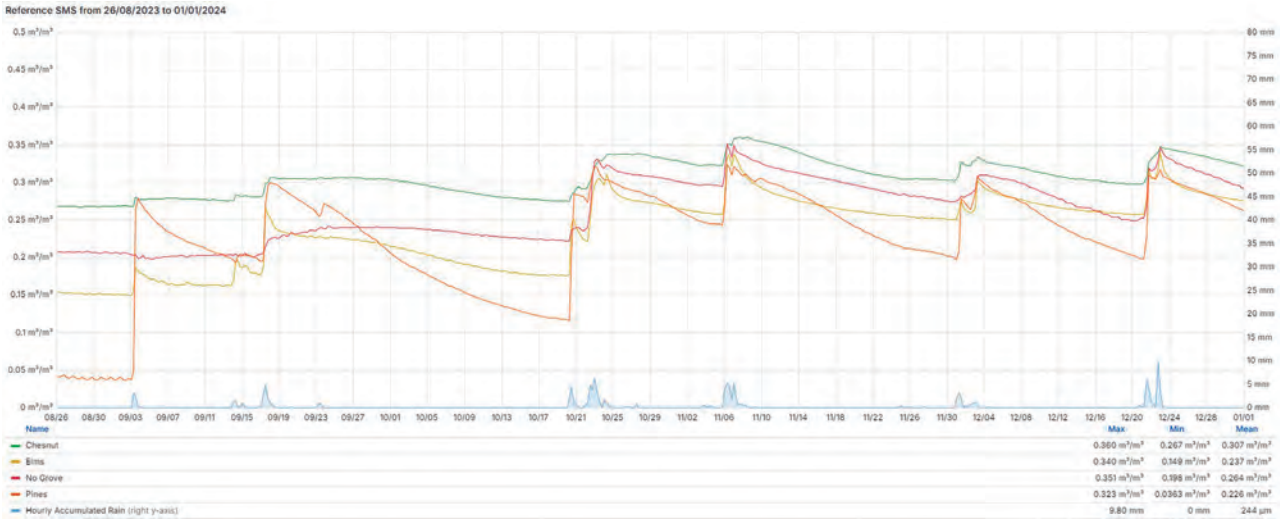
6. Integration into an irrigation management system (optional): Utilise soil moisture data to adjust and optimise irrigation strategies. Target thresholds can be used to trigger irrigation based on actual soil needs, leading to water savings and improved irrigation efficiency.

The following graph shows, for example, two irrigation discharges from an AFDS and their influence on the water content of the irrigated and the non irrigated area.



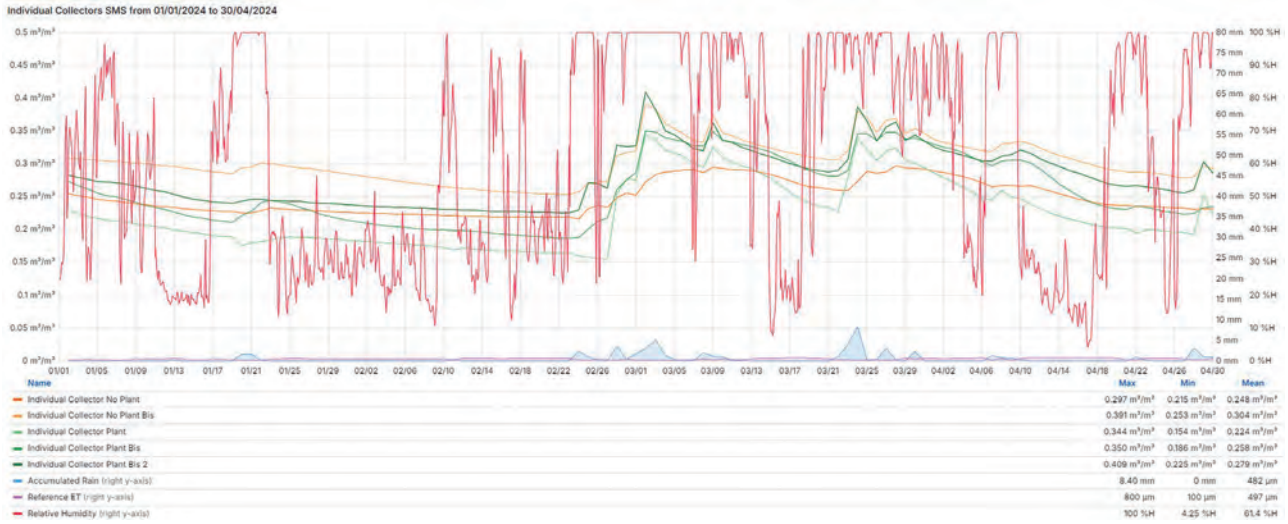
Here is another example where the influence of rain or fog water collection is observed in the reference areas with different types of forest.

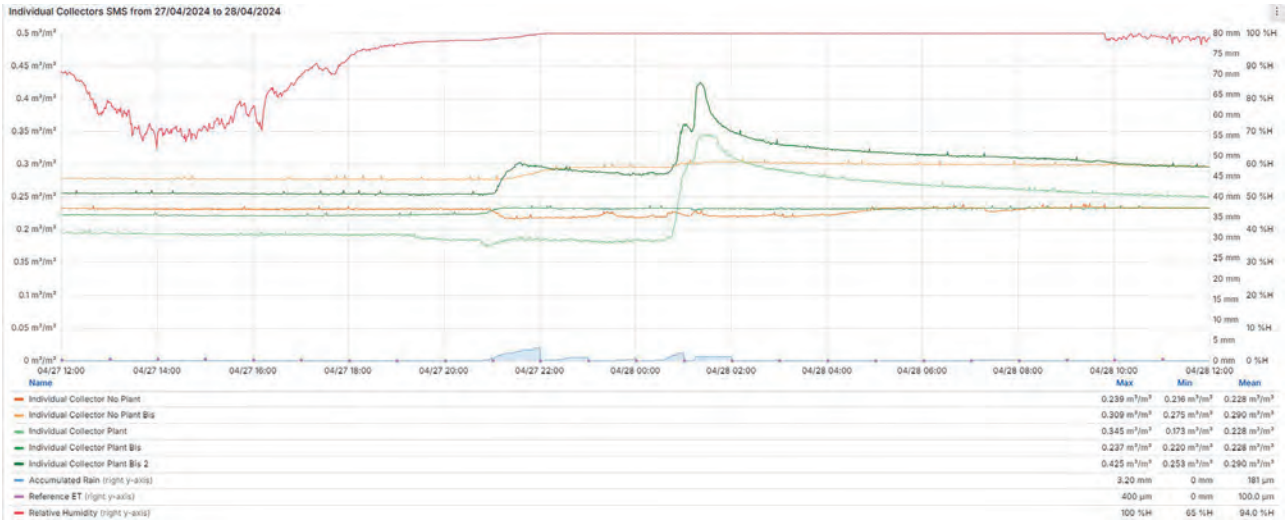
Further insights can be drawn from the graphs below. These illustrate how individual collectors effectively enhance soil moisture under conditions of high relative humidity or during light precipitation:



Results and conclusion

Physico-chemical parameters

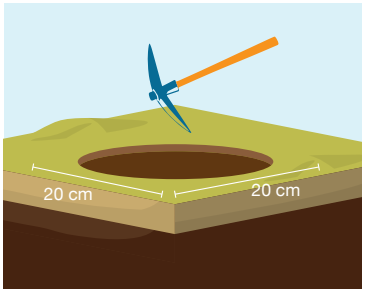




In order to know the tendency in soil quality, a detailed physicochemical characterization is required before planting and at the end of the project. The first characterization will be useful for the discussion of vegetation development results. The second characterization will be focused on determining C storage in soil; a complete characterization is proposed in order to fully benefit from the sampling campaign.

Sampling

Once the site has been selected for soil analysis, you can start sampling.



STEP 1. Excavation method

The method chosen for sampling is based on 20 cm by 20 cm squares. Dig to a depth of 20 cm using a pickaxe.

STEP 2. Removing stones

Once the depth has been reached, manually remove any stones larger than 25 cm. These stones will then be assessed in terms of abundance and weight.

STEP 3. Sieving the soil

The remaining material in the square must be sieved using two sieves. The first sieve has a mesh opening of 7.5 mm. The material that passes through the first sieve must then be sieved through a second sieve with a mesh opening of 2 mm. It is important that all the material passes through these two sieves.

On the 2 mm sieve, it is necessary to manually break up any soil aggregates that are larger than 2 mm. Otherwise; they will be interpreted as stones larger than 2 mm, which is undesirable.

STEP 4. Conditioning the sample

Once all the sample has been sieved, place it in a well-sealed plastic bag to prevent any loss. It is also important to label the package correctly.

STEP 5. Drying the sample

All the samples must be air dried. This can be done in a room or covered area with air circulation, spreading the sample out on a sheet of paper about 1 cm thick. Usually one week is sufficient for the sample to dry completely before analysis.

Granulometry

It is very important to know the granulometry of the soil since this determines soil properties. We distinguish two soil fractions: fine fraction or active fraction, that contains all the particles less than 2 mm diameter; and coarse fraction (stoniness) constituted by gravels and stones of different fractions. Fine fraction contains sand, silt and clay, which is the chemically active fraction of the soil. Texture is the soil property that expresses the proportion in which those fractions are represented. Soil texture determines water retention capacity, plasticity, aeration, nutrient sorption, and organic matter stabilisation, among others.

The composite sample, obtained after mixing the set of samples obtained in the sampling process, will be left to dry spread on a paper or similar for a few days. Once dry, we will sift the sample with a 75 mm sieve. The fractions retained in the sieve consist of stones (fractions with diameters between 250 mm and 75 mm). The sieved material will be sieved again on a 2 mm sieve. Fractions retained will correspond to gravel (75-2 mm) and the material that passed through the sieve will be classified as fine earth (<2 mm). Weighting the three fractions and expressing them according to the overall weight of the composite sample will obtain the proportion of the three different fractions.

Physicochemical characterization

For each sample the following parameters will be determined:

- Texture type
- Clay content (g kg-1)
- Organic matter (g kg-1)
- Carbonates (g kg-1)
- Electrical conductivity, saturated extract (dS m-1)
- Soil pH
- Total nitrogen (%)
- Available phosphorus (mg kg-1)
- Available potassium (mg kg-1)

Erosion

It is important to determine the extent of the concentrated erosion processes that may occur in the area. To do this, it is necessary to delimit the areas where rills are more than 5 cm deep When the affected area is small it can be measured directly in the field, using a measuring tape or similar. When the surface is larger, the area could be delimited using aerial images through photo interpretation. Lateral limits of the affected area will be established by the rills deeper than 5 cm located at the laterals of the area. The upper limit will be delimited by the line that joins the points from which rills started to have a depth of 5 cm. The lower limit will be delimited by the union of the drainage points of the rills. Already stabilised rills will not be included in the accounting. To consider a rill as stabilised, it must have vegetation along its entire length.

- Measurements will be done just after plantations and three years after.

4. Trees monitoring

Seedling identification

Objective

The main objective of seedling identification is to evaluate the effectiveness of the reforestation techniques used in the project. To achieve this, a representative sample of all species and areas must be selected. The goal is to identify at least 50 seedlings per methodology, species, and site, taking into account site-specific variables such as slope, soil types, etc. In cases where there are less than 50 seedlings for a particular species or site, all individuals should be identified.

Identification is crucial to measure and monitor the growth and survival rates of seedlings accurately. The objective is to achieve a minimum survival rate of 30% per parcel, with the selected and identified 30% distributed evenly across the entire parcel.

Identification can be made, for example, using aluminium tags (be aware of using high quality materials in order to avoid degradation of the tags). Moreover, we strongly recommend having a map of the area with a schematic localization of the seedlings, and/or localising them using a GPS and by photo interpretation of high resolution orthoimages (if we do not have a high precision GPS).

It's very important to take measurements on the same plants, so it's important to identify the plants clearly. To do this, you need to use an identification system (such as plastic or metal labels, as shown in the photo below). It is very important to guarantee that the labels will be in place for years and that they will not disintegrate, for which the use of metal labels is recommended.

Methods

For seedling identification, a system of codes should be implemented to assign a unique number to each plant. This system helps keep track of each plant throughout the project's duration. An example of a code format could be "CCN2 – A – 1 - 7-50," where:

- **CCN:** represents the reforestation method used.
- **2:** represents the planted area.
- **A:** refers to different subzones within each parcel to facilitate identification (Variation of irrigation regime, amendment or other variation etc.).
- **1:** indicate the year of planting(it may be the same as the area, in this case it is not necessary).
- **7:** is the species code for each planted species.
- **50:** is a unique number assigned to each individual plant within its species category?

The unique number should be written on durable tags that are resistant to degradation. These tags can be attached to a stake using wire or other suitable materials. Additionally, a map of the area with the localization of seedlings or GPS coordinates and high-resolution orthoimages can aid in accurate identification.

Results

With proper seedling identification, the project can track the growth and survival rates of each species and assess the effectiveness of the various planting methodologies used. This information will help improve reforestation strategies and achieve the project's objectives more effectively.

Quantitative evaluation of seedling

Objective

It is important to control the quantity of the seedlings provided for planting in order to guarantee that the planting actions will be executed according to the working plan. Thus the number of the seedlings delivered and planted for each of the reforestation techniques (control, Cocoon, autonomous discharge system and individual fog collectors) will be recorded. Moreover, the area planted using the different planting techniques will be also delimited in a map. It is important to correctly distribute the planting methodologies (traditional, Cocoon, individual fog collectors, ADS) in the planting area in order to catch all the variables that can affect plant development (orography, soil depth, orientation, etc.), and try to distribute the planting methodologies interspersed (avoid concentrating in unique areas).

Methods

Quantitative evaluation can be performed using tools such as QGIS or Google Earth to create maps and track the planting area. Surface monitoring allows for accurate measurements, and a GPS watch can help trace paths and plot precise boundaries of different planting platforms.

Data on plantings, dates, methodologies used, and exact locations can be recorded in spreadsheets like Excel for efficient tracking and reporting.

Results

The quantitative evaluation results will provide insights into the distribution of planting techniques and their effectiveness in different areas. This data will help identify the most successful methodologies and their impact on plant growth and survival rates.

Qualitative evaluation of seedlings

Biomass

Objective

Moreover, a qualitative evaluation of the seedlings must be done. This evaluation allows detecting shoots and roots health problems (insects, diseases, chlorosis, loss of leaves, stem scars, rotten roots). Moreover, we must pay attention to the presence of spiralized roots or Jrooting. If any of these problems (or others) is detected, it should be noted.

Methods

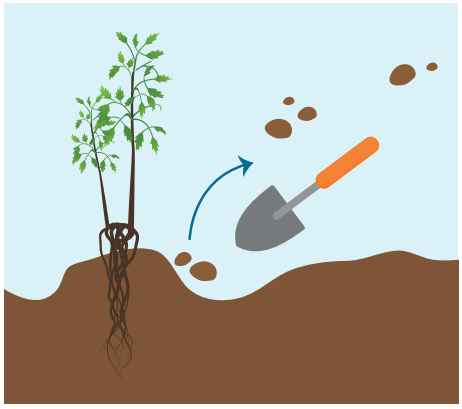
It is necessary to record the type and dimensions of the containers used for the seedlings, as well as the nursery they originate from.

Evaluation of *F. carica* and *O. europaea* planting material. Appropriate root structure in *F. carica* seedlings (right). Bad root development in the container can explain future problems in root growth (left).

To perform a comprehensive evaluation, the length, volume, and biomass of the roots need to be measured for 3 plants per species and per methodology. Height and basal diameter have already been recorded in a previous stage of the project. This analysis should be conducted in the final year of the project.

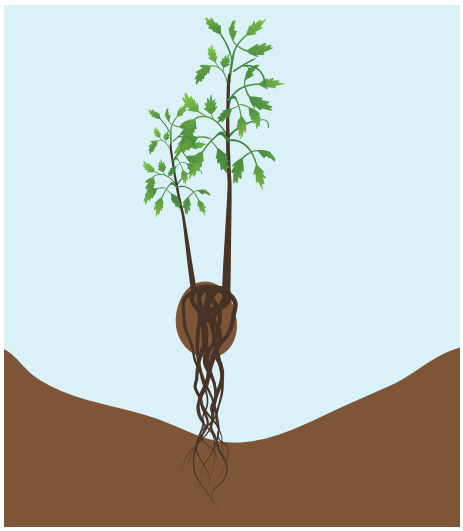
STEP 1. Digging a hole

To gather these data, gently and delicately uproot the plant using a hand trowel or shovel. Caution must be exercised during this step to avoid damaging or cutting the plant's finer roots.



STEP 2. Remove the plant

Remove the plant from the soil, without damaging.

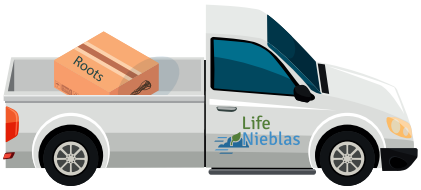


STEP 3. Cut the plant

Once the roots are exposed, trim the above-ground portion of the plant and measure its size and thickness.

STEP 4. Stock the different parts

Store the two plant parts separately in securely sealed bags for transportation. It is crucial to properly identify and note the assigned code for each plant to ensure data integrity and traceability. Labels can be attached to the bags for identification purposes.



STEP 5. Transport

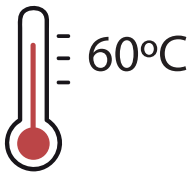
Transport the plants carefully to the laboratory to continue the work.

STEP 6. Wash the plant

To obtain precise measurements of biomass, it is important to wash the roots thoroughly to remove as much soil as possible. However, great care must be taken to avoid breaking or losing the roots. If a root is accidentally severed, it should be set aside and preserved. It will be weighed along with the rest of the plant and included in the overall biomass measurement.

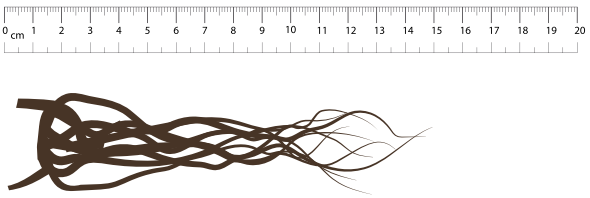
STEP 7. Dry the plants

After washing, the plants need to be dried. Place the above-ground and underground parts in an oven set to 60ºC for 4 days.



STEP 8. Take picture

Once dry, take photographs of the plants. It is essential to include a scale or reference object in the photos, such as a 20cm ruler placed next to the roots. Repeat the same procedure for the above-ground part.

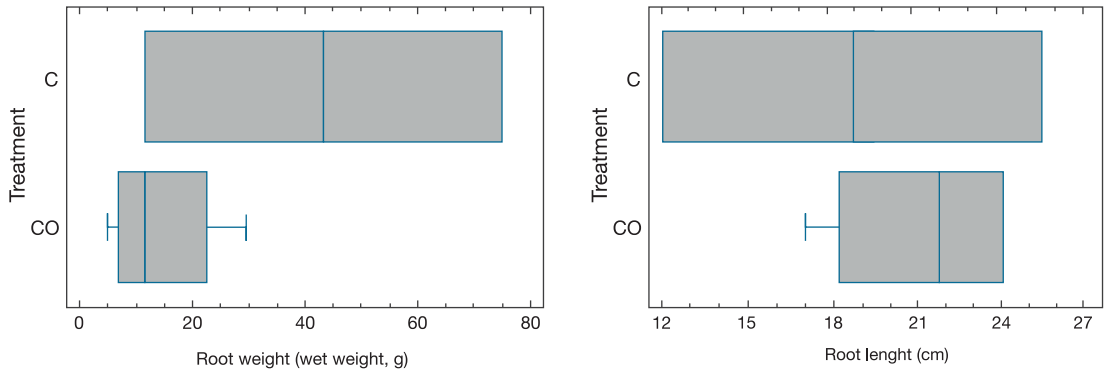


STEP 9. Weigh the plant

Next, weigh the two plant parts separately using a precision balance to obtain the exact weight of the biomass.

Results

This information allows us to monitor the growth of the aerial and subsurface (roots) part of the plant. This ratio is specially important for guaranteeing the survival of the plantations at long term.



Box plots for root growth and weight in *Olea europea var. arbequin* plantations by treatment (C-Control, CO-Cocoon)

Plant vigour

To characterise the planting material, the shoot and root height, as well as seedling base diameter, need to be measured. This should be done for a minimum of 10 seedlings per species prior to planting, immediately after their reception. A measuring tape or stick should be used to measure height, while a calliper is required for measuring diameter.

Tree vigour serves as a health indicator and provides insights into survival rates. During the normal growing period, vigour is assessed using the following semi-quantitative scores:

- **3: Healthy tree**, with more than 75% of green, not wilted leaves. Also active growing points (apices) may be visible.
- **2: Affected tree**, with 25-75% of the leaves being wilted, yellow or brown.
- **1: Severely affected tree** with less than 25% of the leaves being green (i.e. the majority wilted, yellow of brown).
- **0: Presumably dead tree** with no or only wilted leaves. Trees, however, may still recover by reporting after a rain event.
- **R: reported tree.**

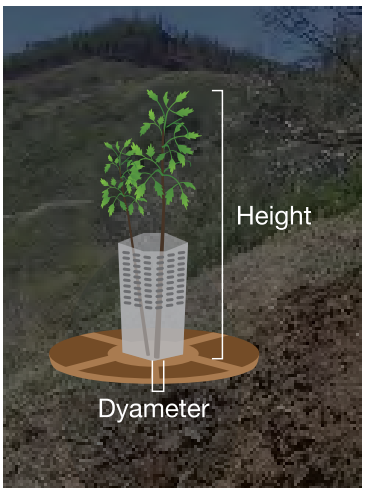
Seedlings vigor (observations)

G: Grazing B: Broken stem P: Plague infestation T: Tree fall

Plants growth

Tree growth is assessed by measuring tree height and diameter. The growth measurements should be conducted on the same seedlings used for vigour assessment.

- **Tree height** using a measuring tape/stick. Rodents grazing on saplings may cause negative growth.
- **Stem diameter** is measured at the tree base (at 10 cm from soil surface).

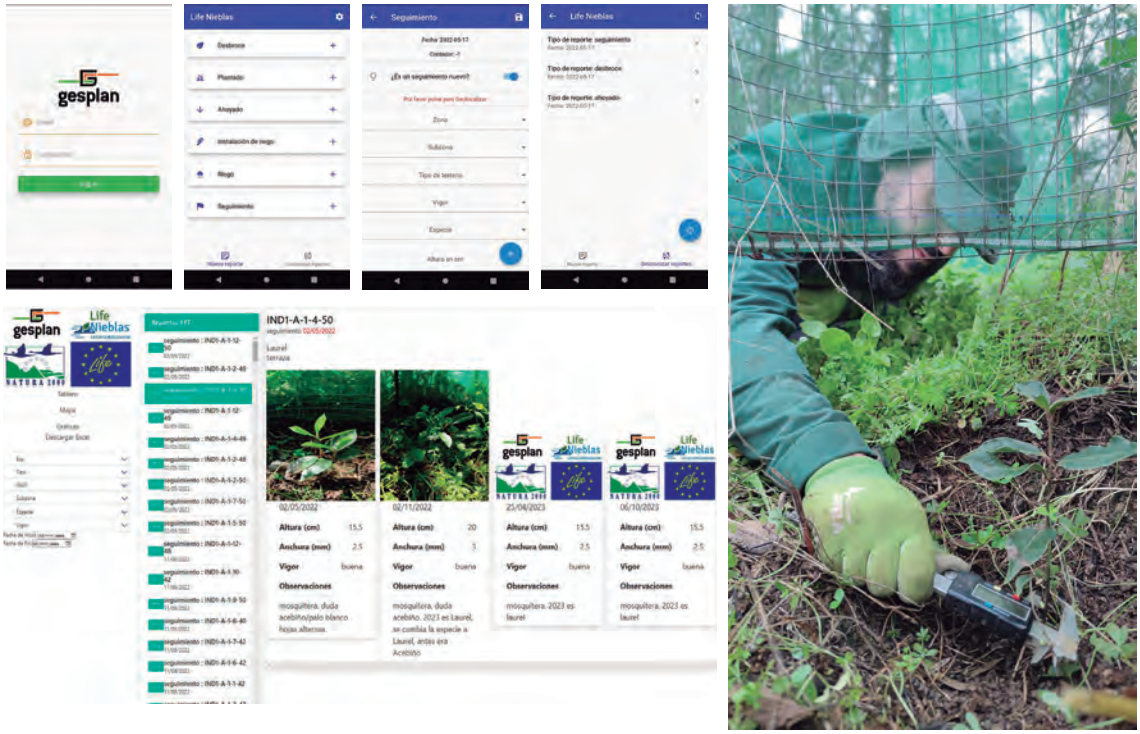


Methods

The following tools are required for the evaluation:

- Measuring tape
- Shovel
- Plastic or metal labels
- Marker pen
- Camera
- Notebook
- Calliper
- GPS

Furthermore, Life Nieblas developed an application to carry out evaluation and to facilitate the monitoring process. This app enables the electronic filing of seedlings' reports, eliminating the need for manual recording techniques. The data generated from the monitoring periods are uploaded to a server, which can be revisited in subsequent years to update seedlings' data.



The app offers several advantages, including the ability to geo-locate each sampled seedling, making it easier to locate them during future visits, thereby saving time and effort during field-work. Another feature that will help technicians is the 'photo' section which provides a visual reference to support measurements taken at every monitoring run throughout the years. Moreover, this App has been designed to also have a field work tracking system that will help field workers register their daily progress i.e. vegetation clearing, planting, ground conditioning, irrigation installation and watering.

The complete monitoring, including photos, height and size measurements, and geolocation, occurs after the plants have been planted. Throughout the project, the vigour of the plants is monitored through more rapid assessments carried out before and after the summer. These assessments involve visual inspection of vigour only.

Monitoring period

Measurements of the physiological state should be taken at annual intervals to see how the plants are developing: Growth measurements are not necessary on an annual basis, but it is necessary to take measurements shortly after planting and to observe their evolution after a few years, when the project is finalised.

The vigour and growth evaluation process involves the following steps:

STEP 1. search for tagged or tagging plants (first monitoring)

Walk through the plot and select a plant. It is important to choose the plant based on representativeness rather than its condition. Refer to the "seedling identification" section for details.

STEP 2. Plant localization (first monitoring)

Provide precise geo-location information about the plant, including its longitude, latitude, and the type of terrain it occupies (e.g., terrace, steep sloped, rocky).

STEP 3. Identify the plant's label (first monitoring)

If the plant does not have a label, create an identification code as explained earlier. It is crucial to be able to identify different species, even in their seedling state. Consider using decision-support tools like identification sheets. Retrieve the plant's identification from the database (application, Excel, or paper) and ensure that all information is correct. Note that there is a risk of losing labels as they may fall off, break, or become illegible. It is important to label a sufficient number of plants as some labelled plants may disappear over time.

STEP 4. Take measurements (first and last monitoring)

Once the plant has been identified and the information is confirmed (plot, species, and reforestation method), measure the plant's height using a measuring tape and its diameter using a calliper. Record each of these measurements carefully.

STEP 5. Take a photo (first and last monitoring)

During the first and the last monitoring of the project you should capture a photo of the plant.

STEP 6. Assess plant vigour (all monitoring)

Assign a vigour score to the tree based on the provided scale:

3: Healthy tree

2: Affected tree

1: Severely affected

0: Presumably dead

R: reported tree

Note that if a tree is initially marked as dead, it should not be removed from the identification system as it may still recover.

STEP 7. Keep information (all monitoring)

It is crucial to have an efficient information management system in place, including backup systems and proper organisation based on the project's scale. As the number of identified plants can be substantial, and as data accumulates over the years, careful consideration should be given to selecting the most suitable long-term system for effective and comprehensive analysis.

Interpretation

The collected data allows for various analyses, including:

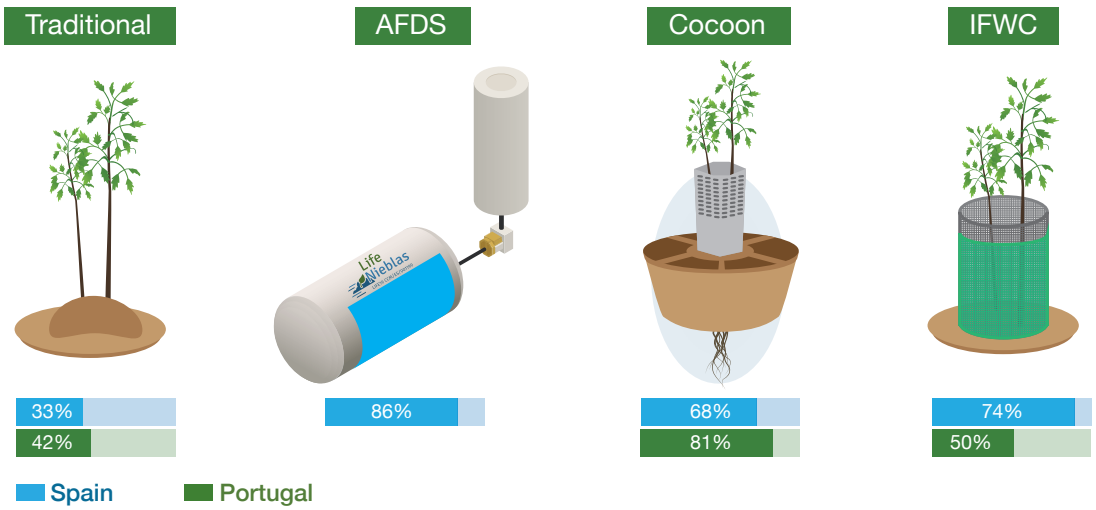
- Overall physiological state assessment.
- Physiological state analysis based on planting methods.
- Physiological state and growth analysis by species.
- Physiological state analysis based on irrigation regime if multiple irrigation strategies were defined.

Results

The results obtained from the evaluation provide insights into the most effective and suitable methodologies, species, and irrigation regimes based on soil type, plant type, and the project's climatic conditions.

From the monitoring results, you can access to different indicator:

- Survival rate:



- Physiological state per planting system.
- Physiological state and growth per species.
- Physiological state per irrigation regime.

Spontaneous vegetation development

It's an important indicator of the passive restoration of the restored site. Passive restoration means simply allowing natural succession to occur in an ecosystem after removing a source of disturbance. The spontaneous regeneration of forests after a wildfire is a classic example of passive restoration. The structure and composition evaluation of this vegetation will allow determining the restoration degree of the site.

Structure will be evaluated quantifying cover types in each site or subsite (differencing site specific variables: flat, slopes, soil type, etc)

It is necessary to distinguish between herbaceous cover, woody species cover (shrubs and trees) and other soil cover types (litter, dry branches, plant remains, etc).

Methods

The method proposed is the point transect, where the evaluator records the main cover every 10 cm in a 25m transect. Woody species must be identified and height measured.

A minimum of 3 transects per hectare and area must be done in order to have representative measures.

In slopes, transects must be done parallel and perpendicular to the slopes.

Transect vertex must be fixed with metal bars in order to repeat measures at the same place.

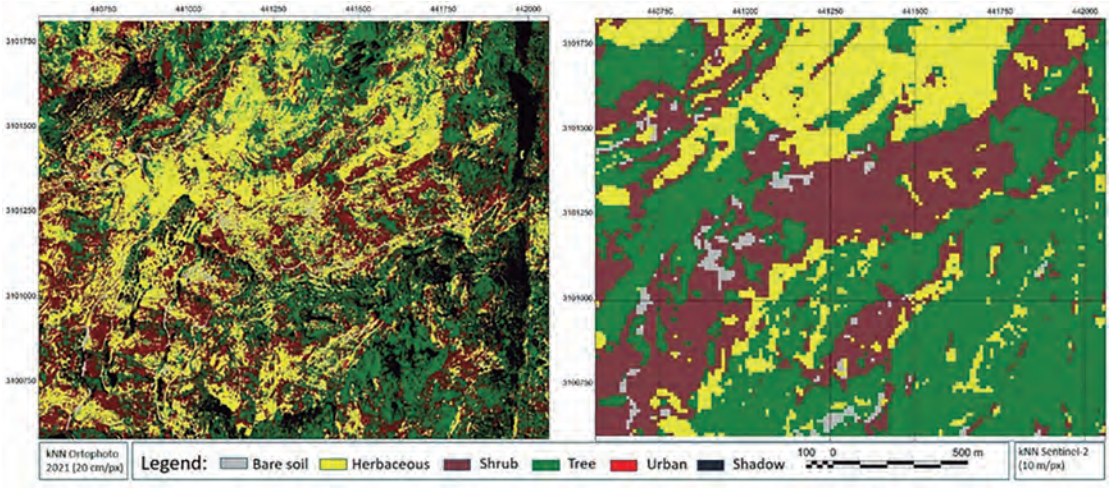
It is recommendable to sketch a map with vertex localization to localise them with precision.

Vegetation structure measures have to be taken just after plantations and at the end of the project.

If some knowledge about remote sensing and geographic information systems is available, the passive restoration of the area can be estimated through using satellite images, orthophoto images or drone images. Those images could be processed using semi-automatic classification methods (K-nearest neighbourhood, KNN) that allows to do land cover maps. The main vegetation types (herbaceous, bushes, trees) could be mapped, and also the presence of bare soil or other land covers.

Results

The results obtained from the evaluation provide mainly insights about the evolution of the spontaneous (non-planted) vegetation, which is the main vegetation in early planted areas. In general, woody vegetation cover must increase after restoration efforts. However, during the first years after planting herbaceous vegetation could be dominant, specially if organic amendments are used.



Land cover classification for El Barranco de la Virgen watershed made using 20 cm resolution orthophotoimages (left) and Sentinel-2 (10m/pixel) images (right), from 2021.

Herbaceous cover predominance in an area restored using sewage sludge as soil organic amendment.

E. IMPACT EVALUATION

1. Environmental impact

Water footprint

Presentation

The water footprint (WF) concept provides a comprehensive measure of water usage and consumption linked to producing goods and services. The WF developed by the Water Footprint Network purports to quantify human appropriation of fresh water resources and is aimed toward a variety of goals such as identification of business, process or product level water consumption, and promoting sustainable use of water resources (Hoekstra et al. 2011). The green WF is typically assumed to be equal to the amount of water that evaporates during the production of a product or service.

However, in the context of natural forest, this concept extends beyond mere consumption, highlighting the intricate role these ecosystems play in the hydrological cycle and their profound impact on global and regional water balances. In general, Forest ecosystems are known to enhance the quality of surface and groundwater by filtering out nutrients and pollutants from the air and minimising erosion and sediment transport (Brauman et al. 2007; Neary et al. 2009). Moreover, forests contribute to the formation of clouds and rainfall, a phenomenon particularly observed in tropical rainforests. The transpiration of water from leaves into the atmosphere also plays a vital role in the climate regulation, affecting patterns of precipitation and temperature both locally and globally. Hence, the water footprint of natural forests includes, not only the water transpired and evaporated in the ecosystem, but also the importance in maintaining and improving the hydrological cycle. Assessing the WF of natural forests entails evaluating both their direct and indirect water usage. Direct usage encompasses the water transpired by the forest's vegetation and evaporated from the soil and plant surfaces, essential for their survival and development. Indirect use, on the other hand, includes forests'

contributions to recharging groundwater, retaining soil moisture, and providing ecosystem services like water purification and climate regulation (Launiainen et al., 2014).

The goal of calculating a forest's water footprint could be addressed in terms of computing the vegetation water use efficiency. Thereby WF it can be defined as the ratio of the green water that returns to the atmosphere through evapotranspiration against the biomass production. Taking advantage of the availability of an eddy covariance system, biomass production can be indirectly expressed as carbon storage, while evapotranspiration can be calculated from the latent heat flux measurements.

Methodology for determining the water footprint

In order to carry out the procedure to obtain the forest water footprint from the eddy covariance system, additional variables (Biomet variables) are necessary: soil flux (G), average soil temperature (T_s) and net radiation (R_n). Additional sensors are thus required: a soil heat flux plate, thermocouple probes, and a net radiometer. These were connected to a CR1000 logger that is networked with the SmartFlux system such that these data can be downloaded from the latter. Details of the instrumentation and methods used are described in the next section. The forest water consumption is derived from the latent heat flux as described below:

1. **Data download**

   In situ download of the data recorded at high frequency (10 Hz) by the SmartFlux system. These data correspond to the following variables: CO_2 and water vapour densities, ambient temperature, atmospheric pressure, dew point, sonic wind speed in the three directions (x, y, z), sonic air temperature, and the Biomet data (G, T_s, and R_n).

2. **Data processing**

   The software EddyPro can process the raw data downloaded from the SmartFlux system and provides the resulting fluxes at the specified time interval (30 min). This implies computing the different covariances necessary to find the turbulent fluxes. It is necessary to previously specify the processing options (i.e., which rotation method to apply: the double rotation or planar fit methods; if quality tests should be used: the *RNcov* and *ITC* quality tests).

   The number of raw data files that can be processed at one time depends on the processing time. It is not productive to process a month of data in one run as the processing time can exceed 8 hours on a conventional personal computer. Therefore, it is recommended to process intervals of 15 days data.

3. **Data post-processing**

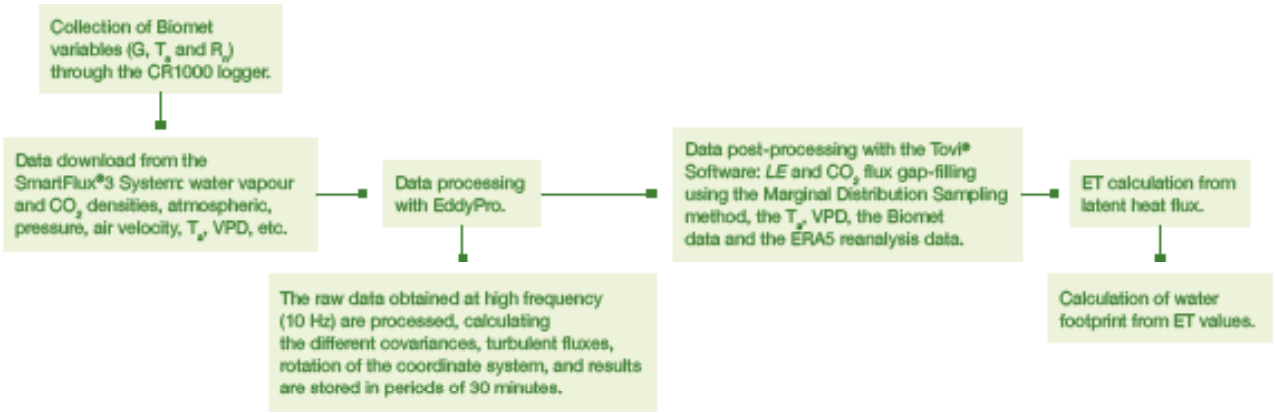
Using the Tovi software (available at licor.com/tovi) the latent heat flux (*LE*) time series is verified, and missing values are replaced by estimations computed using the marginal distribution technique (Falge et al., 2001) based on the Biomet data and reanalysis data. Tovi uses the reanalysis data to be more precise and fill in the missing meteorological variables in the Biomet data. It connects directly to the database and incorporates them into the marginal distribution calculation process. Due to the size of the islands, it can only apply the data from ERA5 (www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5), because it is the database available with the smallest grid (30 km). This fact will slow down the calculation of the gap filling. In addition, through Tovi we also apply the quality control tests, RN_{cov} and *ITC*.

4. **ET calculation**

Using the code developed for processing the data file obtained in step 3, the forest evapotranspiration (*ET*) is computed from the latent heat flux (*LE*), divided by the latent heat of evaporation of water ($2.45 \cdot 10^6$ J kg^{-1}) and the density of water (1000 kg m^{-3}). Since *LE* is provided in W m^{-2}, in order to express *ET* in mm units, the factor 1800 s every 30 minutes must be used: ET (mm)= $7.347 \cdot 10^{-4}$ *LE*(W m^{-2})

5. **Forest water footprint calculation**

The green water footprint (*WF*) represents the volume of soil water consumed during the growing process. It is computed from the evapotranspiration values (*ET*) as illustrated in the following flow diagram.



Flow diagram of the procedure for calculating forest water footprint.

In a forest stand, an important fraction of precipitation is intercepted by trees and understory and evaporated back to the atmosphere. The rest infiltrates into the soil increasing green water supply in the root zone. Tree roots take up green water, which is transported to leaves where it is transpired to the atmosphere. However, in cloud forests, during foggy conditions the small water droplets driven by the wind can impact the vegetation, accumulate and eventually drip to the soil surface. In order to determine the frequency and intensity of canopy wetting due to fog water capture, a system was developed to remotely monitor leaf wetness sensor readings. The technical requirements of the prototype were: Leaf wetness reading, autonomous power system, low power consumption, communications via radio in free band, local data acquisition and storage, IP65, and reduced dimensions.

The system can work with an undetermined number of sensor nodes that send their data to a central system or Gateway (LORAWAN receiver), which gives the data a readable format and transmits it via RS232 to a computer or an acquisition and storage system. In this case, a CR1000X data logger (Campbell Scientific Inc.) was used. The sensor used is the PHYTOS 31 (METER Group Inc.), which can detect small amounts of water if deposited on the surface of the sensor. This sensor has the following advantages: i) imitates the leaf shape; ii) detects traces of water on its surface; and iii) does not require calibration.

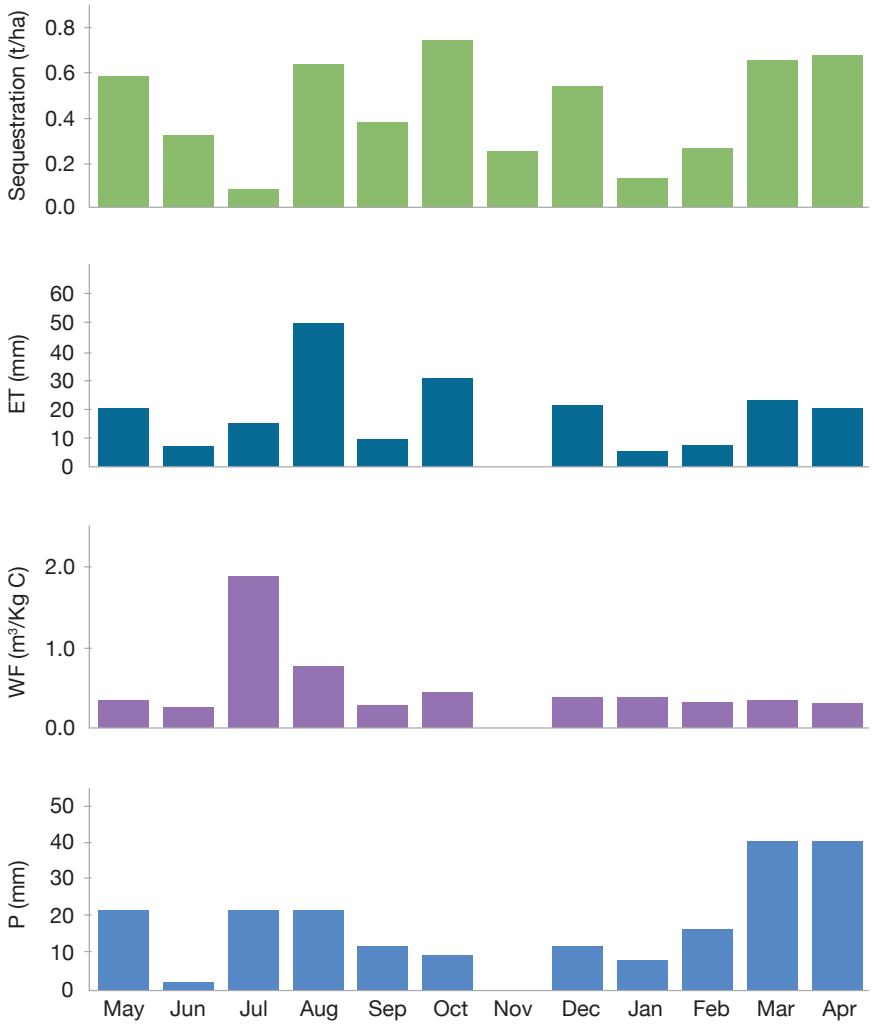
Canopy wetness monitoring system. Sensor node with leaf moisture sensor (left), CR1000X datalogger (centre), and Gateway or LORAWAN receiver (right).

In order to detect foggy conditions suitable for fog water collection, a cylindrical wire-harp gauge of the ASRC (Atmospheric Science Research Center, State University of New York) string collector type as described in Ritter and Regalado (2017) was used. In addition, visibility reductions were detected with a forward scatter infrared visibilimeter (CS120, Campbell Scientific Ltd.), which provides values in the meteorological observable range (MOR).

Canopy wetness detection system installed (left) and fog water collector (right).

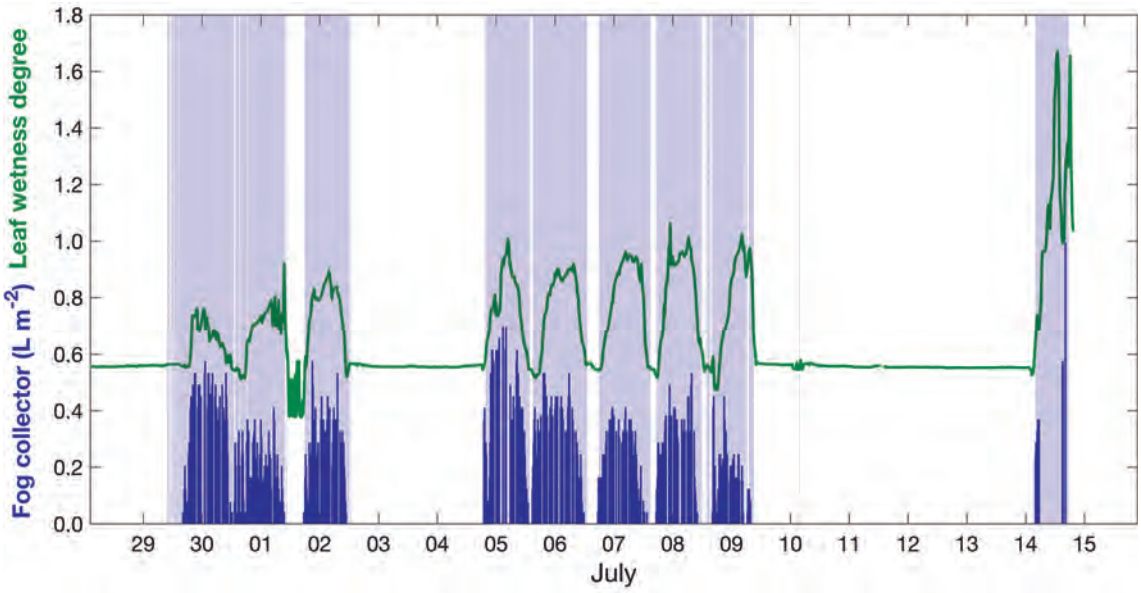
Results

The following figure illustrates monthly values of the variables used for calculating the forest green water footprint (WF), derived from eddy covariance measurements during May 2022 and April 2023. These exhibit variability without any observable seasonality. The resulting WF values are generally between 0.25 – 0.41 m^3 per kg of carbon sequestered. However, in the middle of the summer values are much higher due to reduced sequestration (in July) and larger evapotranspiration (in August). Additionally, monthly rainfall data are also presented to illustrate the extent to which this water contributes to the WF.



Seasonal variation of monthly carbon sequestration, evapotranspiration, water footprint and rainfall from May 2022 to April 2023.

Canopy wetness detection and monitoring was successful. The following figure illustrates different measurements during the end of June and the first half of July. During this period, several episodes of fog incidence occurred (i.e., when visibility was less than 1000 m). The fog characteristics, and probably also the wind speed, were generally suitable for collecting water droplets. Thereby, both the fog collector and the vegetation were able to capture water. In this way, both the fog collector and the vegetation were able to capture water. Consequently, and as shown by the leaf wetting sensor data, the canopy likely experienced progressive wetting processes, each followed by an evaporative drying process.



Evolution of canopy wetting and concomitant fog water collection during 29th June – 15th July. Shaded areas indicate visibility reduction to values below 1000 m.

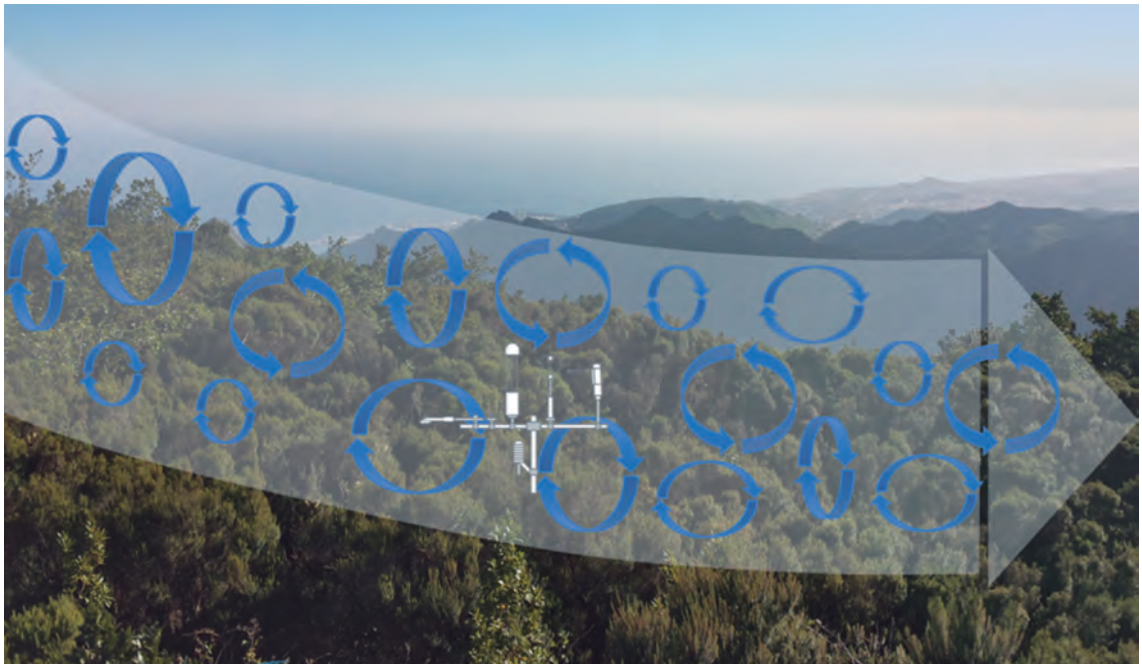
Conclusion

The eddy covariance system proved to be useful to quantify the forest water footprint expressed as the ratio of evapotranspiration to the carbon sequestration. The system designed to detect and monitor when the forest canopy becomes wet worked successfully in the presence of fog. This can be useful in determining how collected fog water contributes to the forest's water footprint.

Carbon storage

To estimate the potential environmental effects that the reforestation of the degraded landscape area in Gran Canaria will have in terms of carbon sequestration, an eddy covariance measurement system to monitor the CO_2 fluxes was installed in well-conserved laurisilva cloud forest in Anaga (Tenerife). This is an elfin cloud forest that represents an evolved state of the degraded area that is being reforested in El Pinillo (Valleseco, Gran Canaria). The extrapolation of results obtained in the studied site in Anaga to the reforested area in El Pinillo is based on the following similarities: i) steep areas and exposed to the wind, ii) frequently affected by fog, and iii) at similar altitude (1015 *vs.* 1200 m a.s.l.).

The eddy covariance system is currently the most preferred method for assessing the ecosystem carbon exchange. It is based on conceiving airflow as a horizontal flow of numerous rotating eddies. These eddies carry air "packages" to and from the vegetation. Each "package" has a vertical wind speed, temperature, humidity, and gas concentrations (H_2O, CO_2) that can be determined. The sum of the eddy covariance CO_2 flux above the forest canopy provides the net ecosystem production, which represents the total amount of organic carbon available for storage.

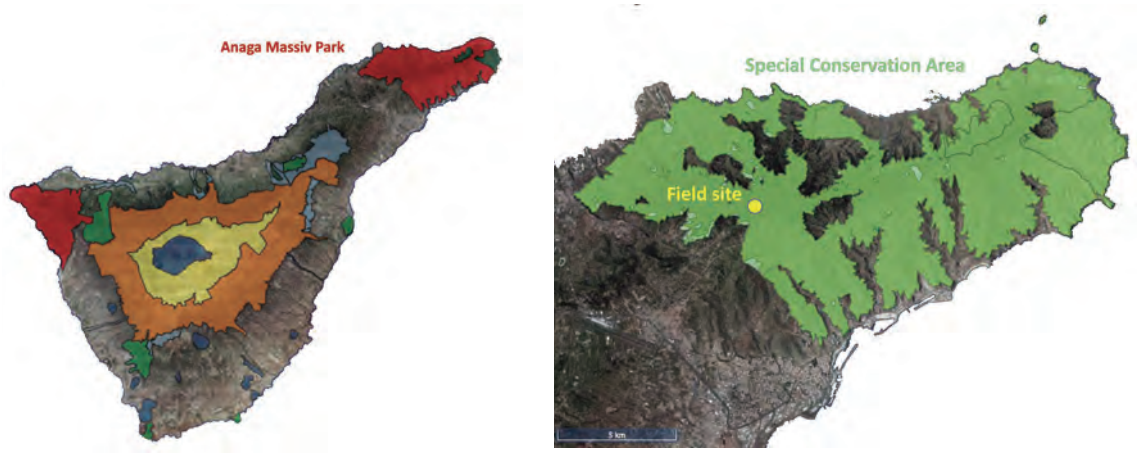


Sketch of the eddy covariance system main instrumentation and the rotating eddies in the airflow above the forest canopy at the measurement site.

It is considered the best technique for CO_2 flux analysis due to its reliability and advantages, i.e., a non-intrusive technique, which provides ecosystem representative data, based on continuous measurements at only a single site.

Study area

The selected site in the Anaga Massif Biosphere Reserve (Tenerife, Canary Islands) is located in the highest part of the park (1015 m a.s.l., 28º 32′ 9.61′′ N, 16º 16′ 8.91′′ W) and exposed to the prevailing NE trade winds, so it is frequently affected by fogs. The vegetation of the area is typical of hyperhumid conditions and windy crests. *Myrica faya* Ait. (wax myrtle) and *Erica platycodon* (Webb et Berth.) Rivas-Mart. et al. (formerly *E. scoparia* spp. platycodon) are the main tree species present.



Field site in the Anaga Biosphere Reserve situated in the north of Tenerife (Source: Grafcan).

Vegetation at the measurement site.

The eddy covariance system uses precision instrumentation to be installed on top of a scaffolding tower. The tower was placed inside the forest and it was anchored to concrete bases with steel tension cables in order to ensure its stability.

For operating the measurement equipment, an electrical supply was provided. This required the determination of the length (40 m) and type (3-wire) of cable necessary. In order to minimise energy losses, it was decided to use alternating current (AC). Previous to the cable installation vegetation clearing was performed. Additionally, a security circuit breaker and an AC to DC converter were connected.

Eddy covariance system installation

The equipment was installed at the scaffolding tower placed inside the forest and was connected to the electrical supply. While the sensors were placed at the top of the scaffolding tower, the cabinet with the data storage device and accessories was mounted at its lower part. The main sensors are an open-path gas (CO_2/H_2O) analyser (LI-7500DS, Li-Cor, USA), and an integrated three-dimensional sonic anemometer (WindMaster, Gill Instruments, England).

Instrumentation placed above the canopy.

For data storing a SmartFlux®3 System was used. This was placed inside the outdoor cabinet and connected using further accessory devices and elements: the AC/DC power supply, an 8-port network switch, cables and connection terminals, and the security circuit breaker.

Control system in the outdoor cabinet placed inside the forest. SmartFlux®3 data storage system (white), network switch (yellow), AC/DC power supply (blue) and other electronic components.

Additional instrumentation

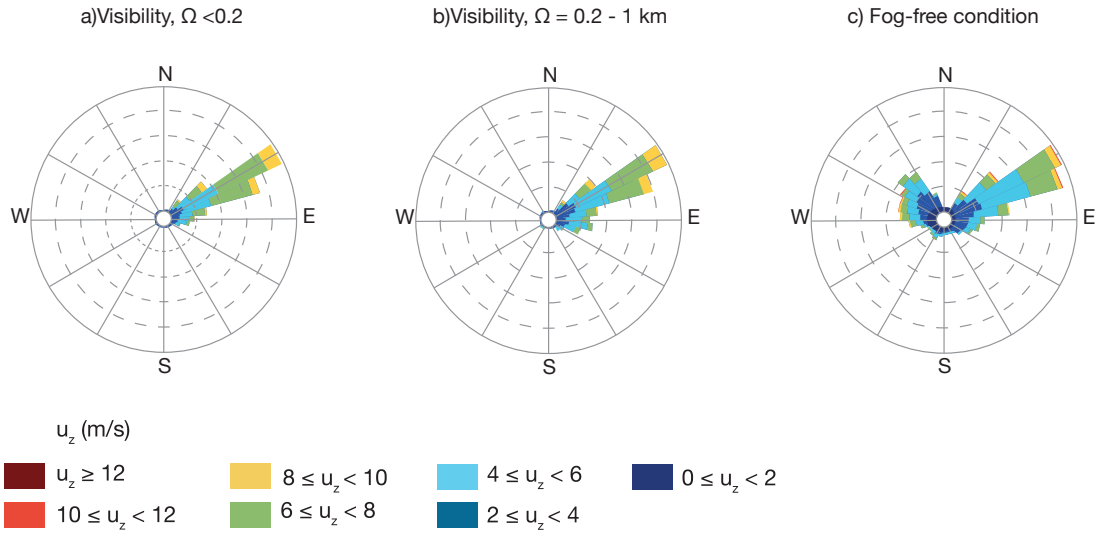
Net and global solar radiations were measured with a NRLite2 radiometer (Kipp&Zonen, The Netherlands). Soil heat flow, temperature and water content were measured, respectively, with a heat flux plate (HFP01SC, Campbell Scientific Ltd., USA), two thermocouple pairs (TCAV, Campbell Scientific Ltd., USA), and a TDR Trime-EZ probe (Imko GmbH, Ettlingen, Germany) at about 10 cm soil depth close to the eddy covariance tower. These were connected to a CR1000 logger that was networked with the SmartFlux®3 System such that these data can be downloaded from the latter.

Performing the eddy covariance system configuration

Configuring the software of this equipment requires parameters to be entered in the software as well as the proper orientation of the system with respect to the orientation of the wind.

The parameters to be determined before starting the system to collect data are the following. For the sonic anemometer: Serial number, north offset, and anemometer height (measured at the centre of the measurement volume). For the gas analyser: North separation, east separation and vertical separation.

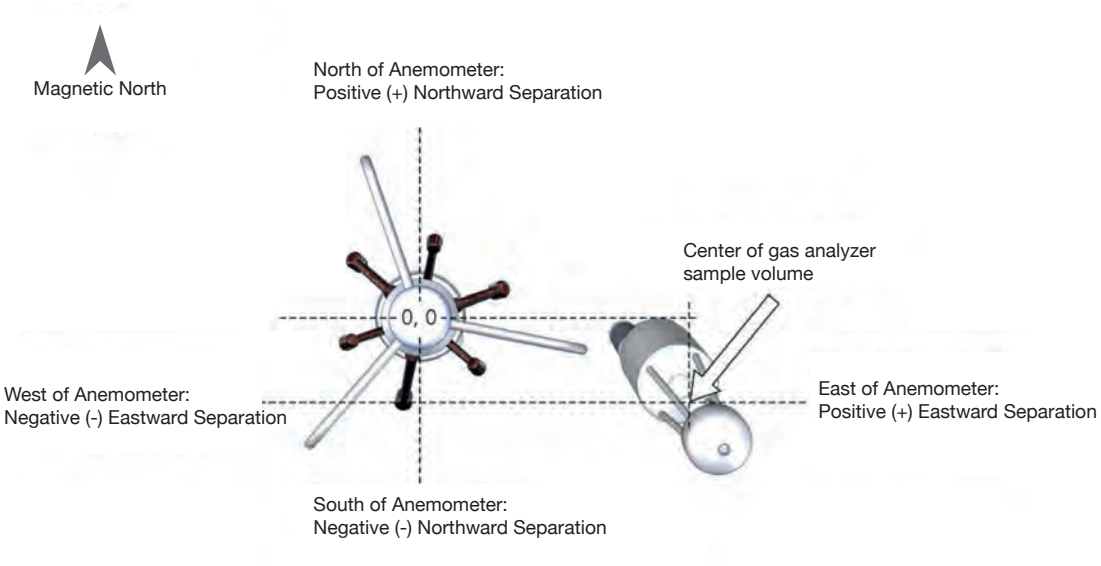
In order to determine the displacement to the north in Cruz de Taborno (measurement site), it was necessary to know the prevailing wind direction in this area. For this purpose, the wind regime based on data from 2014 to 2019 was used. These data were analysed based on visibility conditions in the area due to frequent fog presence in the site. The following fog conditions were considered: conditions of dense fog (visibility < 0.2 km), light fog (0.2 to 1 km visibility), and free of fog (visibility > 1 km).



Wind rose diagrams under dense fog (a), light fog (b), and fog-free conditions (c) in the study site.

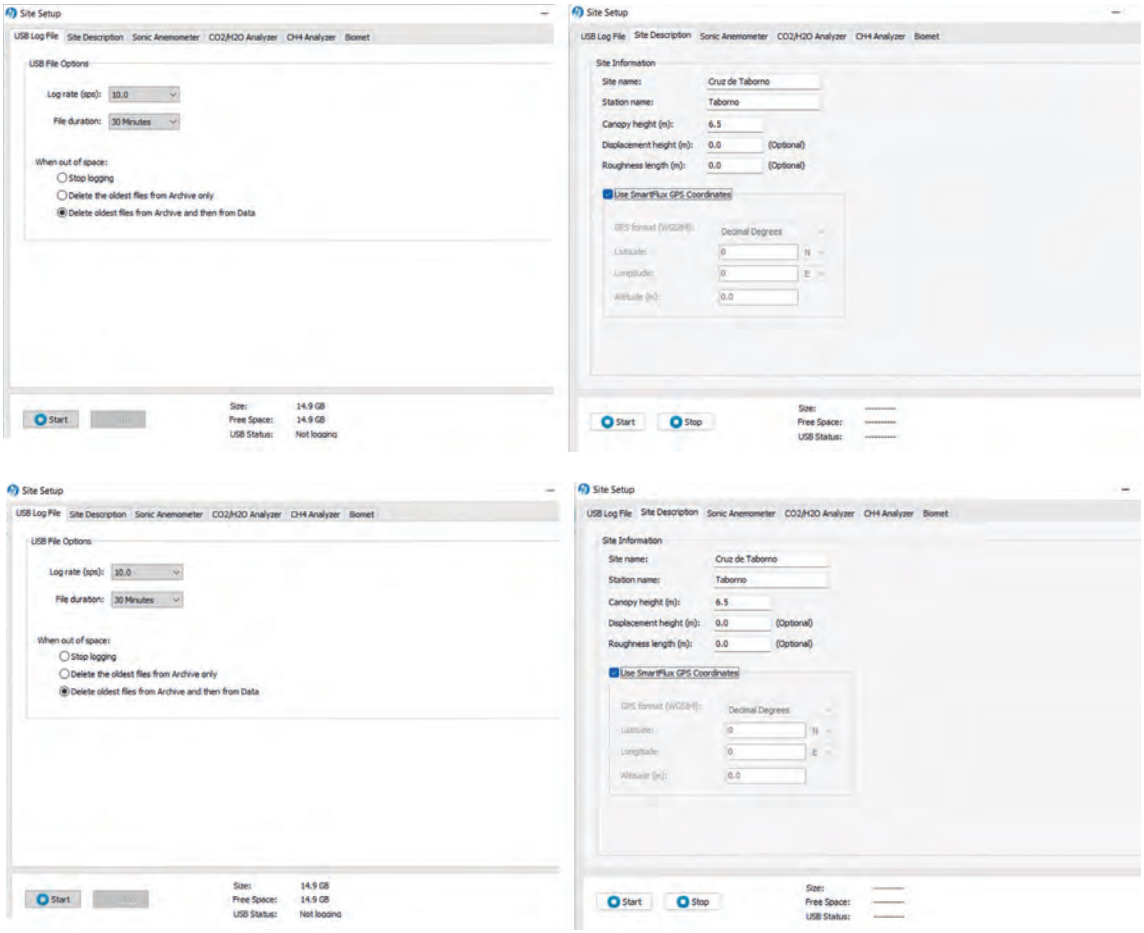
In addition, the correct rotation of the eddy covariance system was studied for its ENE orientation in the field.

By following the steps outlined in the equipment-mounting manual, various tests were carried out. Since the manual specifies that the analyser should go alongside one of the anemometer stringers, in a first attempt to meet this condition, the base of the analyser was rotated without rotating the anemometer and fixed, noting that the analyser protrudes the anemometer in the predominant direction of the wind. This mounting was considered invalid and was dismounted again to test other options. Once all the possible configurations that fulfil the initial condition were tested, it was possible to place the system as described in the assembly manual, leaving the separation distances to the east and north (-19 cm and -4.1 cm, respectively). The centre of the anemometer was assumed to be at the origin (0,0) of the reference system as illustrated in the following figure.



Arrangement of the coordinate axis in the system for determining the separation distances to the north (+ to the north and - to the south) and to the east (+ to the east and - to the west).

Once the installation was finished, the system was configured through the connection program Li-7x00 A RS DS version 8.9.0. Specifically, the following parameters were introduced: size of the files (30 min), the site coordinates and elevation, the height of the vegetation (6.5 m), the height were the anemometer was mounted (8.25 m), the north offset (240º)**,** eastward and northward separation (-19 cm and -4.1 cm, respectively) and the variables to log.



Configuration of the eddy covariance system using the SmartFlux®3 System.

Procedures applied to correct the eddy covariance results

Correction due to the orientation of the anemometer

Eddy covariance systems are based on the theory that, as the wind moves, it does not move in one direction but in eddies, that is, in three-dimensional circular patterns (Burba, 2013). In its movement, the wind transports molecules of water vapour and other gases such as carbon dioxide. If the speed in all three directions of these eddies can be measured, then it is possible to determine the net exchange of these molecules between the surface and the atmosphere (Moorhead et al., 2019).

Under ideal conditions of stability and homogeneity, it is considered that the concentration of the atmospheric component to be studied (carbon dioxide or water vapour) does not experience a significant variation over time and that the vertical speed of the wind in the atmosphere is stable. Therefore, the existence of a non-zero mean vertical wind speed, due to the orientation of the anemometer, alters the calculation of the fluxes of the atmospheric components. To nullify this apparent speed, it is necessary to change the coordinate system of the anemometer to that of the main wind direction.

Two methods have been studied for this, double coordinate rotation and planar fit:

a) Double coordinate rotation method. A double rotation for each turbulent period, making the vertical and lateral components of the velocity equal to zero (Tanner and Thurtell, 1969) is the most common technique to transform from the coordinate system of the anemometer to that of the main wind direction.

b) Planar fit method. This approach is based on finding the mean plane of the local streamline parallel to the irregular surface of the terrain and rotating the coordinate system of the anemometer to the plane that best fits it, considering the axis of the anemometer perpendicular to the slope of the terrain (Wilczak et al., 2001).

After studying these two techniques, it was decided to apply the double rotation method, and if it were necessary, the planar fit would be used. The application of the double rotation was carried out through the EddyPro® Software, a program of the LI-COR data processing company.

Correction of gas fluxes for air density fluctuation effects when using open-path gas analysers

The system uses an open-path InfraRed Gas Analyser (IRGA) for measuring fluctuations in the density of CO_2 and water vapour. On uniform surfaces with active heat and water vapour exchanges, fluctuations in air density appear with this type of gas analyser. To avoid the errors associated with this phenomenon, the Webb-Pearman-Leuning correction (*WPL*) derived from the variations in water vapour density can be applied. This allows for rectifying the measurements of the CO_2 and water vapour fluxes provided by the IRGA (Webb et al., 1980). This correction is necessary, since otherwise errors may occur in the measurement of turbulent flows of up to 40% (Leuning et al., 1982) and was applied to the CO_2 flux and the latent heat flux (*LE*) estimations.

Sonic temperature correction

The sonic anemometer is capable of measuring a temperature determined by the speed of sound. This temperature, called sonic temperature, undergoes fluctuations and must be corrected for humidity so that its value corresponds to that of air temperature (Kaimal and Gaynor, 1991).

Control tests to verify data quality

Procedure applied for data quality control

Due to the complexity of the calculation procedure required for applying the eddy covariance technique, it is recommended to perform three tests for data quality control (Foken et al., 2012).

a) Relative Non-stationarity test. In order to apply the theory of turbulent fluxes, the stability condition must be fulfilled. Under these conditions, the structure of the flux surface layer should be independent of the measurement time interval. This test computes an index, expressed in percentage, which represents the average of the difference between the 5-minutes and 30-minutes covariance values of a scalar and the vertical wind speed with respect to the 30-minutes covariance. Based on the value of this index, a quality grade was assigned from the interval 1 – 9, such that 1 represents the greatest stability and therefore determines the highest quality data, and 9 corresponds to the worst quality data.

b) Integral Turbulence Characteristics Test. An index (*ITC*) is computed, which expresses the turbulence degree (in percentage) as the relative difference of the standard deviation of a fluctuating scalar (such as temperature, wind speed or momentum) normalised by a scale factor (such as the friction speed, or the scale temperature) (Tillmann, 1972) with respect to the *ITC* computed in conditions of fully developed turbulence. Based on the resulting index value, a quality grade was assigned from the interval 1–9, such that 1 represents the highest grade of stability.

c) Wind direction in the coordinates of the sonic anemometer. According to Foken et al. (2012) under conditions with wind flowing with angles 180 ±10º, data should be considered as low quality data and are rated with grade 3. The data considered as of average quality (rated with grade 2), are those that are outside this range but within the interval of 29 - 180º. Good quality data (rated with grade 1) correspond to conditions when the wind flows with any other angle.

These three quality tests can be combined to obtain an overall quality grade (1 – 9) according to the following table (Aubinet et al., 2012).

Overall quality assessment system considering the grades obtained in each test.

Quality tests breakdown

Overall quality	Relative Nonstationarity Test index	Integral Turbulence Characteristics Test index	Wind direction index
1 (best)	1	1-2	1
2	2	1-2	1
3	1-2	3-4	1
4	3-4	1-2	1
5	1-4	3-5	1
6	5	5	2
7	6	6	2
8	7-8	7-8	2
9 (worst)	9	9	3

The above scheme allows determining which data should be used. Data associated with overall quality 1 to 3 can be used for fundamental research, such as the development of parameterizations. Those with values between 4 and 6 can be used for systems in continuous operation of the FLUXNET software. Classes 7 and 8 are only indicative, and may be used for a filling procedure when dealing with missing values, but these data should not differ significantly from the data before and after the time series. Class 9 data must be excluded in all circumstances.

Based on this procedure, the fluxes were calculated here only with good quality measurements, corresponding to restrictive overall quality grades ≤ 3 and the data rated with grades > 3 were discarded.

In addition, those flux values that were outside the range established by CARBOEUROPE were also rejected: CO_2 flux [-50 – 50 µmol m^{-2} s^{-1}]; Latent and sensible heat fluxes [-250 – 800 W m^{-2}]; Momentum flux [0 – 5 kg m s^{-1}].

Night-time CO_2 flux values were also discarded if the friction velocity was less than the velocity limit ($u^* < 0.2$ m s^{-1}). This was because, during the night, under conditions of atmospheric stability, when u^* is below the established limit, the existing small turbulence will not be able to mix the air from the adjacent layers. Thus, the CO_2 exchanges that take place between the surface and the immediate atmosphere will not reach the height of the measuring instruments.

Data processing

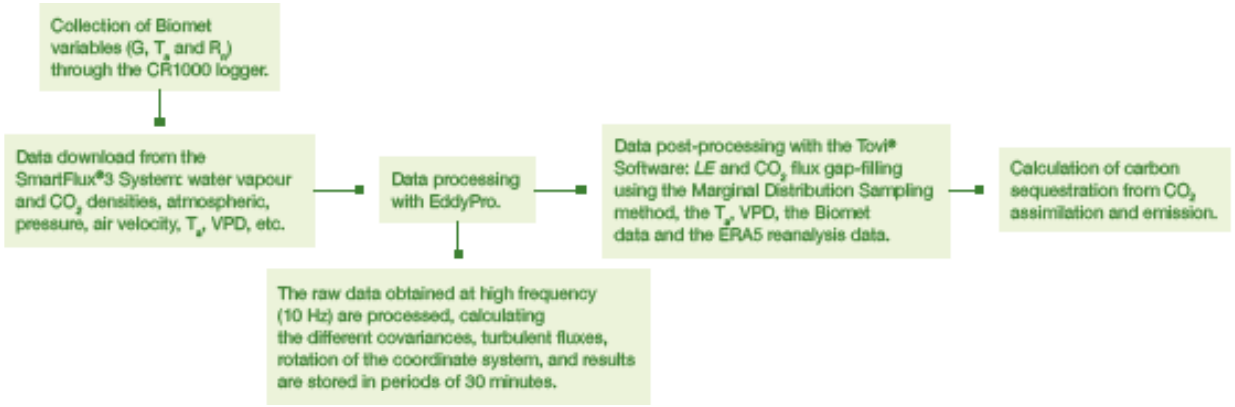
The data downloaded from the SmartFlux®3 System were recorded at high frequency (10 Hz) and correspond to the following variables: CO_2 and water vapour densities, ambient temperature, atmospheric pressure, dew point, sonic wind speed in the three directions (x, y, z), and sonic air temperature.

The EddyPro® Software, provided by the LI-COR Company, was used for processing the raw data and for obtaining the resulting fluxes at the specified time interval (30 min). In addition, this software allows for applying the double rotation and flat adjustment methods. The number of raw data files to process at one time depends on the required time. On a conventional personal computer, the processing time for a month of data can exceed 8 hours. Therefore, it is recommended to periodically process sets of 15 days data.

The Tovi® Software (available at www.licor.com/tovi) was used to verify the time series applying the *Relative Non-stationarity* and *Integral Turbulence Characteristics* quality tests. In addition, this software was used also for filling missing values in the variables based on the Marginal Distribution Sampling (MDS) gap filling method (Reichstein et al., 2005), which requires the Biomet data (soil heat flux, soil temperature and net solar radiation), ambient temperature, vapour pressure deficit, and reanalysis data. Tovi® uses the reanalysis data to be more precise and fill in the missing meteorological variables. It connects directly to the database and incorporates them into the marginal distribution calculation process. Due to the size of the island, it can only apply the data from ERA5 (www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5), because it is the database available with the smallest grid (30 km).

Carbon sequestration computation

The procedure for obtaining the forest carbon sequestration from the eddy covariance system and the additional instrumentation is sketched in the following figure. In the last step the processed CO_2 flux time series is partitioned into CO_2 the assimilation and the emission. Assimilation occurs during the daylight hours and is represented by positive values of CO_2 flux or net carbon exchange (*NEE*), while carbon release occurs at night and is the nocturnal *NEE* (negative CO_2 flux values). Thus, carbon retention or sequestration can be obtained as the difference between the diurnal NEE and the nocturnal *NEE* ($NEE_{diurnal} - |NEE_{night}|$), which can be expressed in terms of carbons amounts by multiplying by a factor of 0.216 to convert mmol m^{-2} s^{-1} units to kg C ha^{-1} each 30 minutes.



Flow diagram of the procedure for calculating carbon sequestration.

Maintenance and routine inspection of the equipment

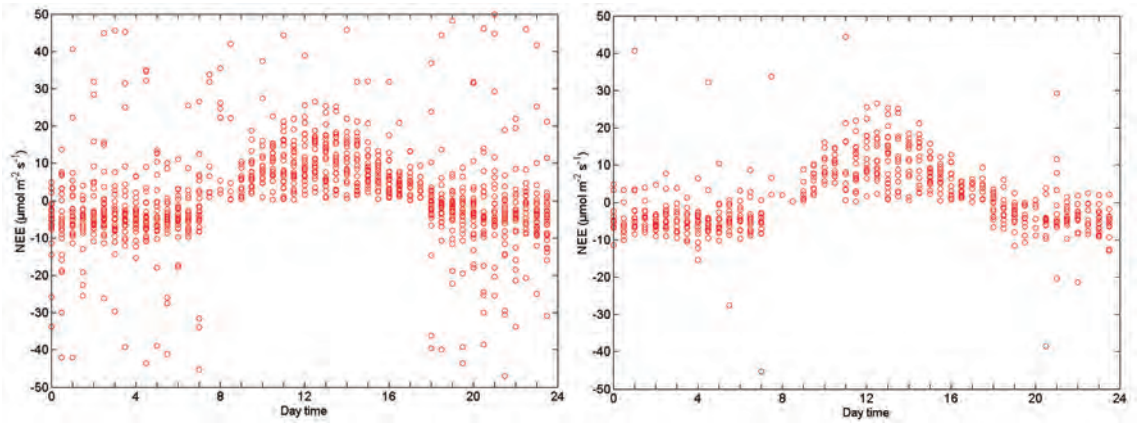
Field visits were performed at least once a week for downloading the data, and for verifying the correct operation of the equipment, that is, verifying that the data were saved to the storage disk and that the SmartFlux®3 System correctly recognized this disk.

In order to minimise measurement failures under saturated ambient conditions (relative humidity≈100%) for long periods of time (more than one day) due to the condensation of water droplets on the sensors, the lenses were periodically treated with a water repellent (Rain-X® Rain Repellent). It should be noted that although the analyser lens can tolerate droplets to a certain extent, if the droplets coalesce and grow large enough, the readings could be affected.

Once connected to the SmartFlux®3 System, instantaneous CO_2 readings were observed, and if these were in a range greater than 450 ppm, the lenses were cleaned and the anti-rain product was applied. Due to the weather conditions in the area, frequently affected by trade winds and foggy conditions, this process was necessary almost every week.

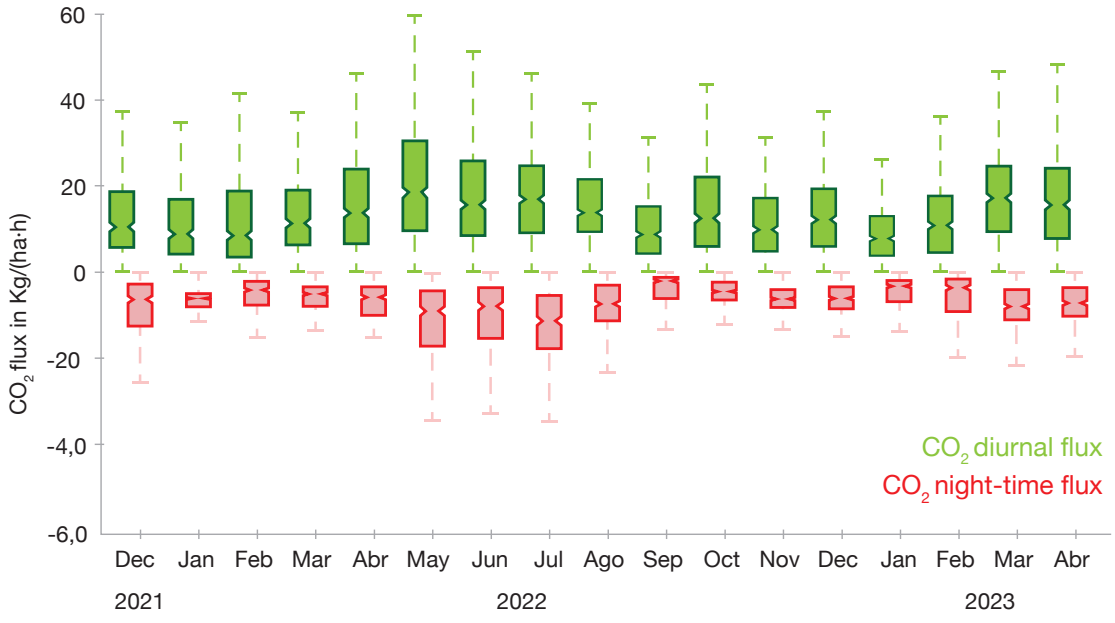
Results

The following figure illustrates how the data filtering eliminates a large number of *NEE* values. This reduction of data occurs not only with the carbon flux or *NEE*, but also with the fluxes calculated from the variables recorded by the eddy covariance system.



NEE for January 2022 without considering before (left) and after the data filtering (right).

The monthly variability of the hourly NEE or CO_2 flux can be represented with boxplots, where the boxes are delimited at the top and bottom by the third and the first quartiles, respectively, while the median divides the rectangle. The dashed lines extend to non-outlier maximum and minimum values. The variability of CO_2 fluxes can differ significantly between months and the median and maximum values exhibit an apparent cyclical trend (see figure below). CO_2 night-time (emission) fluxes are lower than diurnal (assimilation) fluxes.



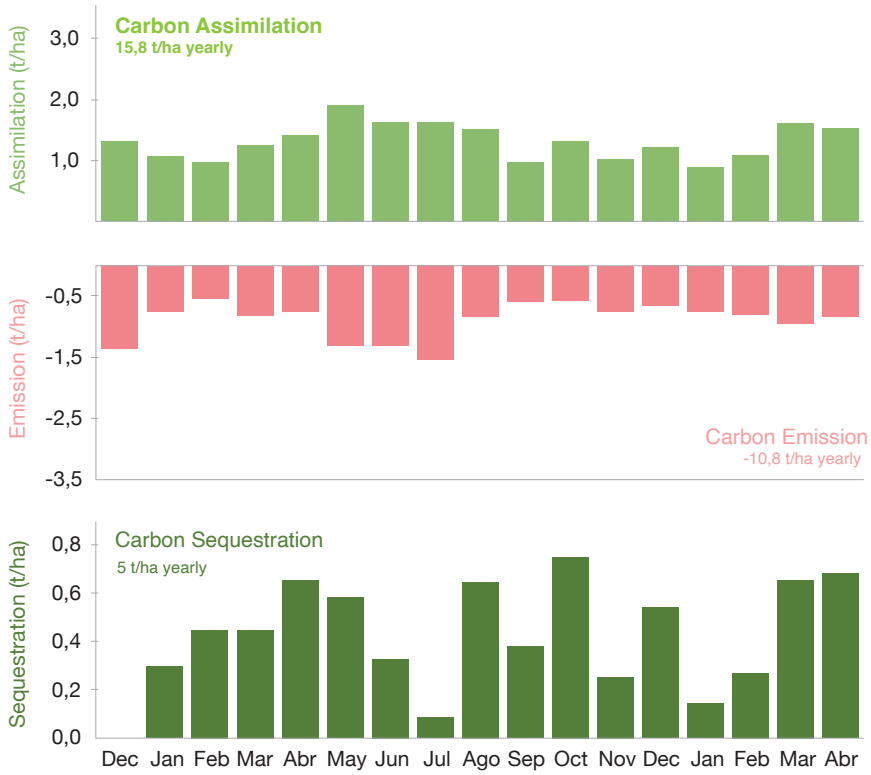
Monthly variability of the hourly CO_2 assimilation and emission fluxes.

From May to August, CO_2 assimilation and emission present high median and maximum values. However, the greater vegetative activity in these months is accompanied by a larger CO_2 emission due to higher temperature and lower ambient humidity.

Variation of monthly carbon assimilation, emission and sequestration is illustrated in the next figure. Since monthly carbon storage was mainly positive during the study period, it is assumed that this ecosystem behaves as a CO_2 sink. On a yearly basis the carbon storage was estimated at 5 t/ha. Compared to other forest types, this laurisilva cloud forest has a capacity to sequester carbon similar to temperate (0.7 – 7.5 t/ha) or tropical (3.2 – 10.0 t/ha) forests. According to the Köppen climatic characterization carried out for Cruz de Taborno, this area has a humid temperate climate. Net carbon exchange studies carried out for different types of forests have shown that humid temperate forests are net sinks of CO_2 (Baldocchi et al., 2001; Bernal et al., 2018). The amount obtained here is close to those reported by Bernal et al. (2018) for temperate humid pine forests (5.8 t/ha) and for young tropical humid broadleaf forests (6.9 t/ha), as well as for evergreen needle-leaf forests, specifically *Picea sitchensis* (5.7 – 6.7 t/ha) (Valentini et al., 2000) and temperate forests broad-leaved deciduous (5.0 – 7.0 t/ha) (Baldocchi et al., 2001).

Conclusions

The eddy covariance system installed in the Special Conservation Area of the Anaga Massif is currently the only system of its kind in the Canary Islands and based on this first long-term study of carbon exchange in laurisilva cloud forests, it was observed that monthly forest carbon sequestration is variable according to the intensity of CO_2 assimilation *versus* emission. However, this laurisilva cloud forest behaves as a sink of atmospheric CO_2 throughout the year. Assuming that the reforested area in Gran Canaria reaches a state of development similar to that of the forest studied, a 35 ha area will contribute to mitigate anthropogenic climate change with an annual carbon sequestration of the order of 5 t/ha, that is, approximately 175 tons of carbon per year.



Seasonal variation of monthly carbon assimilation, emission and sequestration.

Carbon footprint

Presentation

The main objective of Life Cycle Analisys (LCA) is to obtain an estimate of the environmental impact generated in the life cycle of the different reforestation systems and irrigation methods installed by the project.

A complete stroke is performed, in a cradle to gate manner, that is, it is considered the reception of materials necessary for the construction and installation of the different systems and the operation and maintenance of them.

Functional unit

This LCA can contribute to making project decisions, for that it's necessary to define a functional unit (FU). The objective of this FU is to provide a reference for system inputs and outputs. In this case, **1 surviving tree** has been chosen since this way the different irrigation systems can be compared, since not all of them supply the same number of trees. Likewise, considerations will be made between the quantities necessary to ensure the survival of each tree and the origins of water.

System boundaries

The phases taken into account in this analysis are the follows:

- Construction materials for the facilities and their transport.
- Assembly phase taking into account the required energy/fuel consumption.
- Maintenance and operation of the material.

It excluded from the analysis id dismantling of equipment and the impact of the treatment and final disposal of waste in landfill or to be treated as by-products.

Also, emissions related to the manufacture of machinery and infrastructure necessary for extraction and transport of any raw material or material used in the construction of the systems are not included, wastes or by-products resulting from their maintenance.

Requirements

Different requirements before starting the LCA, here the data use for life Nieblas project LCA:

- **Geographical:** the study system is located in Barranco de La Virgen, in Valleseco, composed of 35.8 ha. The selected hill is partially located within the Special Conservation Area (SAC) ES7010038 being, in turn, part of the Natura 2000 network included in the priority 4050 habitat 'Macaronesian Heath'.
- **Time:** the project starts in 2020 and has a duration of four years. For the LCA realisation, a time limit of one year is proposed. It's the time necessary for the plants to stabilise and transfer the irrigation system to other areas.
- **Data validity range:** the service life of each system varies. Certain materials under optimal conditions will be reused. Due to the limited time devoted to the project, the average durability time of each system is not yet known.
- **Data cutting rules:** it is necessary to establish equal cutting rules for all cases. For this reason, only components weighing more than 10 kg are analysed, provided the important parts of each system are not omitted.

Inventory

In this section, the inventory of each system must be performed. To do this, it is necessary to describe the facilities and know all the processes that compose it.

This inventory will consist of two parts. One associated with the construction phase, where all materials and processes necessary for the construction of the systems will be collected and the second associated with the maintenance and the operation (with costs involved, necessary personnel, transport, water mobilisation and associated energy).

Tool

For a correct LCA, you have to use adapted software. The basic function of this software is to perform the material and energy balances on the specific process and to allocate emissions, energy uses, etc. normalised to common basis. For Life Nieblas, the tool SimaPro version 9.1.1 is used with the ecoinvent v3.7.1 database with more than 4.000 processes.

Assembly phase

Impact category	IPCC GWP 20a kg CO_2 eq.
3_AFDS supplied with 3 fog collectors (400)	18.931
3_AFDS supplied with 3 fog collectors (500)	15.145
6_Cocoon supplied with tank and fog collectors	13.725
9_Traditional irrigation with fog collectors	12.232
Individual Fog Collector	7.483
2_AFDS with pumped water from spring (1200)	5.046
1_AFDS supplied with water tank truck	3.798
5_Cocoon supplied with pumped water from spring	2.851
4_Cocoons supplied with water tank truck	1.522
8_Traditional irrigation with spring pump	1.192
7_Traditional irrigation with water tank truck	0.409

Exploration and maintenance phase

Impact category	IPCC GWP 20a kg CO_2 eq.
Cocoon Sector 3 truck support	1.336
Cocoon Sector 2 truck support	1.084
Cocoon Sector 3 pump support_rev	0.855
Cocoon Sector 3 fog support	0.700
Traditional Sectors truck support	0.591
AFDS3 (400) water truck support (May-Sep)_rev	0.575
Cocoon Sector 2 pump support_rev	0.572
Cocoon Sector 2 fog support	0.560
Individual FWC Sector 3, truck support (May-Sep)	0.542
Cocoon Sector 1 truck support	0.455
Traditional Sectors pump support	0.427
Traditional Sectors fog support	0.420
AFDS1 (400) water truck support (July-Sep)_rev	0.387
Individual FWC Sector 2, truck support (July-Sep)	0.365
Cocoon Sector 1 pump support_rev	0.318
Cocoon Sector 1 fog support	0.315
AFDS3 (400) pump support (May-Sep)_rev	0.240
Individual FWC Sector 3, pump support (May-Sep)	0.240
Individual FWC Sector 3, fog support (May-Sep)	0.233
AFDS2 (400) fog collectors supply, truck support_r	0.194
AFDS1 (400) pump support (July-Sep)_rev	0.171
Individual FWC Sector 2, pump support (July-Sep)_r	0.171
Individual FWC Sector 2, fog support (July-Sep)	0.166
AFDS2 (400) fog collectors supply, pump support_re	0.102
Individual FWC Sector 1, truck support_rev	0.102
Individual FWC Sector 1, pump support_rev	0.067
Individual FWC Sector 1, fog support	0.066

CONCLUSION

After analysis of the data, some conclusions of the study can be established:

- As systems incorporate more elements to ensure their autonomy from operation and maintenance, equivalent emissions are higher at the assembly stage.
- The system with the highest equivalent emissions per tree planted is the one that incorporates 3 FWCT, a 5,000 litre tank, an AFDS and a drip irrigation network for 500 trees (15 kg CO_2 equivalent/tree). In this case, 77.4% of the emissions are due to fog collectors, 18.4% to AFDS and irrigation network, and 2.5% to interconnection pipelines between the mist water collection area and the reservoir for AFDS. This deposit accounts for 1.6% of emissions. On the contrary, this same system in the operation and maintenance phase is the one with the least associated emissions. Considering a survival rate of 80%, emissions would be 0.1 kg of equiv/surviving tree CO_2, taking into account that an emergency irrigation is carried out in August (in this case with water pumped from the spring).
- The system that incorporates less emissions in the assembly phase is the traditional irrigation system that will later be irrigated in the exploitation phase by transporting water tanks from the outside (0.4 kg CO_2 equiv./tree). It requires the installation of minimal infrastructure in the field, basically water storage tanks and hoses. On the contrary, it would be one of the systems that would generate more emissions in the exploitation phase, depending on the irrigation system, with about 0.6 kg of equiv/tree CO_2.
- The system with the highest emissions associated in the operation and maintenance phase is the Cocoon system in sector 3 with Cocoon fillers from May to September, and with water transport from the outside 1.3 kg of equiv/tree CO_2. This is because it is the system that requires the most water transported and more visits by operators to carry out irrigation.
- The system with the least emissions in the maintenance phase is associated with the IFWC and use of fog water as support irrigation.

2. Social impact

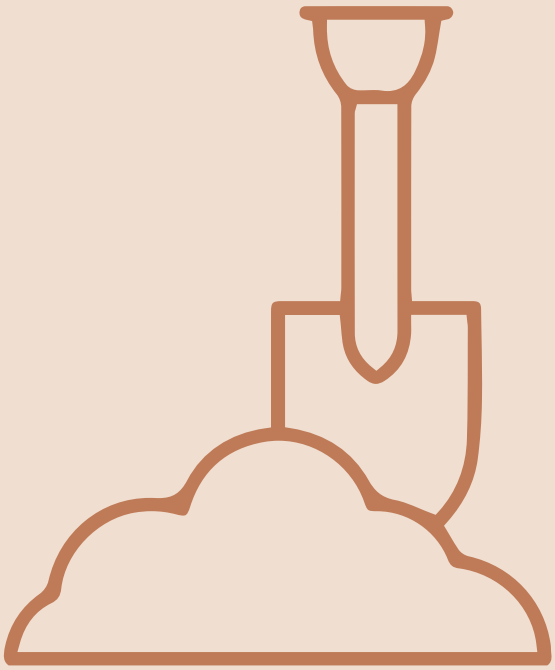
Monitoring the socioeconomic impact of the project is as important as monitoring the environmental impact. The effect of the project on end-users will be essential to evaluate the overall benefits. The socioeconomic perception of the actions in all areas will be monitored during the project's lifetime on the local economy and population. Ongoing assessment of the social and economic effects of reforestation on the local population will enable action to be taken in the event of any deviations that could compromise the achievement of the project's objectives and expected results.

The following indicators are proposed:

- Job generated directly by the project (personnel hired for the tasks execution).
- Job directly affected by the project (beneficiaries pre-existent personnel with tasks assigned).
- Job indirectly affected by the project (total personnel of each beneficiary).
- Number of companies and professionals directly hired as an external assistance by the project.
- Number of jobs created or affected by replications of the different reforestations typologies tested on the project.
- People reached during talks, workshops or different events.
- Farmers and workers who will receive direct information of the project.

To bring all this information together, several graphs, tables, maps and charts will be produced in order to visually represent the socioeconomic activity that the project generated.

Life Nieblas long story short.



The Natura 2000 Network is the largest conservation action undertaken by the European Union and we all play a part on its goals fulfillment

III. Summary

CONCLUSIONS:

- Considering the extremely dry conditions during the project years and the challenges that had to be faced, the results obtained from the testing of the innovative planting techniques are remarkably successful, significantly increasing survival rates

- Reduced water usage when applying innovative reforestation methods

- Although Cocoon system degrades faster in soils with high soil fauna activity, the higher survival rates and reduced water usage have yielded excellent results also in those environments

- Individual Fog Water Collectors (IFWC) that incorporate a capture net are the most successful method of reforestation due to its multifunctionality: light rain, dew and fog collection, evapotranpiration reduction and herbivores prevention

- IFWC is a reusable reforestation method which will see its carbon footprint reduced after every reimplementation in the field

- It was proved that the AFDS method and irrgation network are reusable which will see its carbon footprint reduced after every reimplementation in the field.

- Fog water collection could be a useful tool when considering water sourcing in isolated areas where fog is present, especially those under risk of desertification which may take advantage of foggy conditions

-In terms of the fog collection infrastructure installed, Life Nieblas has developed a structurally stable against wind gust, easily deployable and more efficient fog water collector: the innovative FWC (i-FWC)

- i-FWC yielded up to 3 times more fog water per collecting screen surface compared to the other two fog collectors tested at El Pinillo

- Fog collection is highly dependent in location as proven by other locations studied in Life Nieblas (Osorio and Fuente Fría, both in Gran Canaria)

- External stakeholders see a benefit in the replicability of this project as proven by several case studies currently under development

- Passive restoration is working well in El Pinillo and Vouzela, despite plantations and vegetation management is needed in order to guide it to the targeted habitats

- The experience obtained during the project execution and monitoring provides practical guidelines for future reforestation efforts.

- The carbon sequestration observed in the cloud forest of the Anaga Massif in Tenerife—acting as a year-round sink and sequestering approximately 5 tons of carbon per hectare annually—could be extrapolated to the reforested areas in Gran Canaria, provided they reach a similar level of development.

Traditional FWC Performance in Life Nieblas

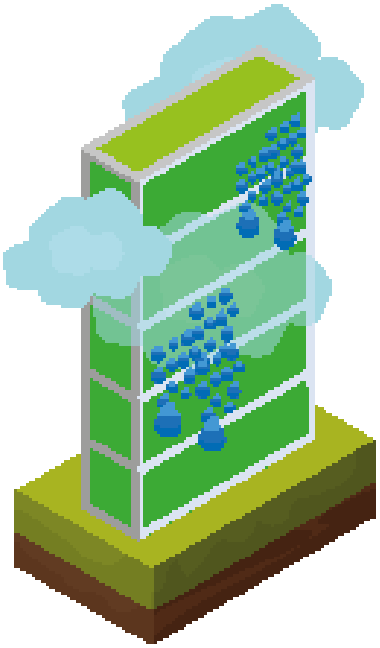
Number of collectors

9 FWC

Size

8 m²

of impact surface



Price

4670 €

Weight

400 kg

Litres collected per m²

292 L/m²/ year

Carbon footprint

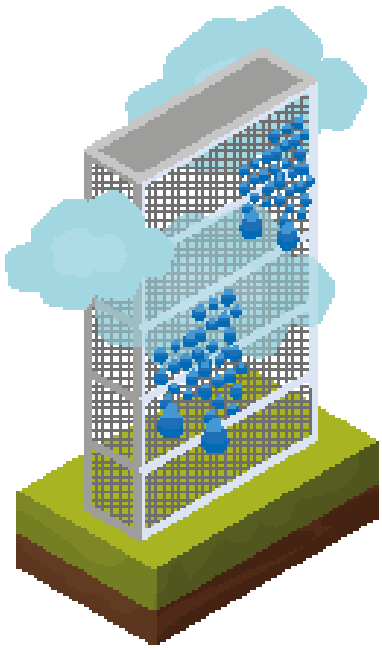
1953 kg

CO_2 eq.

Volumetric FWC Performance in Life Niebl

Number of collectors

3 V-FWC



Size

8 m²

of impact surface

Price

4700 €

Weight

400 kg

Litres collected per m²

470 L/m²/ year

Carbon footprint

1953 kg CO_2 eq.

Innovative FWC Performance in Life Nieblas

Number of collectors

24 i-FWC

Size

0,63 m²

surface impact

Price

500-1000 €

Weight

26 kg

Litres collected per m²

642 L/m²/ year

Carbon footprint

50 kg CO_2 eq.

Traditional Performance in Life Nieblas

Planted trees

4625 trees

Survival rate

33%

Cost per surviving tree

136 €

Need of water per trees

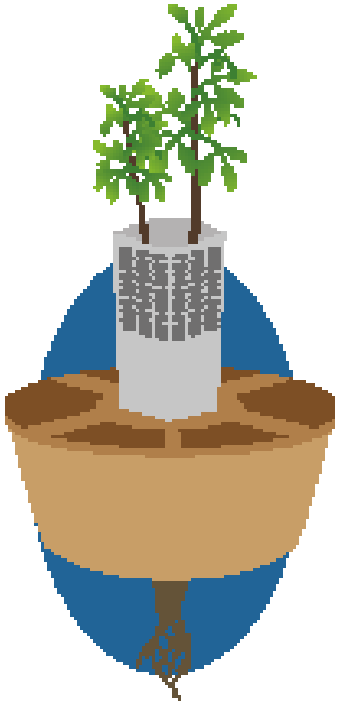
53 L

Carbon footprint / Huella de carbono

FWC **0.42 kg CO_2 eq.**

Tank truck **0.6 kg CO_2 eq.**

Cocoon Performance in Life Nieblas



Planted trees

2215 trees

Survival rate

68%

Cost per surviving tree

109 €

Need of water per trees

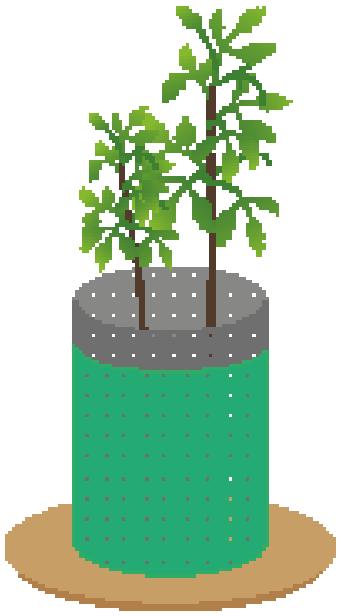
49 L

Carbon footprint / Huella de carbono

FWC **0.31 kg** CO_2 eq.

Tank truck **1.08 kg** CO_2 eq.

i-FWC Performance in Life Nieblas



Planted trees

5964 trees

Survival rate

74%

Cost per surviving tree

59 €

Need of water per trees

30 L

Carbon footprint / Huella de carbono

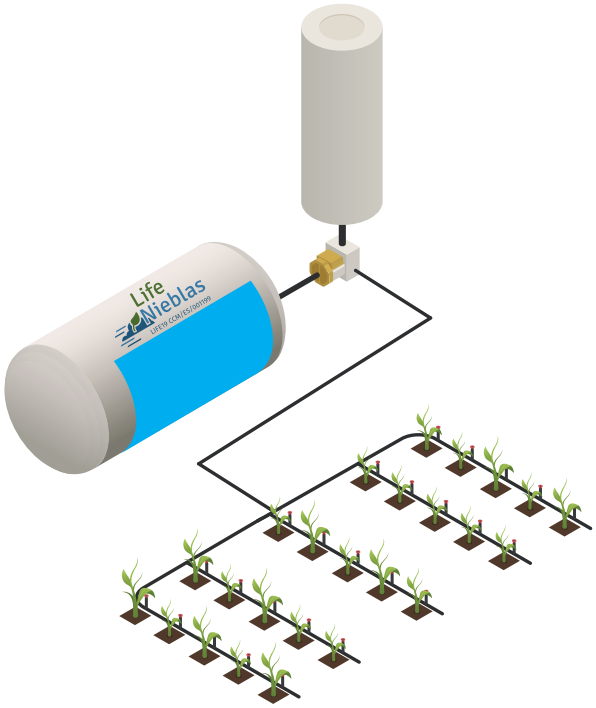
FWC **0.17 kg** CO_2 eq.

Tank truck **0.36 kg** CO_2 eq.

AFDS Performance in Life Nieblas

Planted trees

2132 trees



Survival rate

86%

Cost per surviving tree

115 €

Need of water per trees

75 L

Carbon footprint / Huella de carbono

FWC **0.19 kg** CO_2 eq.

Tank truck **0.4 kg** CO_2 eq.

WHAT'S NEXT...?

The long-term management of the forest following planting is a crucial subject and is essential in order to determine the signs of success or failure of the project. Regular monitoring after planting is essential to assess the success of the project, detect any potential problems. Corrective action can be taken by the reforestation manager if there are signs of degradation or damage risks, such as a second reforestation, watering management, etc. The monitoring also allows us to take lessons from existing plantations and apply them in new planting projects (adaptive restoration).

Ideally, this monitoring should be carried out:

- For at least 4 years, so that a plant growth assessment can be made at the end of those 4 years. However, monitoring would be ideal if it were carried out at midterm (10 years) and long term (20 years).

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Technical Manual

Technical details to replicate the different reforestation and fog collection methodologies tested at Life Nieblas

Climate change is a major challenge facing our planet, with undeniable consequences for water availability. Reduced rainfall and increased periods of drought are having a significant impact on many regions, compromising the water resources essential to human life, agriculture and the environment.

In this technical manual, we look at the challenges posed by the decline in water availability due to climate change and droughts. We will present **two solutions** that can help tackle these issues in a sustainable way.

WATER ACCESS: The first solution concerns regions where access to water is extremely difficult or even impossible. In such places, we will be presenting the use of fog collectors, an emerging technology that collects and uses atmospheric moisture. This ingenious approach offers a valuable alternative to traditional water sources, harnessing the natural resources available to meet water needs.

HABITAT RESTORATION: The second solution is aimed at deteriorated areas where water soil infiltration is reduced, leading to increased runoffs, erosion related issues and loss of biodiversity. To remedy this situation, we propose reforestation actions. By restoring plant cover, we encourage the natural water cycle and reduce the risk of erosion, allowing nature to regain its balance and water to play its essential role in ecosystems.

This technical manual aims to present these two solutions in detail, providing practical information, application examples and recommendations for their implementation. By adopting these innovative approaches, we contribute to the preservation and sustainable management of water resources, adapting to the challenges posed by climate change and promoting a more resilient future for our planet.